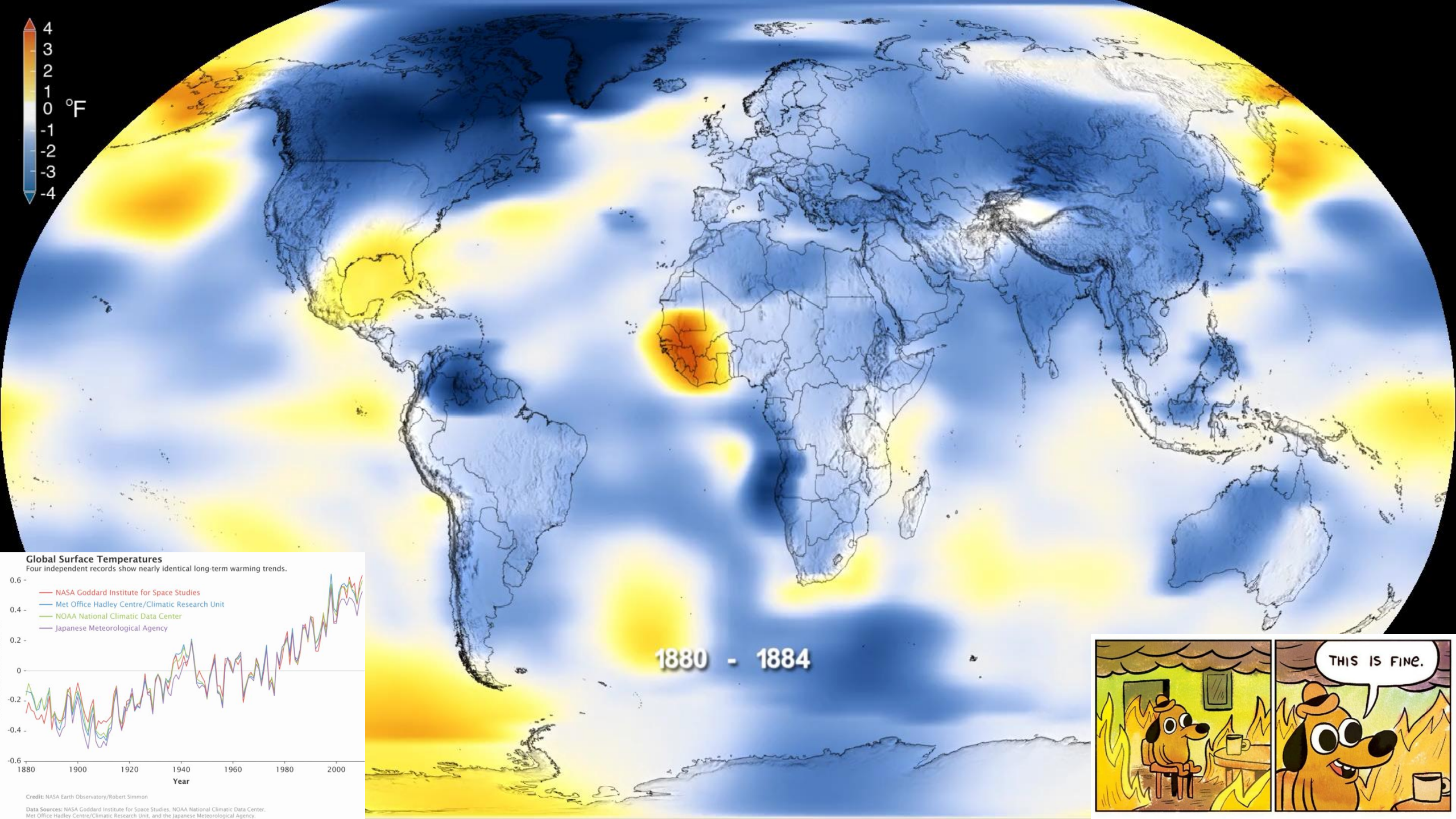

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°F
1880 - 1884

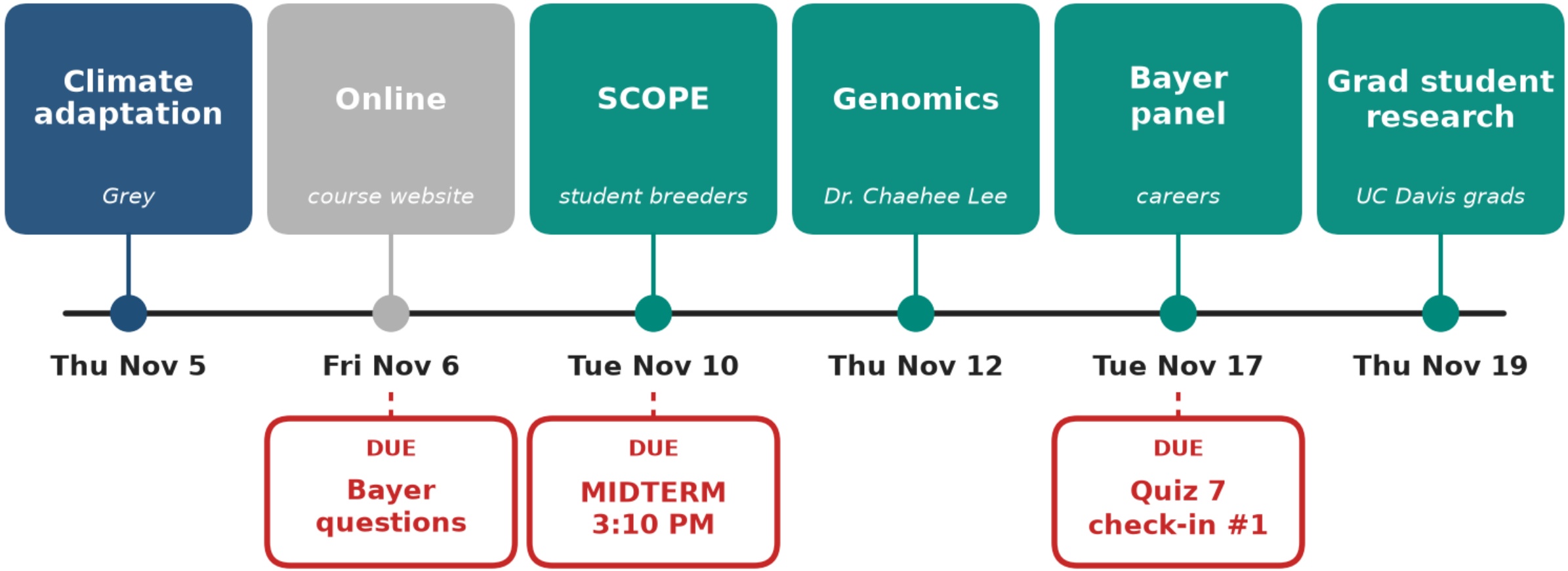
Global Surface Temperatures
Four independent records show nearly identical long-term warming trends.
NASA Goddard Institute for Space Studies
Met Office Hadley Centre/Climatic Research Unit
NOAA National Climatic Data Center
Japanese Meteorological Agency
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
1880
1900
1920
1940
1960
1980
2000
Year
Credit: NASA Earth Observatory/Robert Simmon
Data Sources: NASA Goddard Institute for Space Studies, NOAA National Climatic Data Center, Met Office Hadley Centre/Climatic Research Unit, and the Japanese Meteorological Agency.

THIS IS FINE.



A brief history of climate adaptation research in plants: 1700 – 2100

Coming up: guests + the midterm



Tuesday recap

Kahoot-style warm-up

9 quick questions
Not graded
Fast and correct wins.

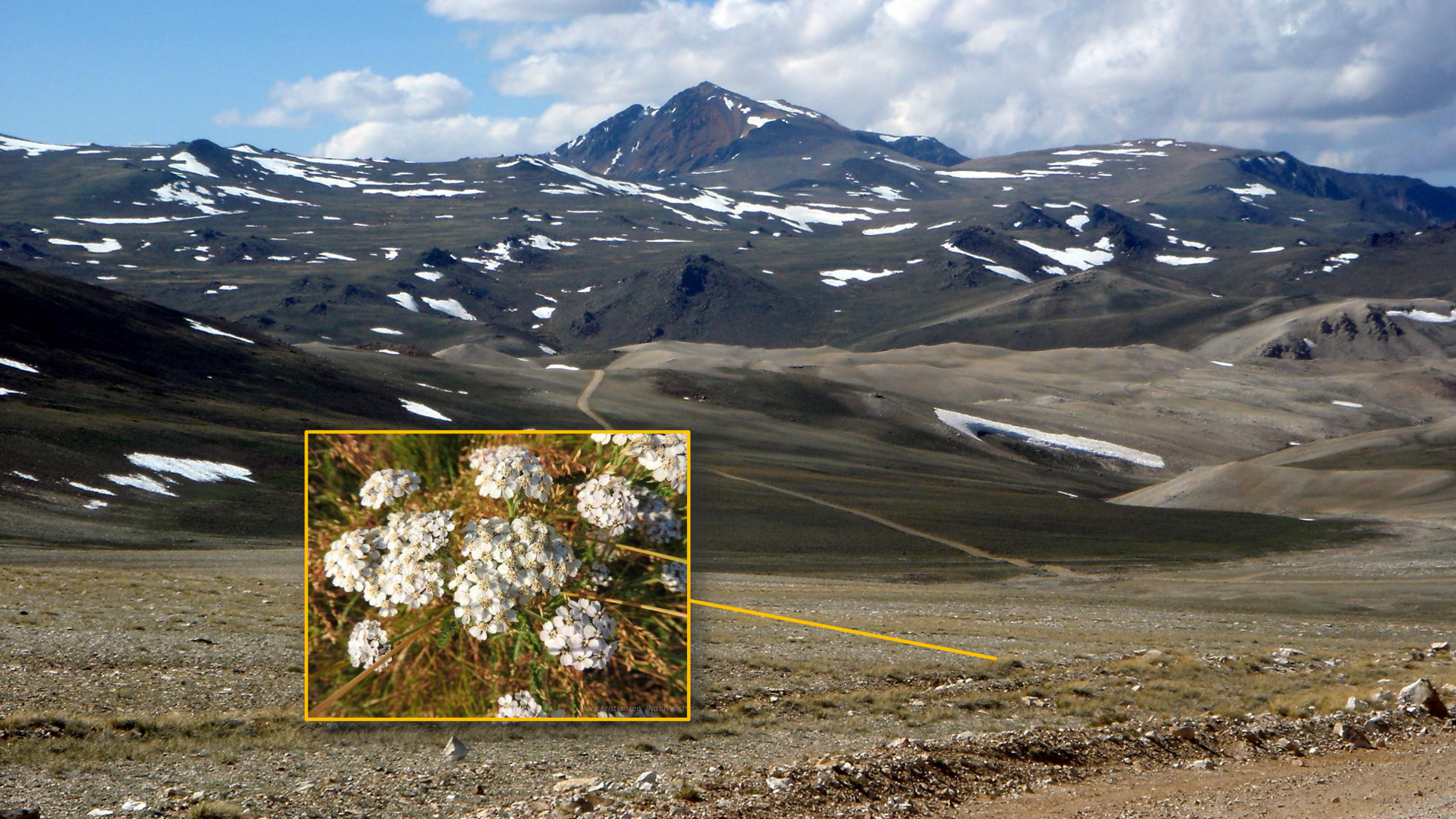


Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-ca-recap](greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-ca-recap)

[host screen]





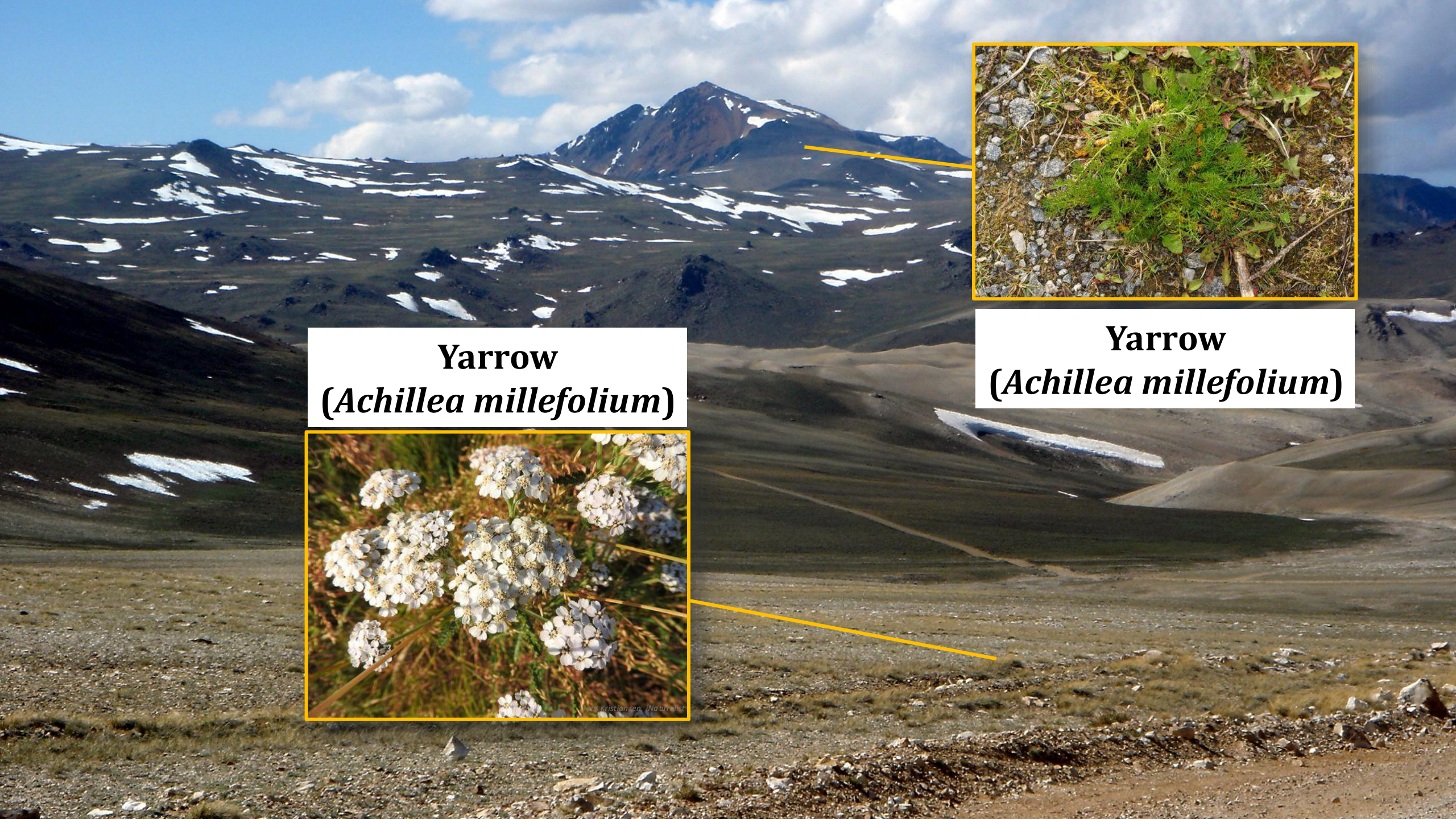
Yarrow
(*Achillea millefolium*)



Yarrow
(*Achillea millefolium*)






Yarrow
(*Achillea millefolium*)
Yarrow
(*Achillea millefolium*)



Is this adaptive genetic variation?



Live poll

Multiple choice

Yarrow plants look different from place to place. Is this adaptive genetic variation?

Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ca-yarrow](greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ca-yarrow)

[host screen]

Today

- Why does it matter if plants are adapted to climate?

Today

- Why does it matter if plants are adapted to climate?
- Early ideas

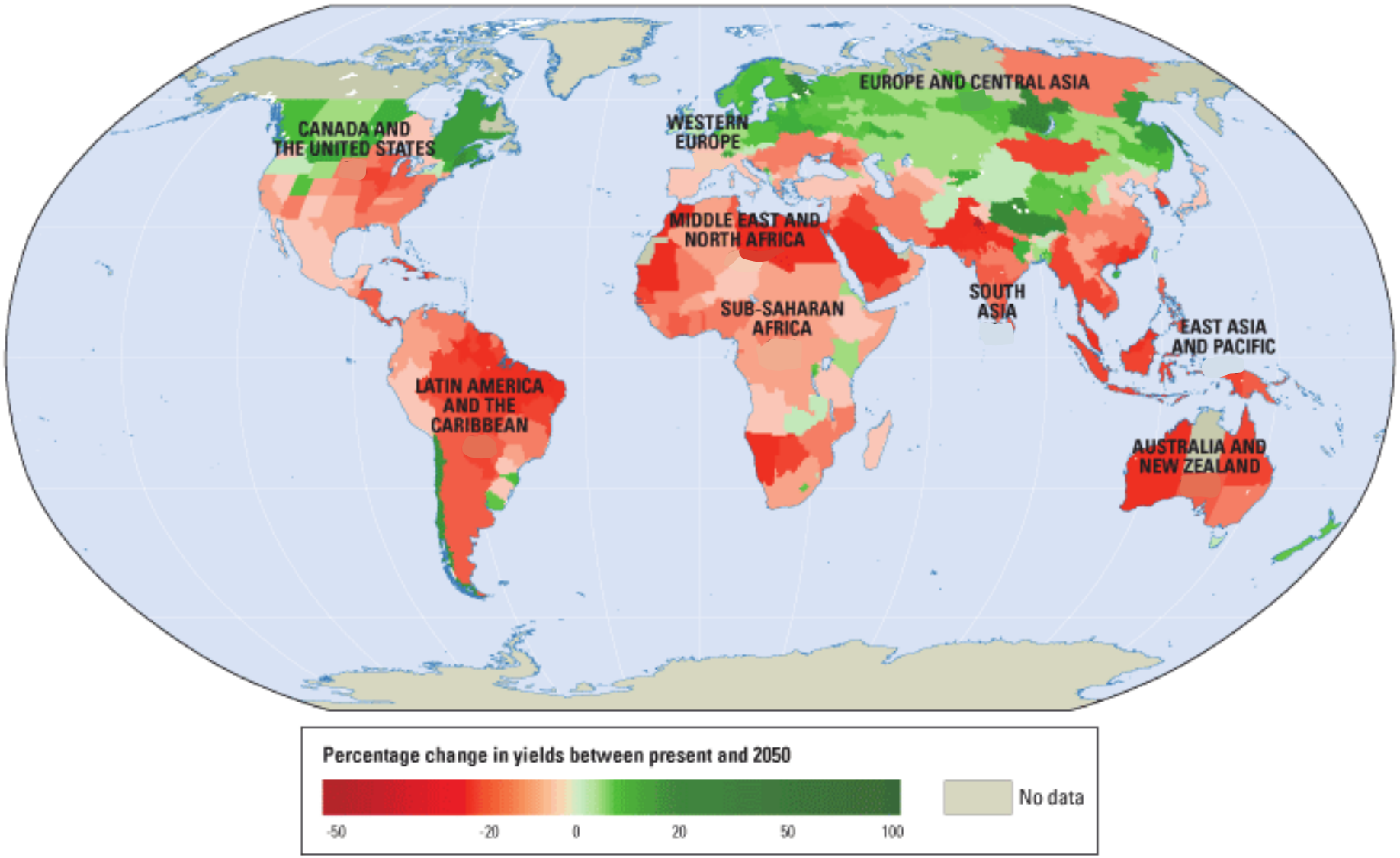
Today

- Why does it matter if plants are adapted to climate?
- Early ideas
- Breakthrough experiments

Today

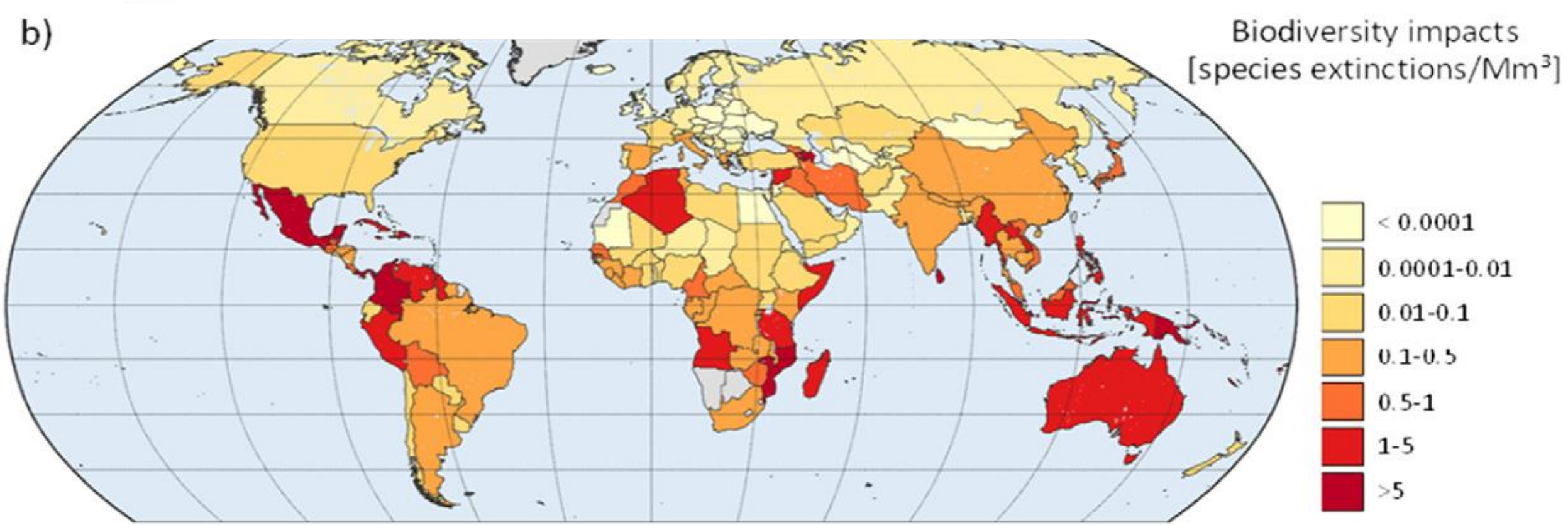
- Why does it matter if plants are adapted to climate?
- Early ideas
- Breakthrough experiments
- Climate change

Climate change threatens crop yields…



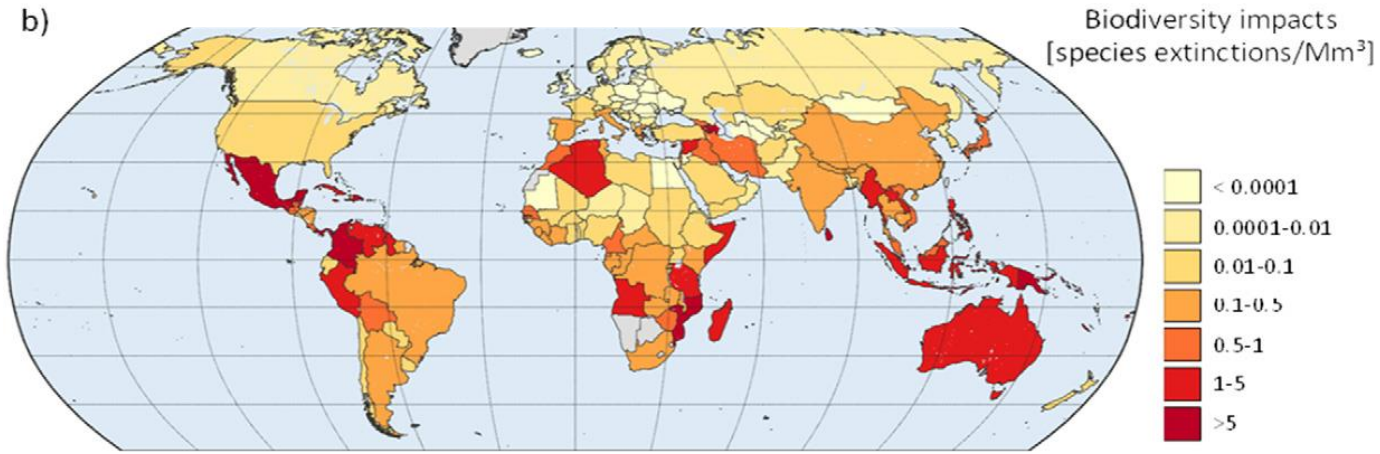
Sources: Müller *et al.* 2008, World Bank 2008

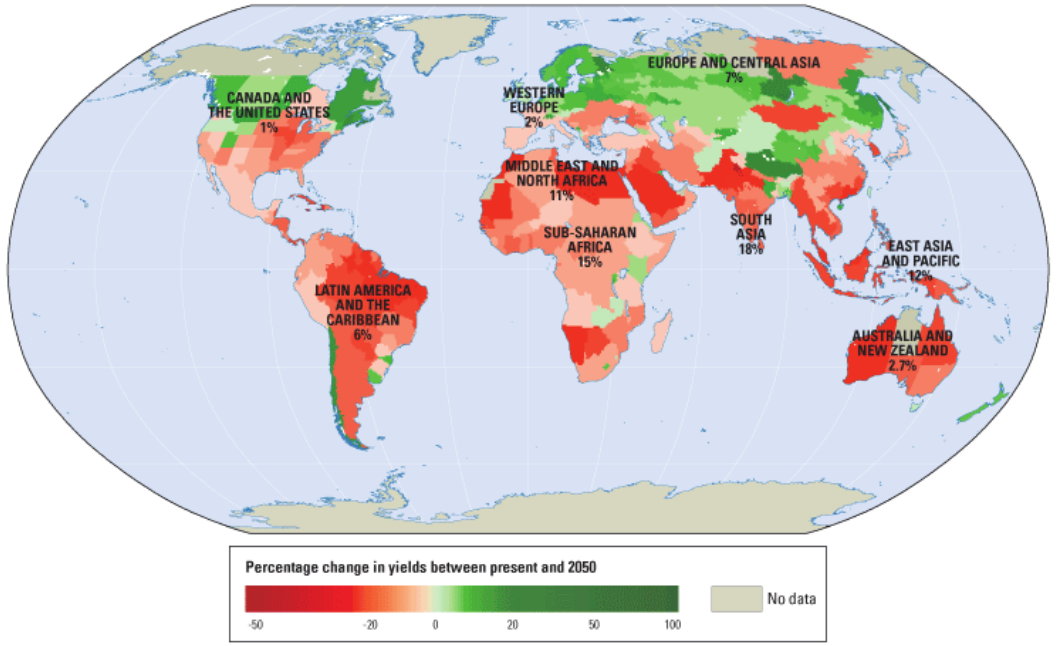
… and biodiversity



Chaudhary *et al.* 2017

Plants are often locally adapted to contemporary climates…

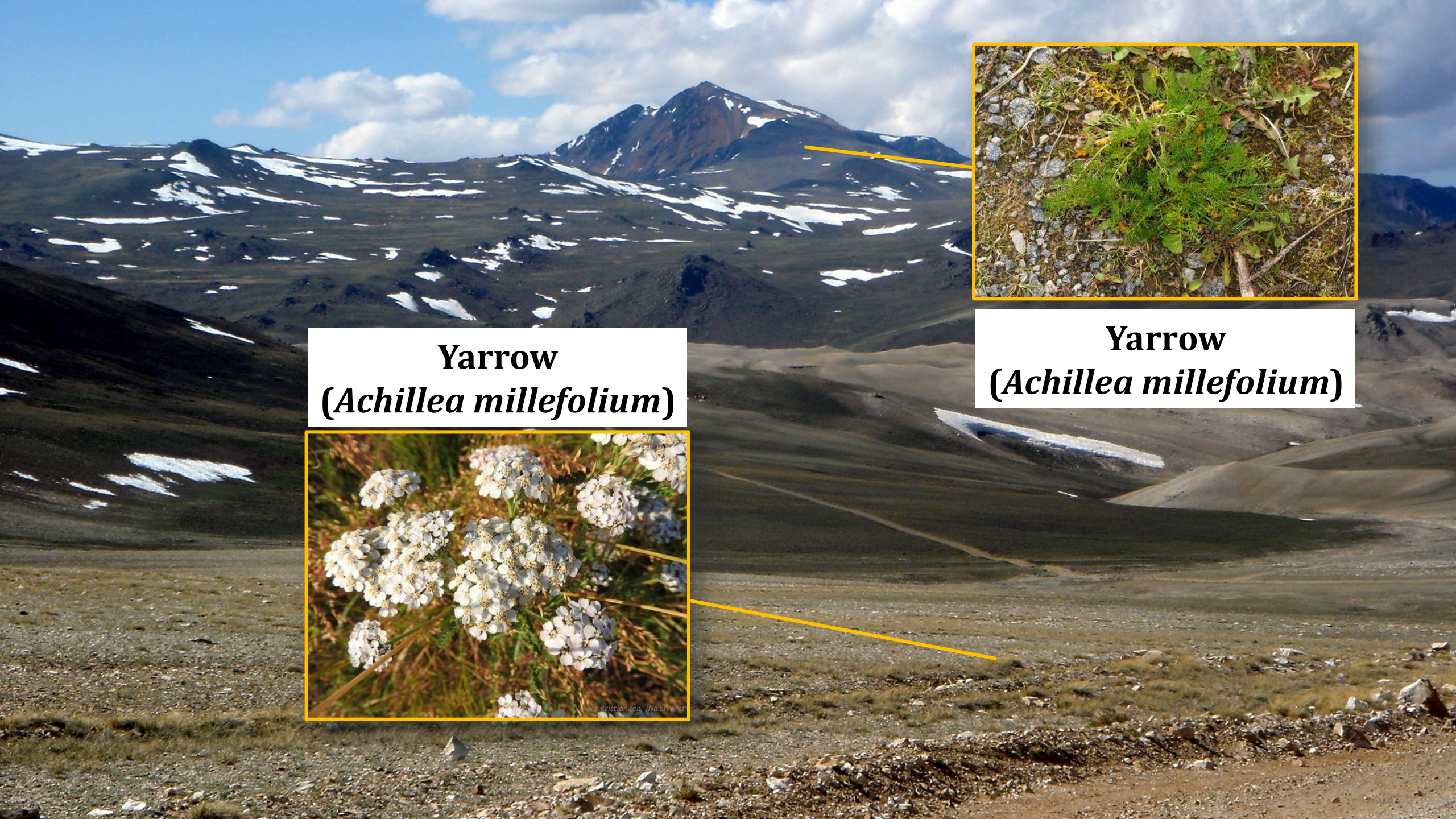




Sources: Müller *et al.* 2008, World Bank 2008

But how do we know this?

Chaudhary *et al.* 2017


Yarrow
(*Achillea millefolium*)
Yarrow
(*Achillea millefolium*)

Geographical variation within species has been documented since at least 1700s





Carl Linnaeus 1700's

Geographical variation within species has been documented since at least 1700s



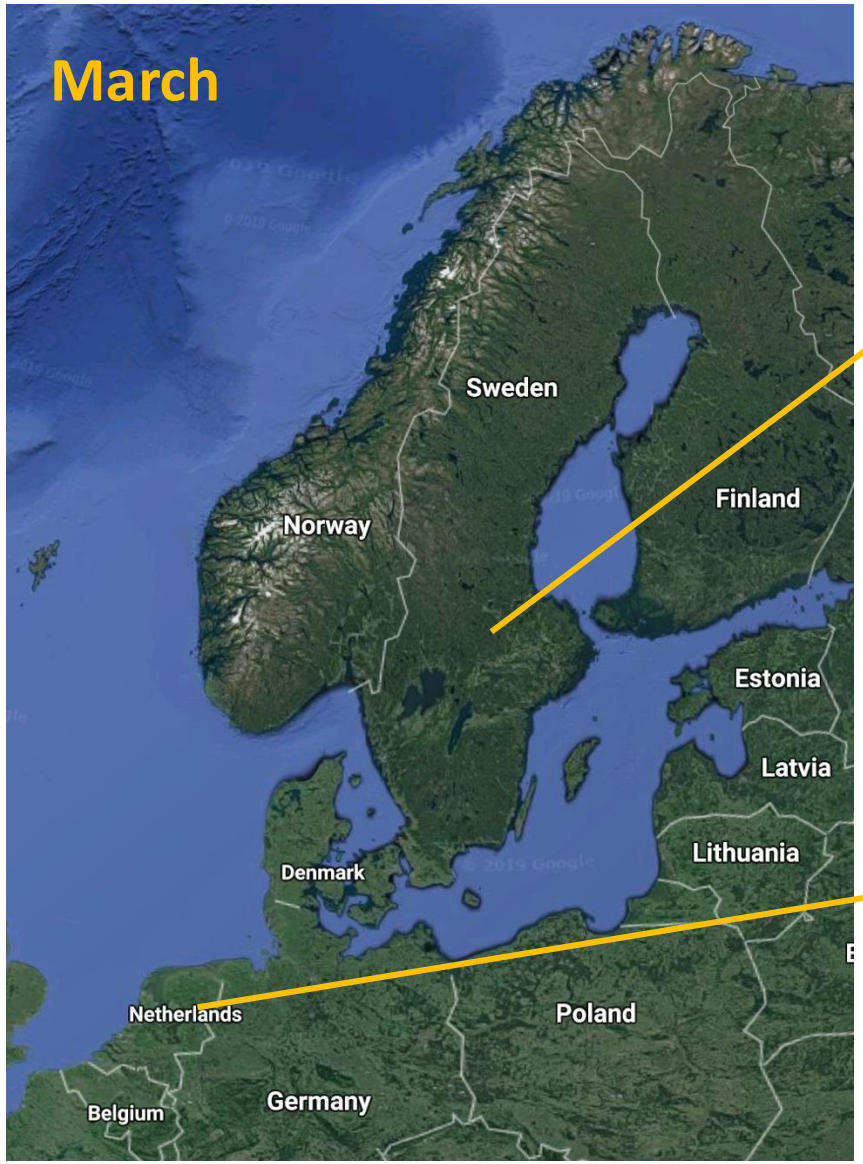
Marsh-marigold (*Caltha palustris*)





Carl Linnaeus 1700's

Geographical variation within species has been documented since at least 1700s



Marsh-marigold (*Caltha palustris*)

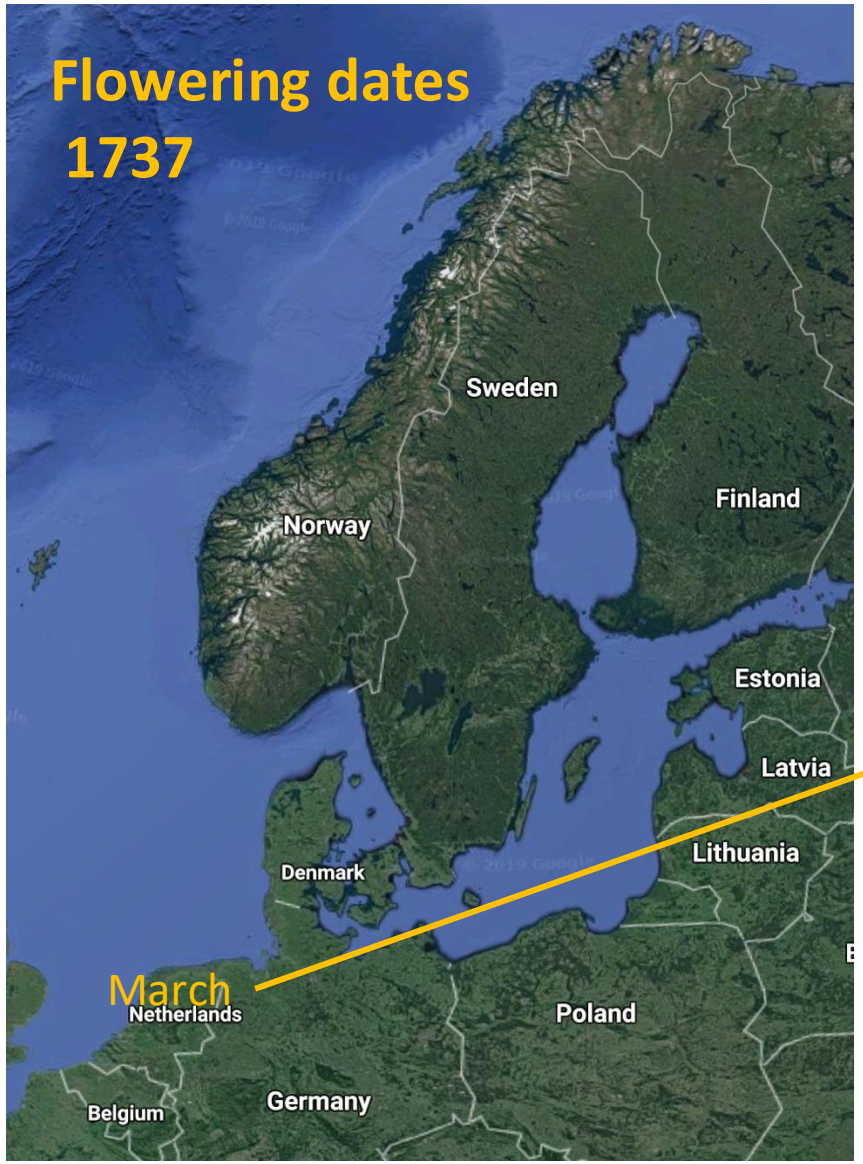






Carl Linnaeus 1700's

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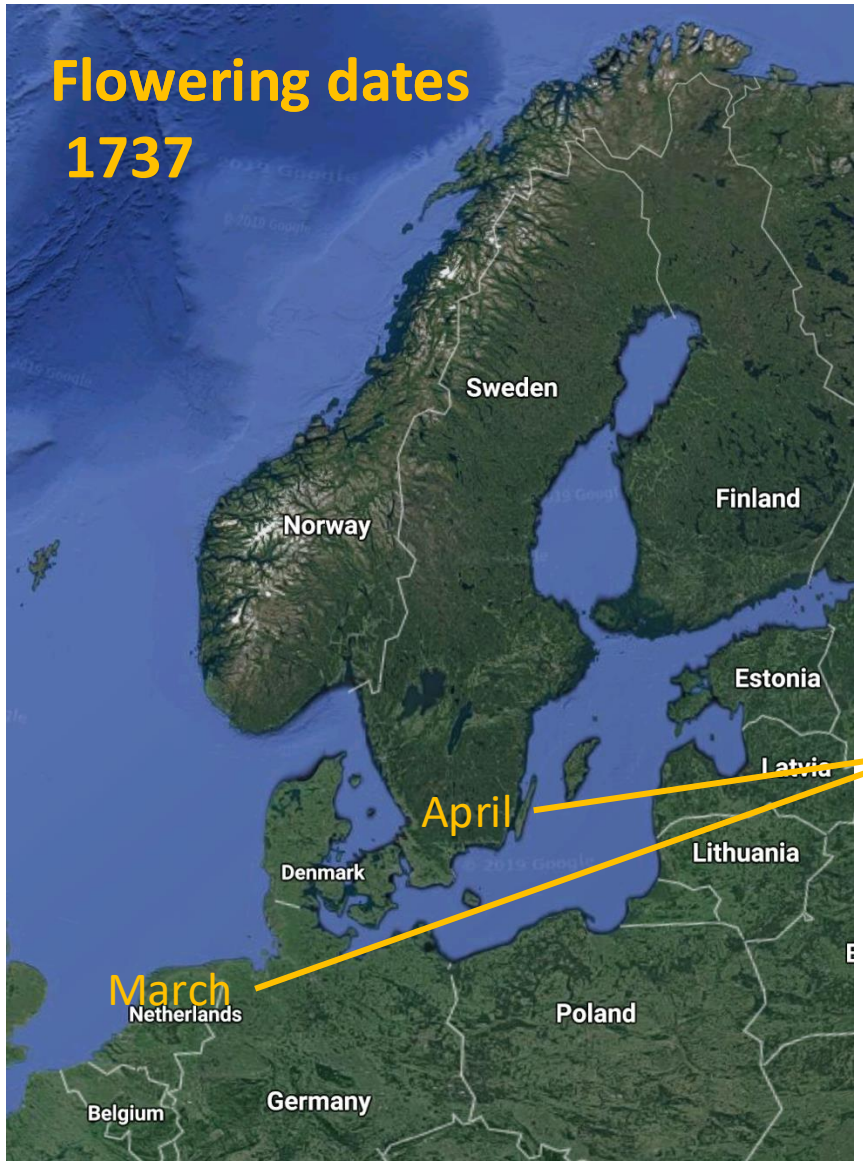
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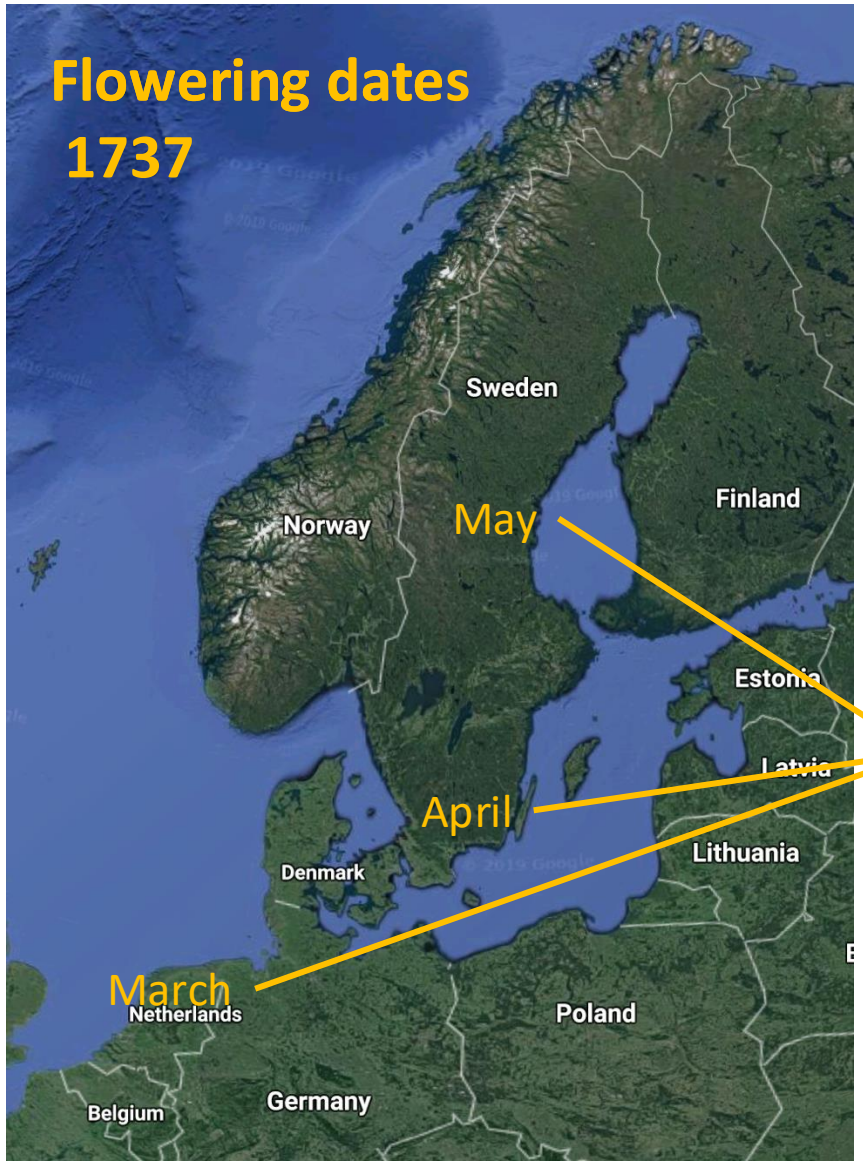
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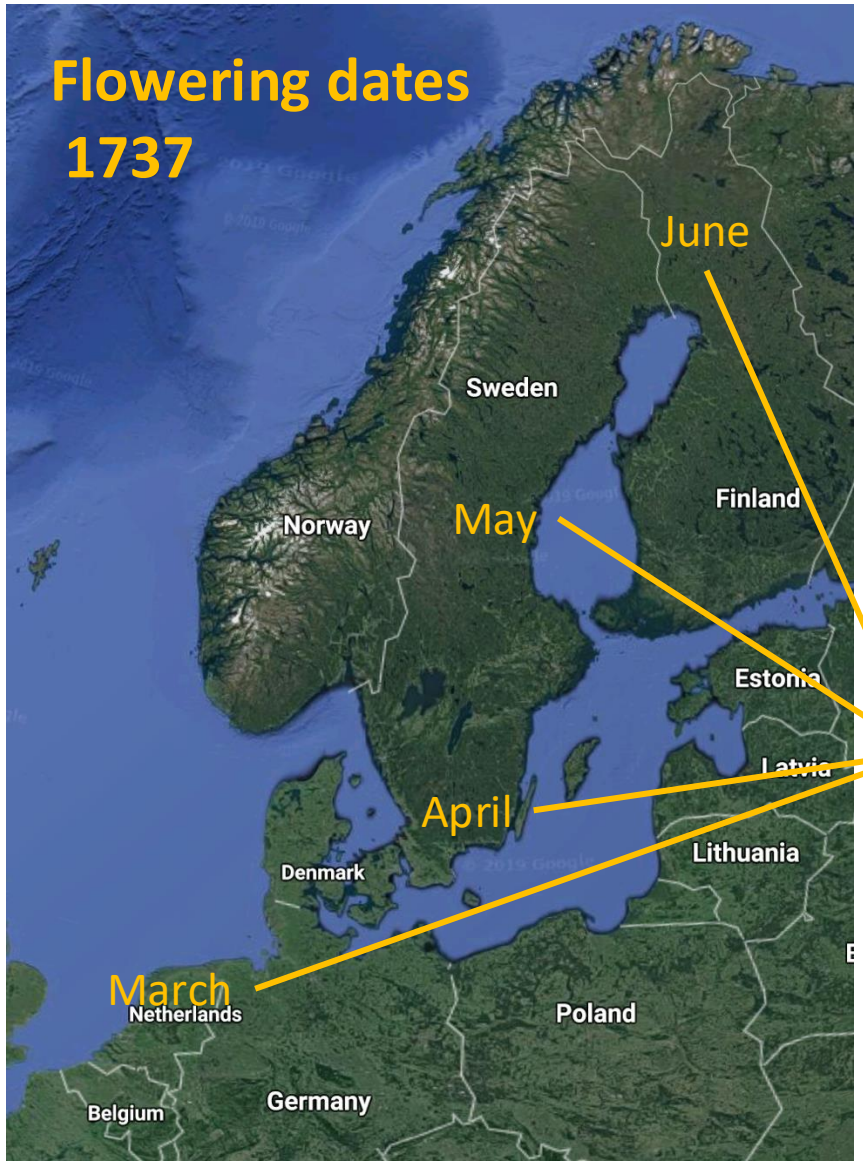
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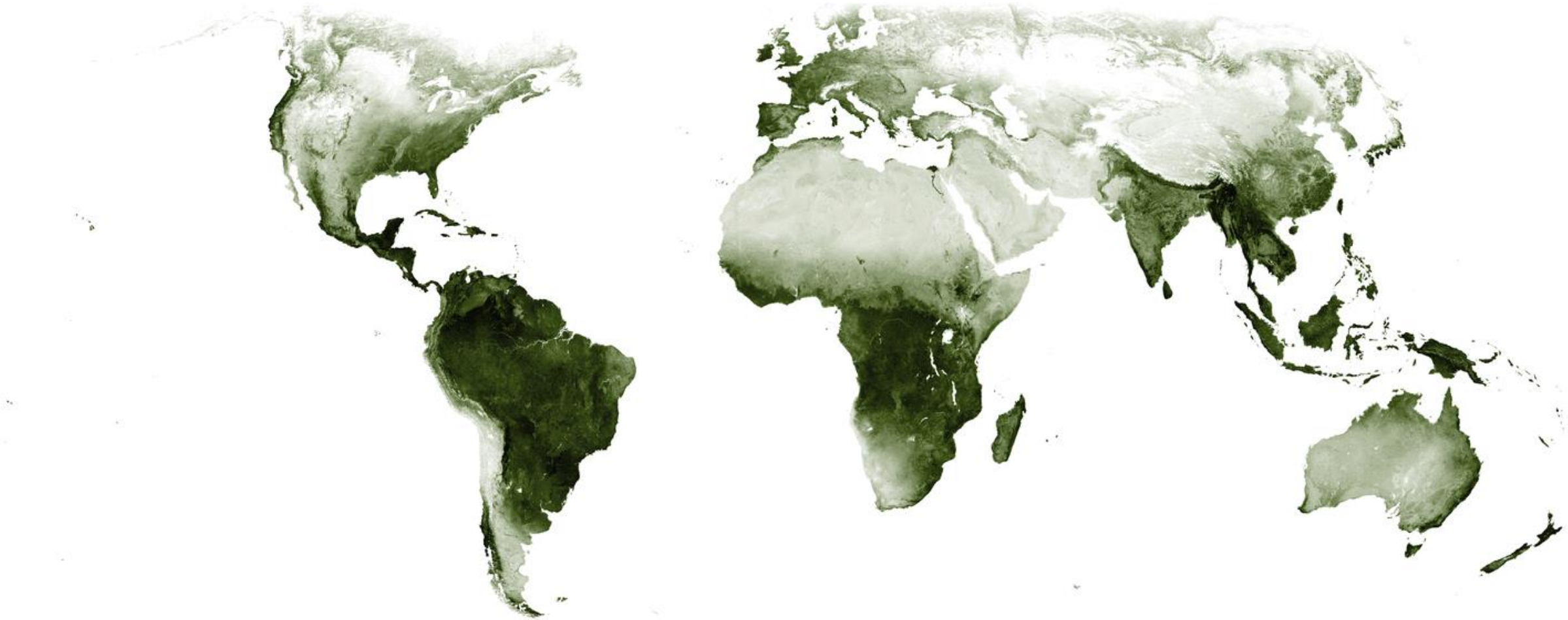
Marsh-marigold (*Caltha palustris*)



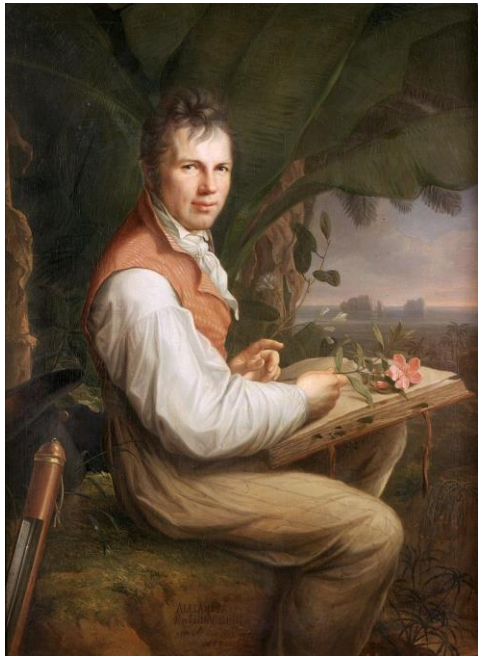


Carl Linnaeus 1700's

Annual vegetation cycle



Plant differences according to habitat



Alexander von Humboldt
(late 1700's)



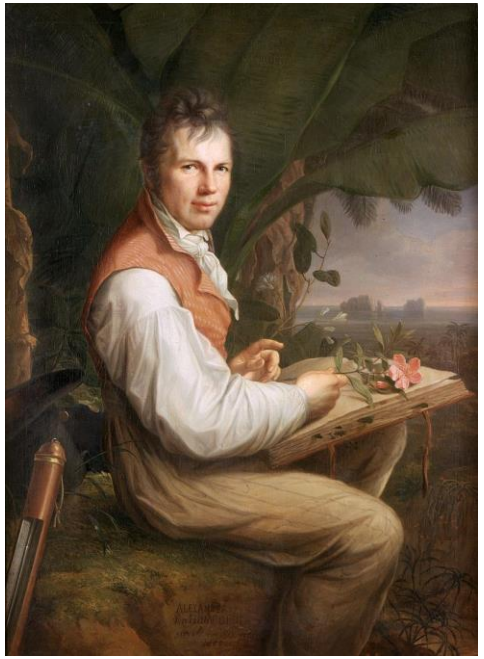
Plant differences according to habitat



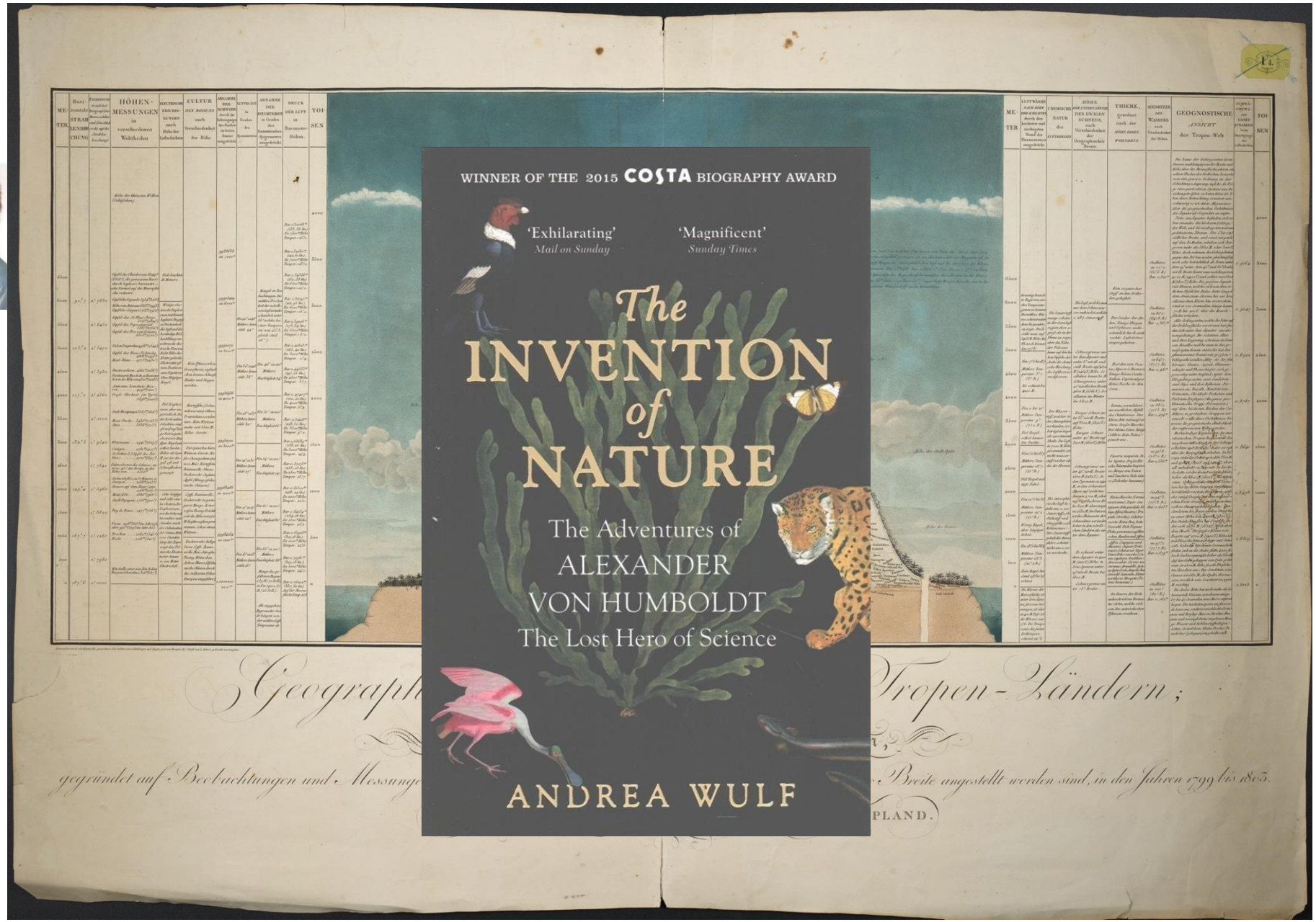




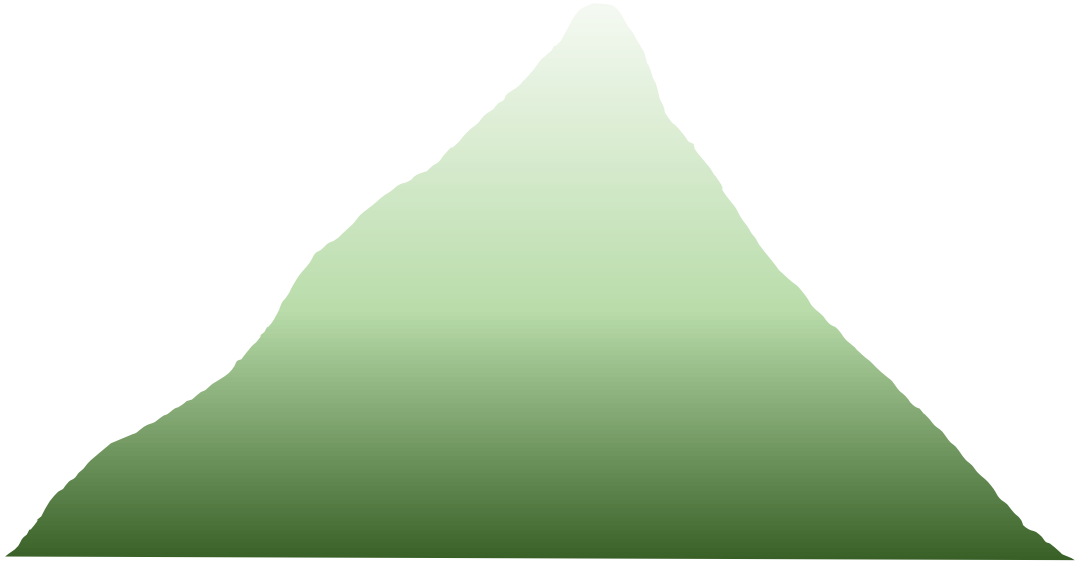




Alexander von Humboldt
(late 1700's)



Early theory: geographical variation caused by environment





Adolphe Quetelet
(early 1800's)



Florence Nightingale
(early 1800's)

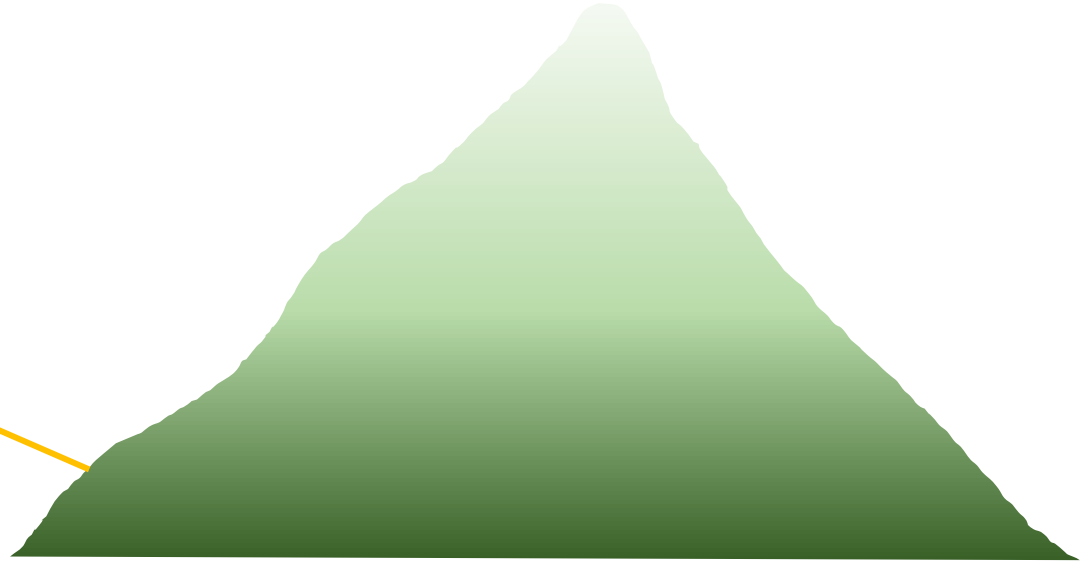
Early theory: geographical variation caused by environment



Adolphe Quetelet
(early 1800's)

Lilac (*Syringa vulgaris*)



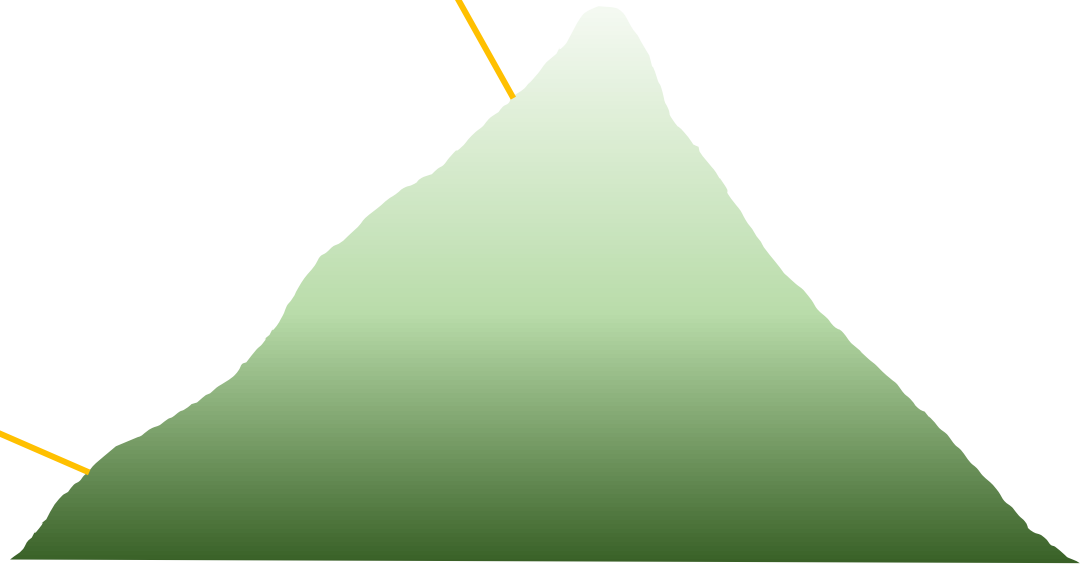




Florence Nightingale
(early 1800's)

Early theory: geographical variation caused by environment





Lilac (*Syringa vulgaris*)





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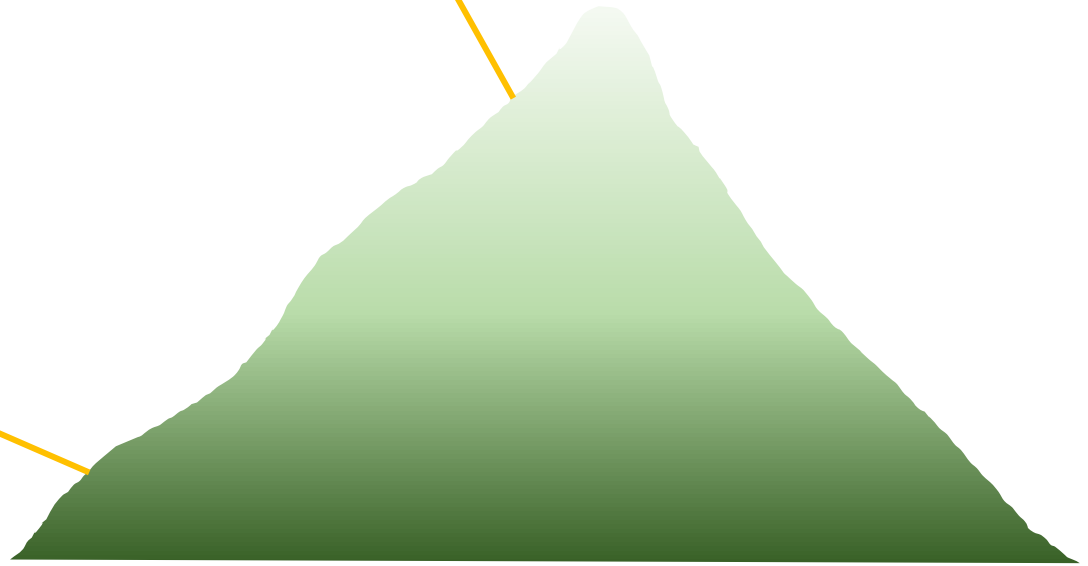




Adolphe Quetelet
(early 1800's)

Lilac (*Syringa vulgaris*)







Florence Nightingale
(early 1800's)

"(Flowering occurs when) the sum of the squares of the mean daily temperature, counted from the end of the frosts, equals 4246 degrees C"

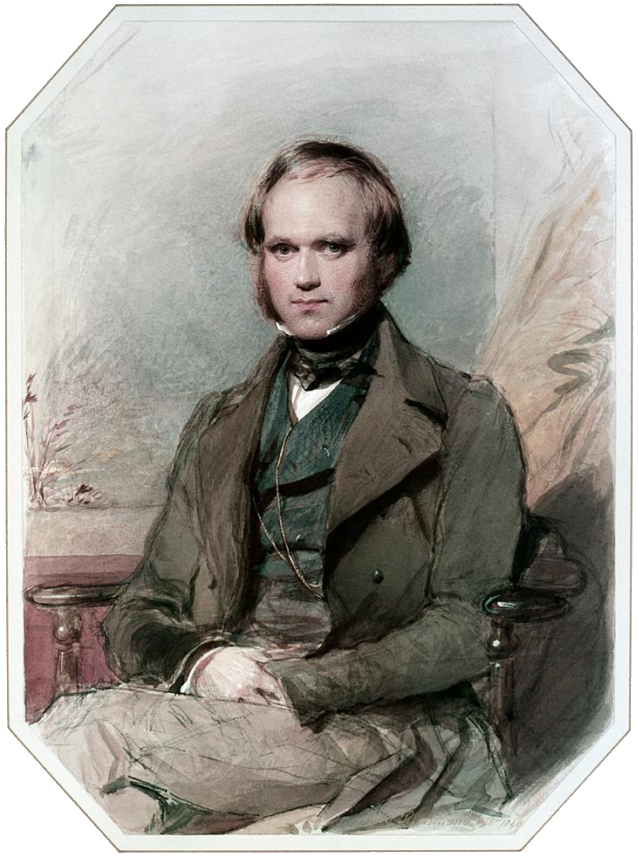
(Cook 1913, vol. 1, pp. 429–30)

Yarrow (*Achillea millefolium*)

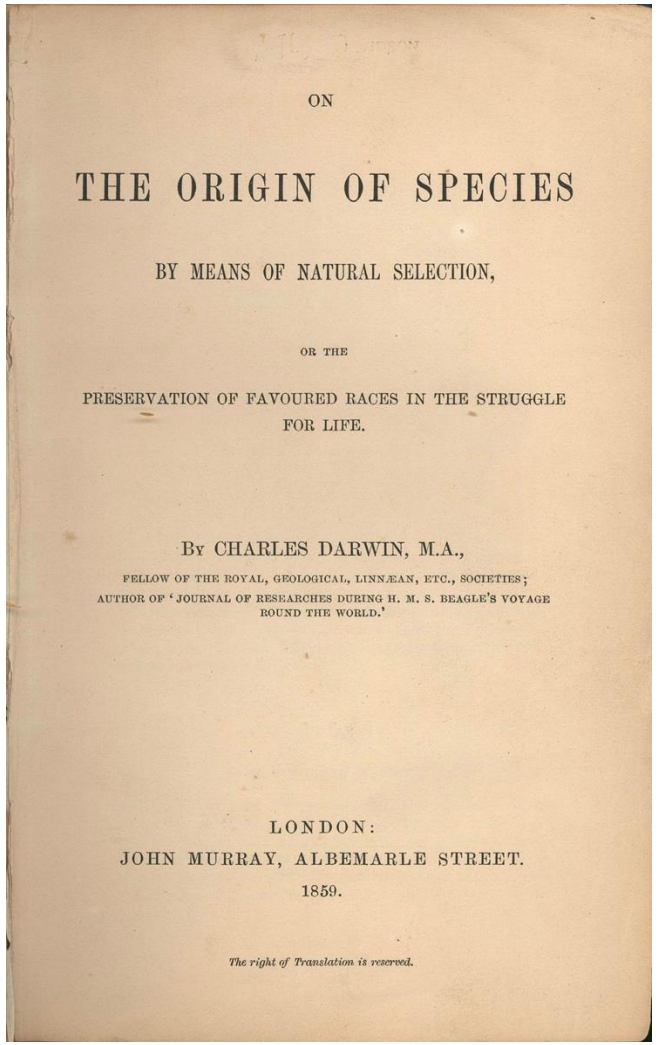




Theory of natural selection: geographical variation could be heritable (genetic) and explained by adaptation.



Charles Darwin



ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE FOR LIFE.

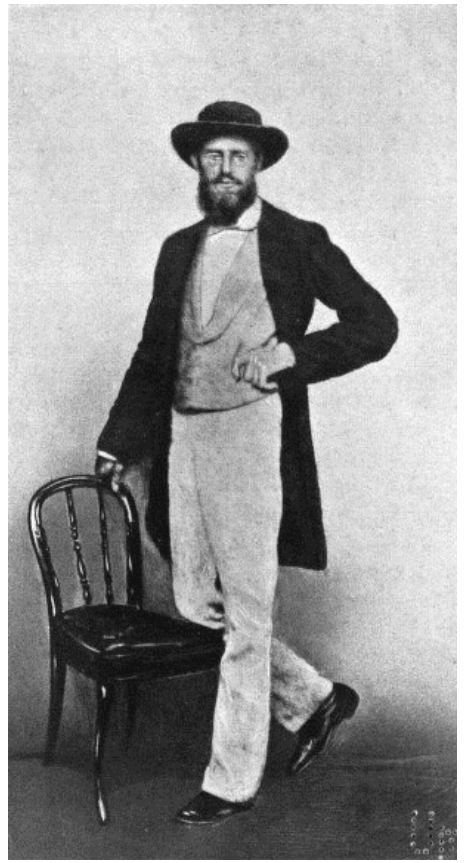
By CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE ROUND THE WORLD.'

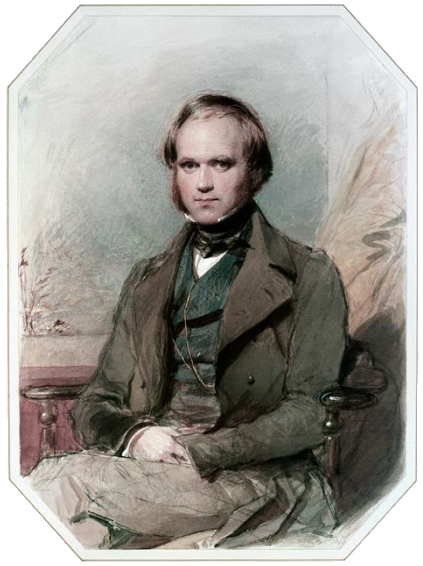
LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

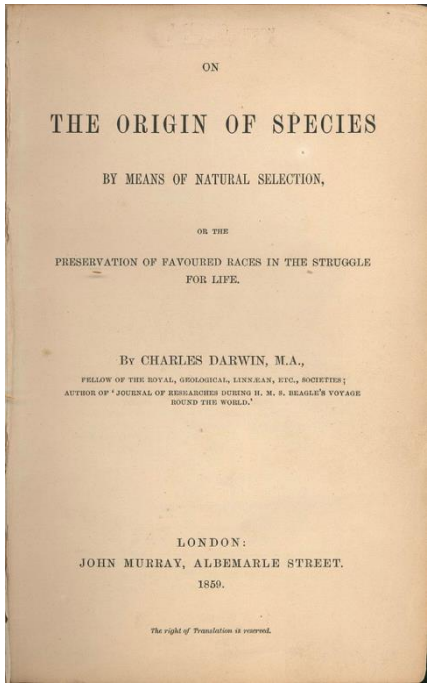
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Mid 1800's



Alfred Russel Wallace





ON

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Chap. V. *Acclimatisation.* 112

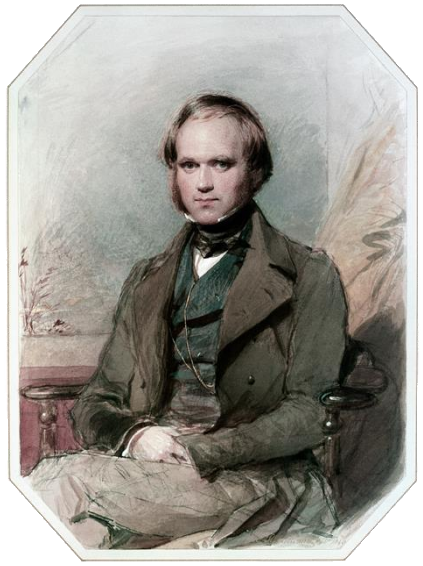
have ranged over both continents, and then have become extinct, excepting in their present secluded abodes. Far from feeling surprise that some of the cave-animals should be very anomalous, as Agassiz has remarked in regard to the blind fish, the Amblyopsis, and as is the case with the blind Proteus, with reference to the reptiles of Europe, I am only surprised that more wrecks of ancient life have not been preserved, owing to the less severe competition to which the scanty inhabitants of these dark abodes will have been exposed.

Acclimatisation.

Habit is hereditary with plants, as in the period of flowering, in the time of sleep, in the amount of rain requisite for seeds to germinate, &c., and this leads me to say a few words on acclimatisation. As it is extremely common for distinct species belonging to the same genus to inhabit hot and cold countries, if it be true that all the species of the same genus are descended from a single parent-form, acclimatisation must be readily effected during a long course of descent. It is notorious that each species is adapted to the climate of its own home: species from an arctic or even from a temperate region cannot endure a tropical climate, or conversely. So again, many succulent plants cannot endure a damp climate. But the degree of adaptation of species to the climates under which they

in historical times, their range from warmer to cooler latitudes.

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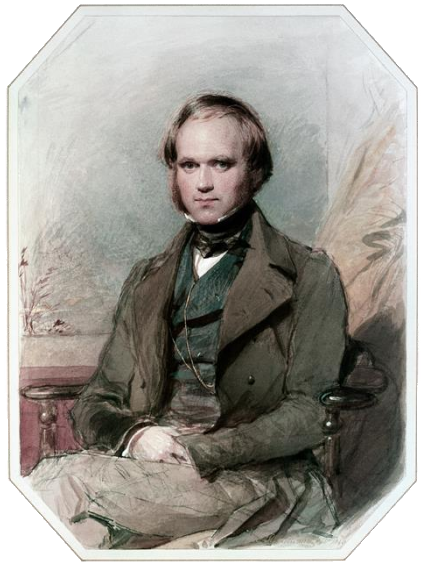
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Seed collected… from the same species growing at different heights on the Himalayas, were found to possess... different constitutional powers of resisting cold



ON

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1st edition

edits

Last edition

(like the elephant on both continents) habitations. since any

in the time of sleep, probably

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and as I believe very

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Himalaya, trees to possess

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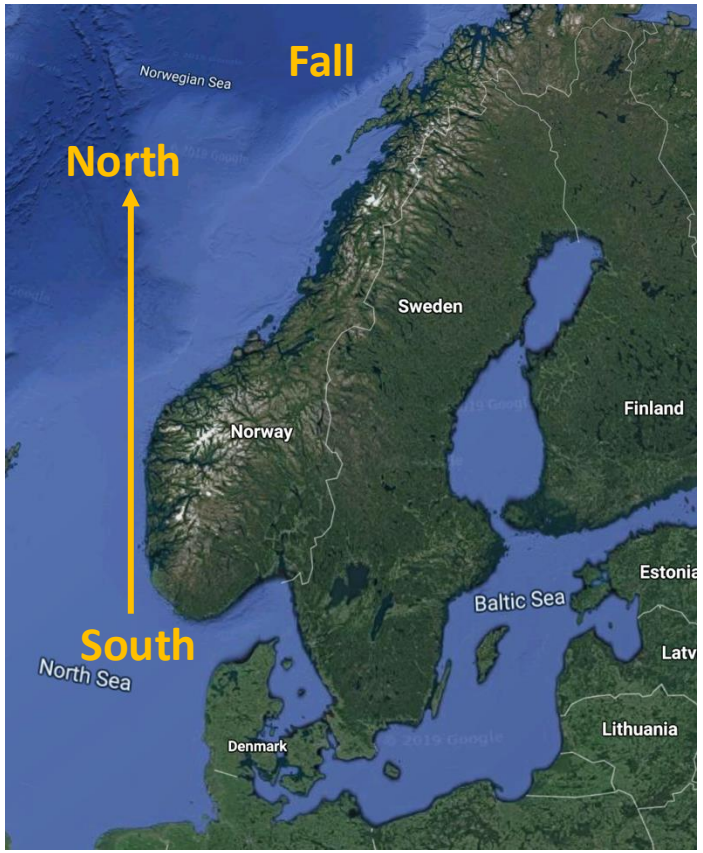
CHAP. V.

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A new generation of scientists sees geographical variation differently

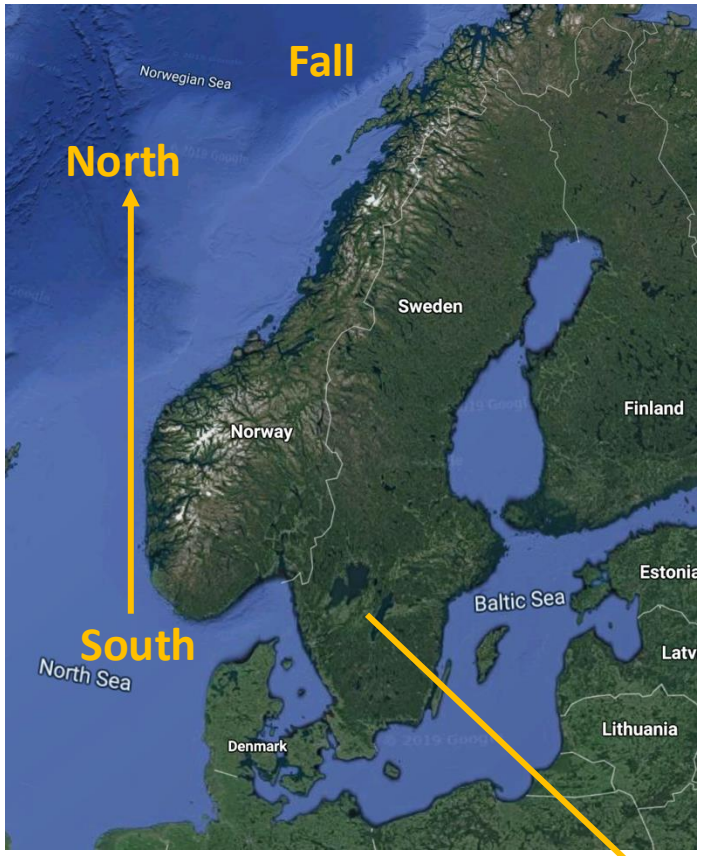


Betula pendula



Göte Turesson early 20th century

A new generation of scientists sees geographical variation differently



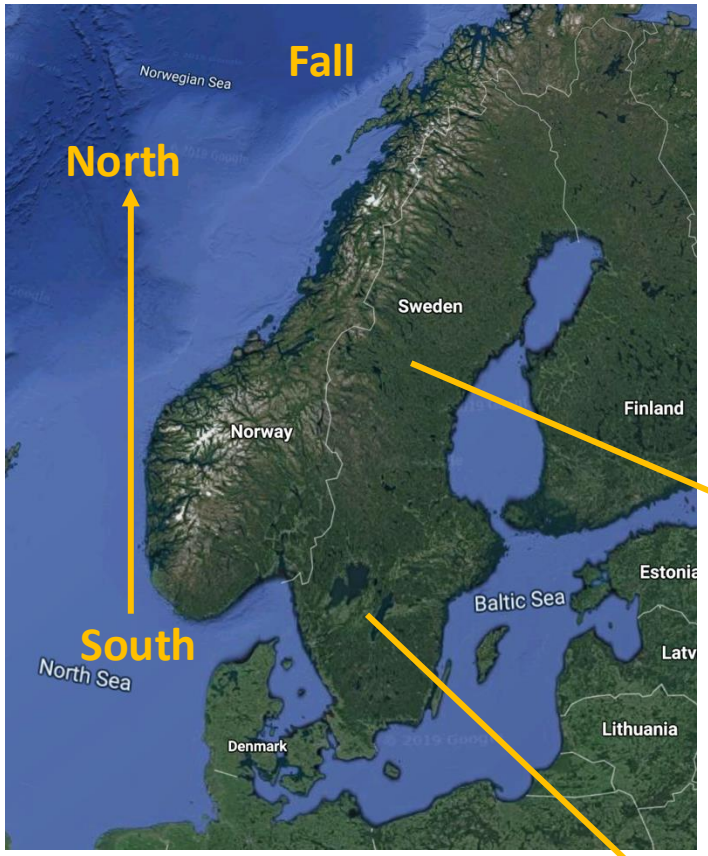
Betula pendula





Göte Turesson early 20th century

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Betula pendula

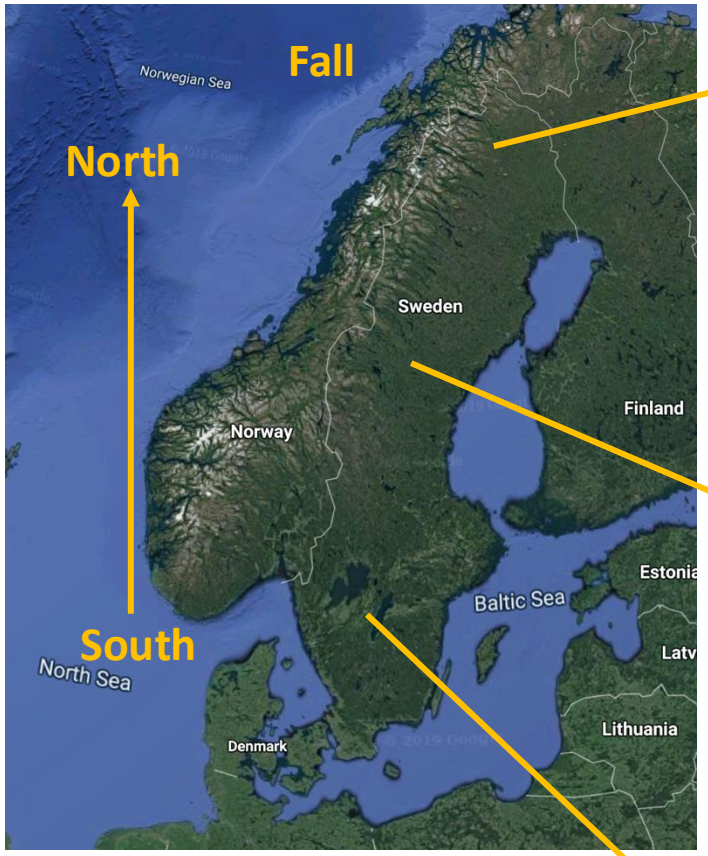






Göte Turesson early 20th century

A new generation of scientists sees geographical variation differently



Betula pendula







Genetic variation?



Göte Turesson early 20th century

“Common garden” experiments reveal genetics vs environment as cause of variation

- **Hypothesis:** phenotypic variation is caused by genetics



“Common garden” experiments reveal genetics vs environment as cause of variation

- **Hypothesis:** phenotypic variation is caused by genetics
- **Prediction:** plants will look different (have different phenotypes) when controlling for environment (grown in the same environment)



“Common garden” experiments reveal genetics vs environment as cause of variation

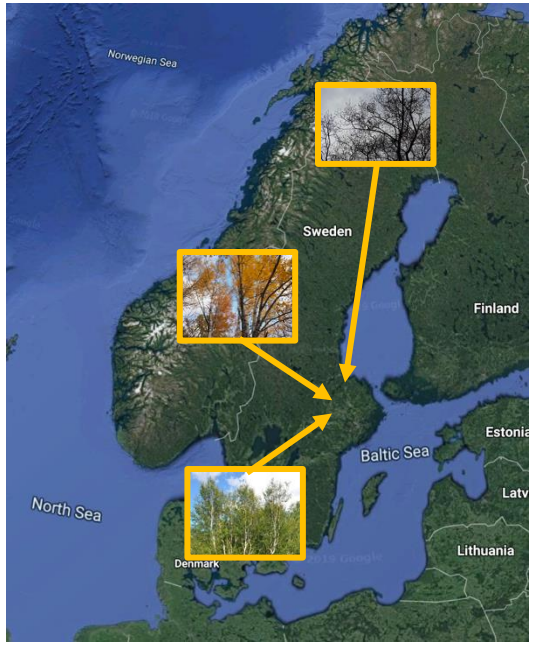
- **Hypothesis:** phenotypic variation is caused by genetics
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- Collect seeds or clones from different populations





“Common garden” experiments reveal genetics vs environment as cause of variation

- **Hypothesis:** phenotypic variation is caused by genetics
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- Collect seeds or clones from different populations
- Grow them in common environment





“Common garden” experiments reveal genetics vs environment as cause of variation

- **Hypothesis:** phenotypic variation is caused by genetics
- **Prediction:** plants will look different (have different phenotypes) when controlling for environment (grown in the same environment)
- Collect seeds or clones from different populations
- Grow them in common environment
- Observe trait differences and similarities between genotypes

Genetic variation!







Turesson's common garden experiments reveal genetic variation in relation to climate

Silene maritima





Genetic variation!



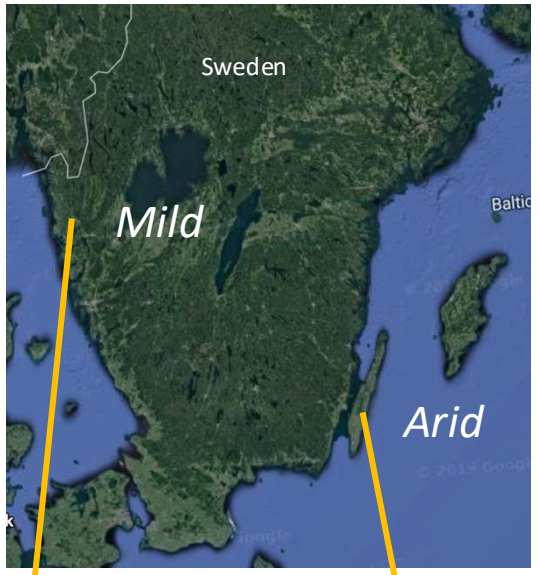
Turesson's common garden experiments reveal genetic variation in relation to climate

Silene maritima





Allitum schoenoprasum

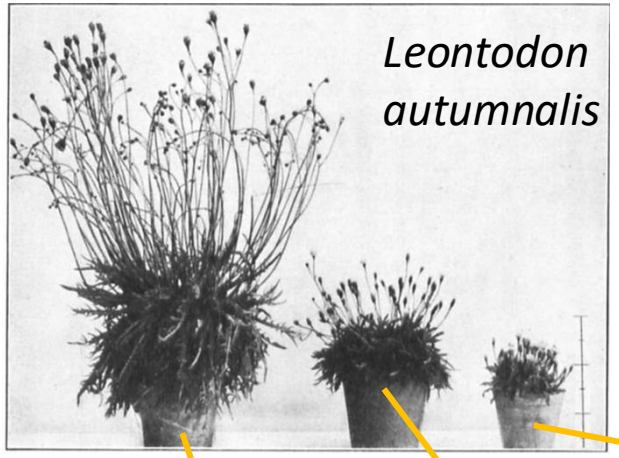


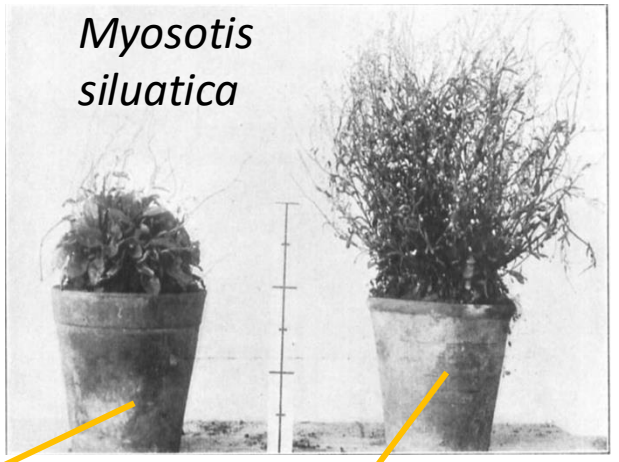


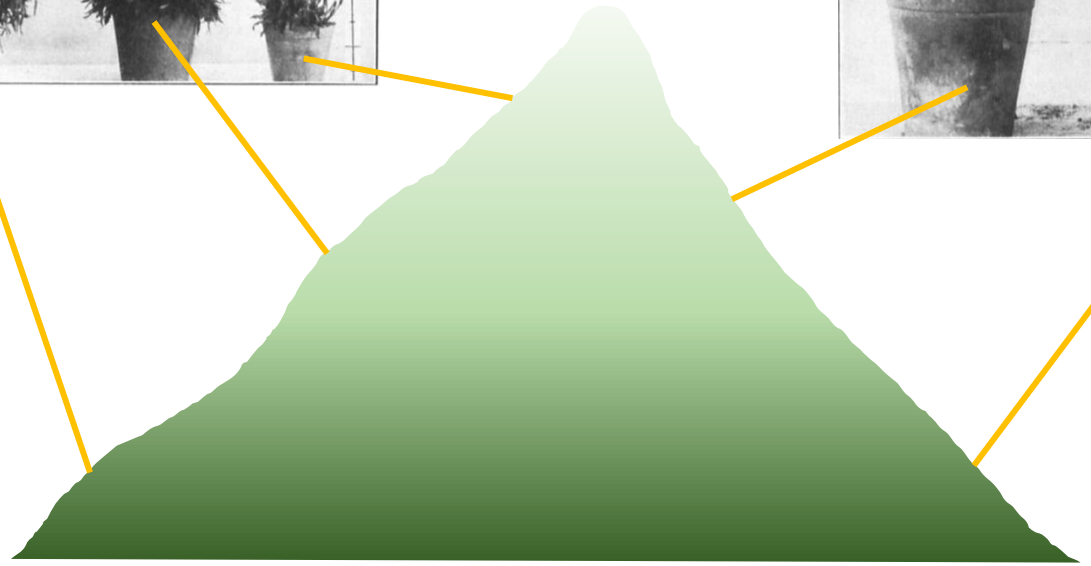
Genetic variation!



Turesson's common garden experiments reveal genetic variation in relation to climate



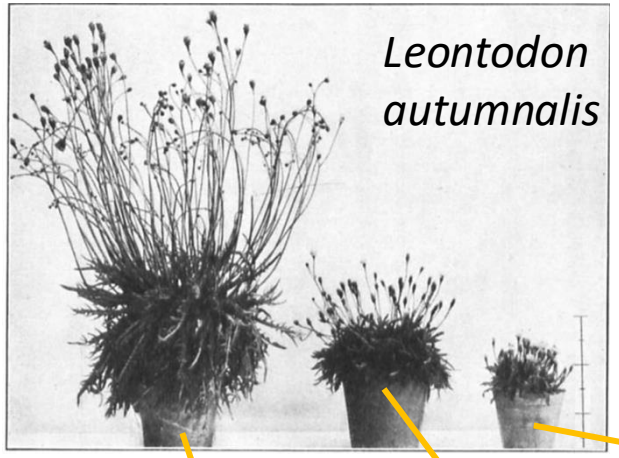


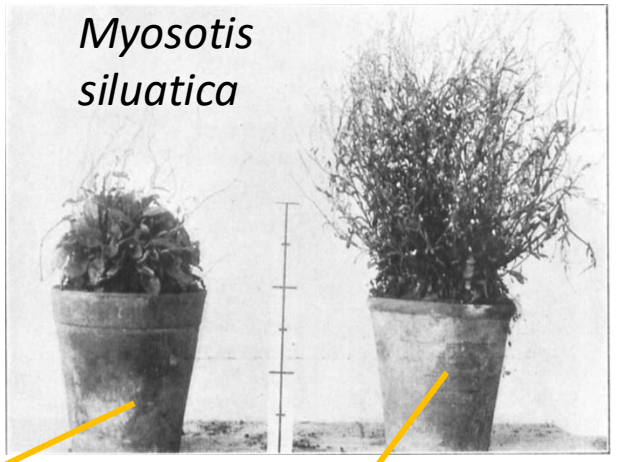


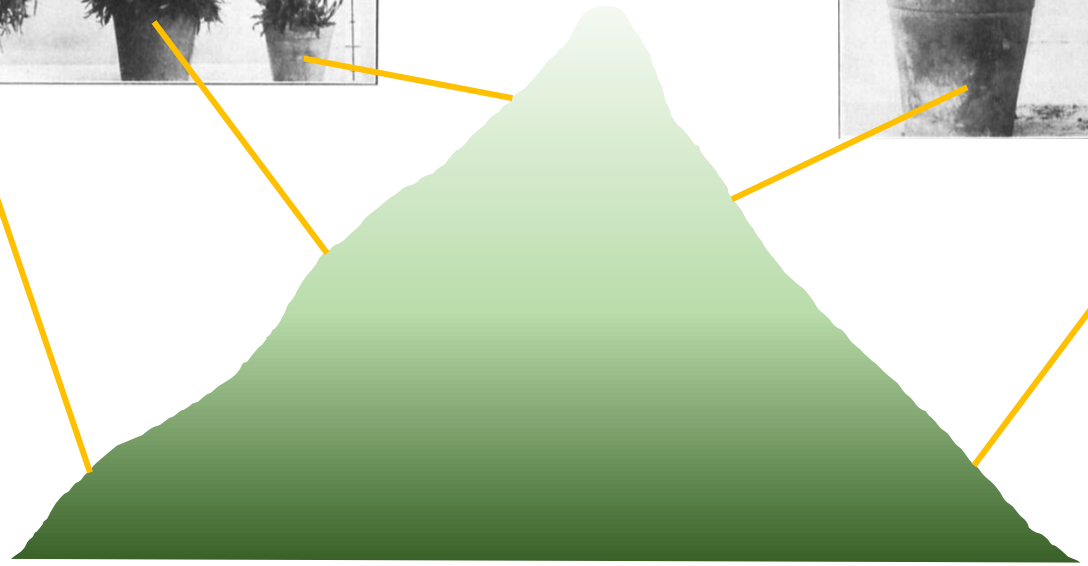
Genetic variation!



Turesson's common garden experiments reveal genetic variation in relation to climate







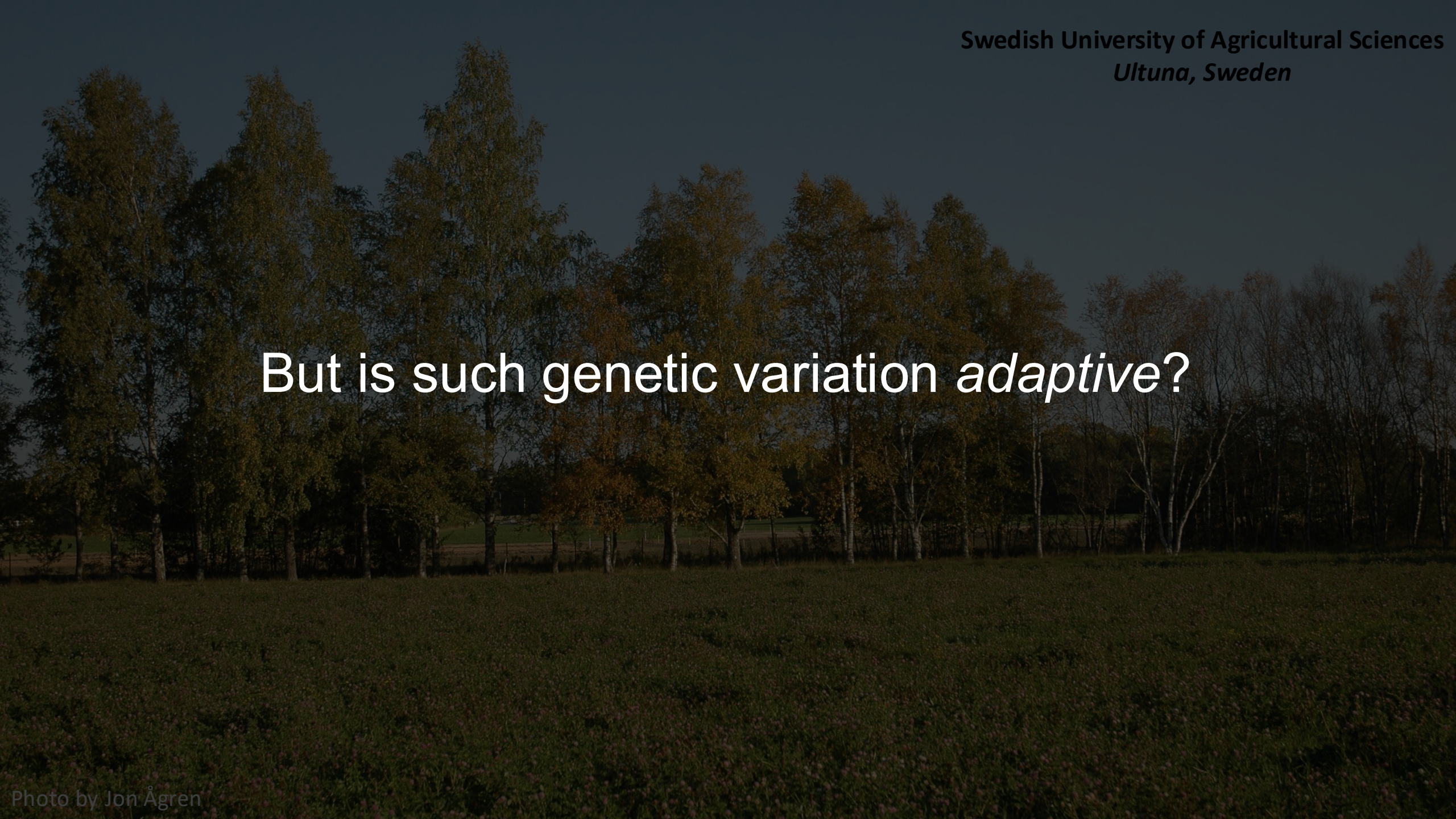
Genetic variation!



"individuals composing the population in extreme alpine situations are no doubt genotypically lower in stature than individuals inhabiting less extreme localities"

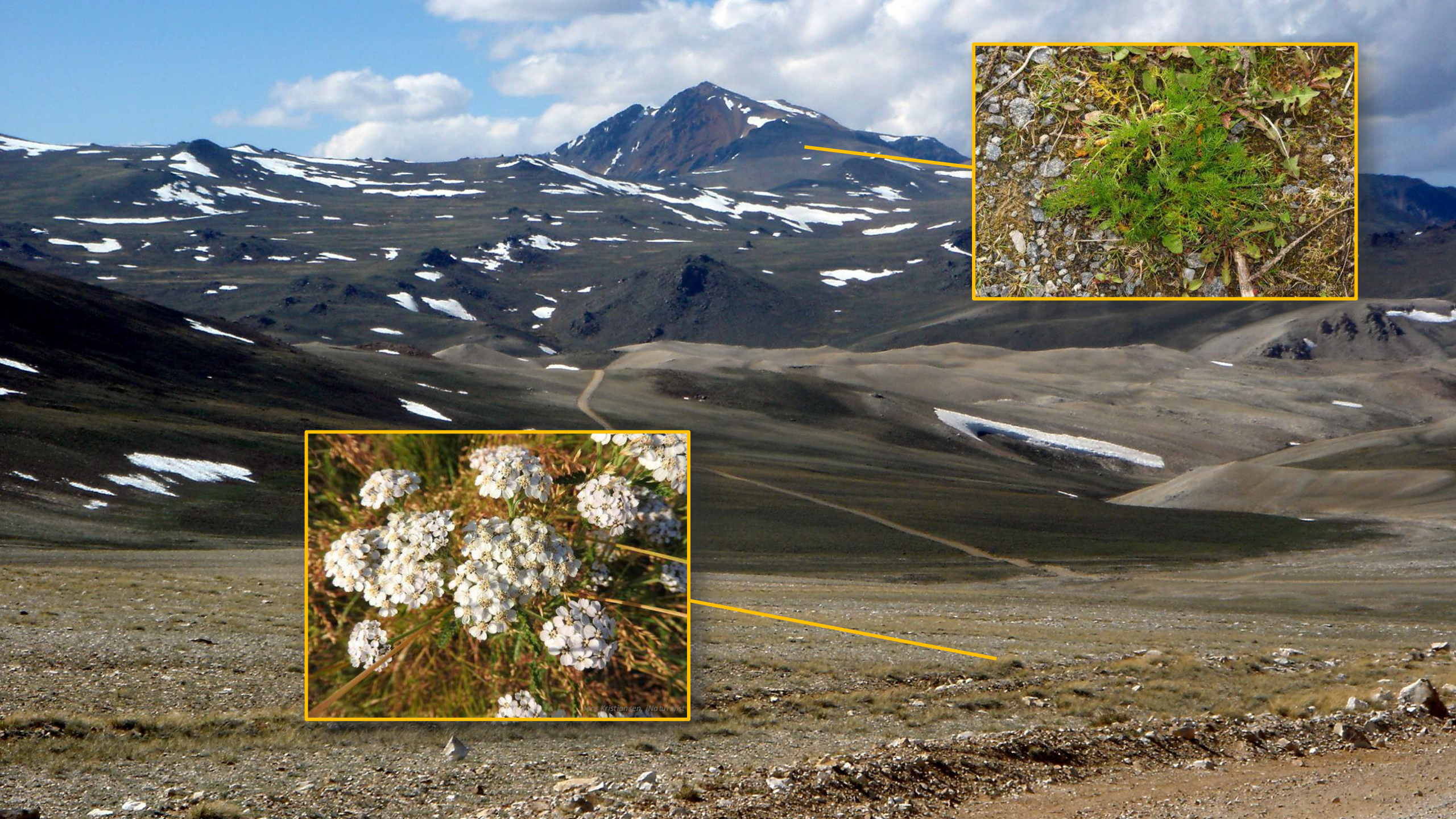

Southern genotypes
Swedish University of Agricultural Sciences
Ultuna, Sweden
Northern genotypes
Photo by Jon Ågren

Swedish University of Agricultural Sciences
Ultuna, Sweden



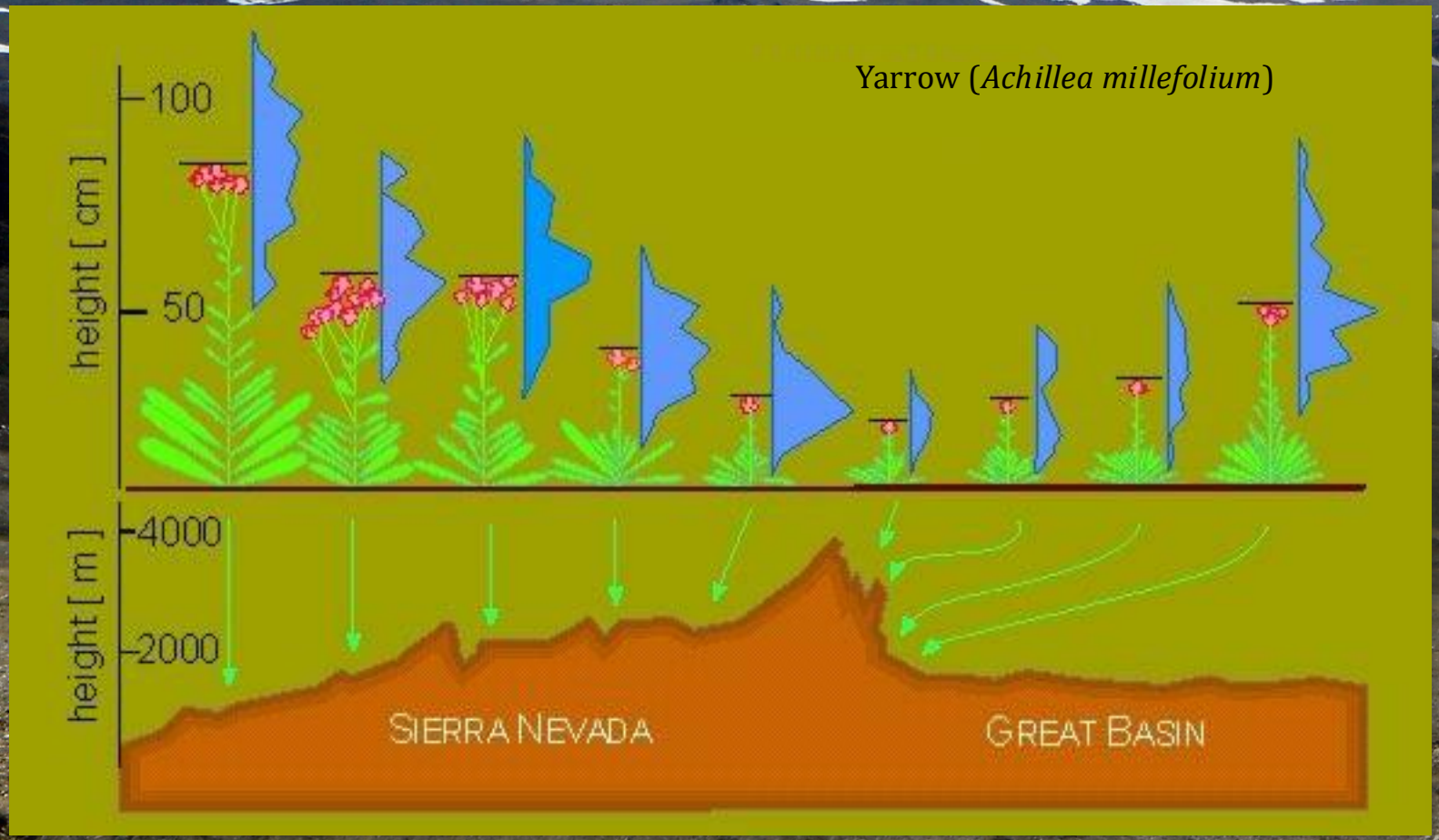
But is such genetic variation *adaptive*?

Photo by Jon Ågren





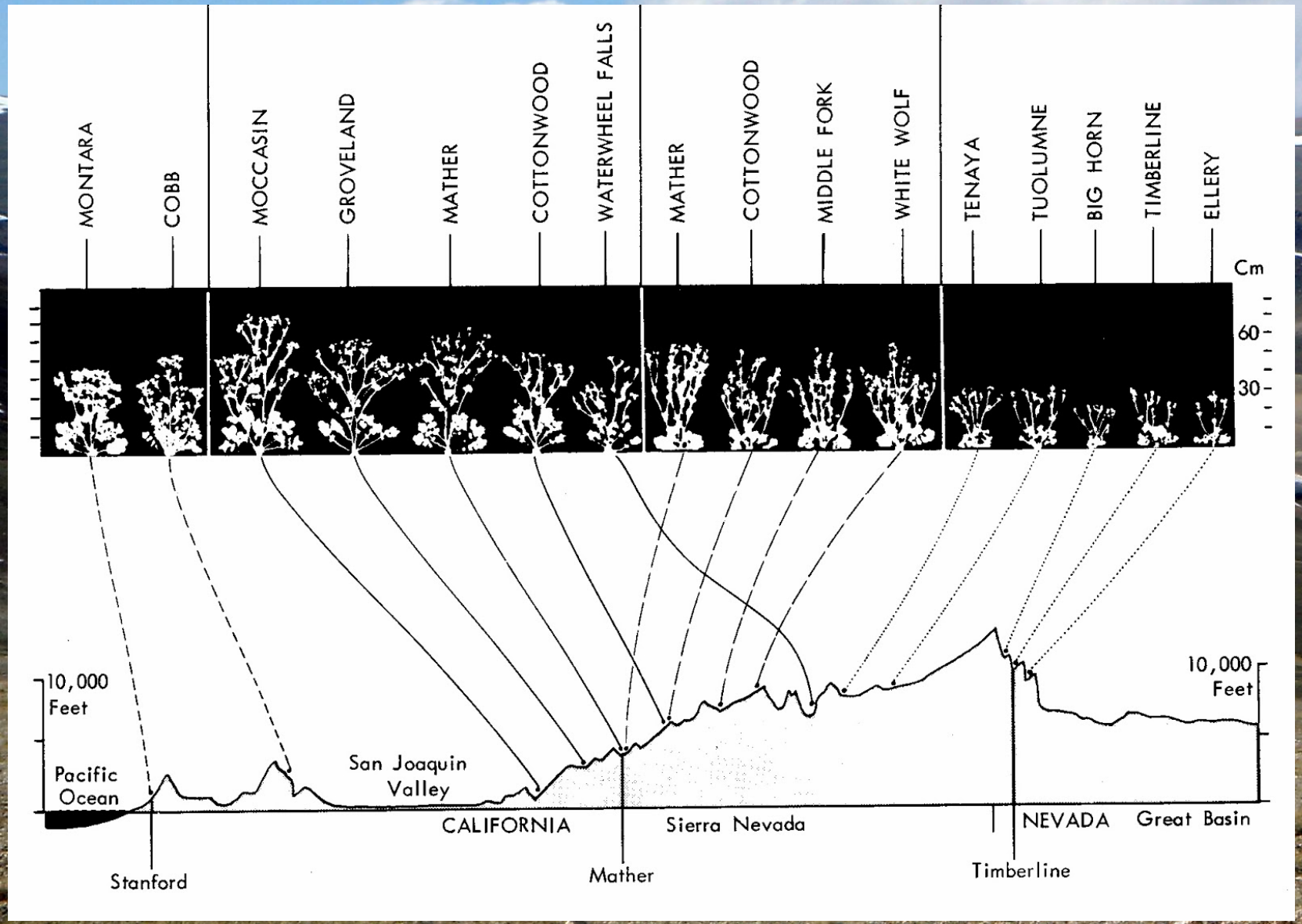
Clausen, Keck, & Heisey
Early-mid 1900s


Yarrow (*Achillea millefolium*)
100
50
height [cm]
4000
2000
height [m]
SIERRA NEVADA
GREAT BASIN

Adaptation?

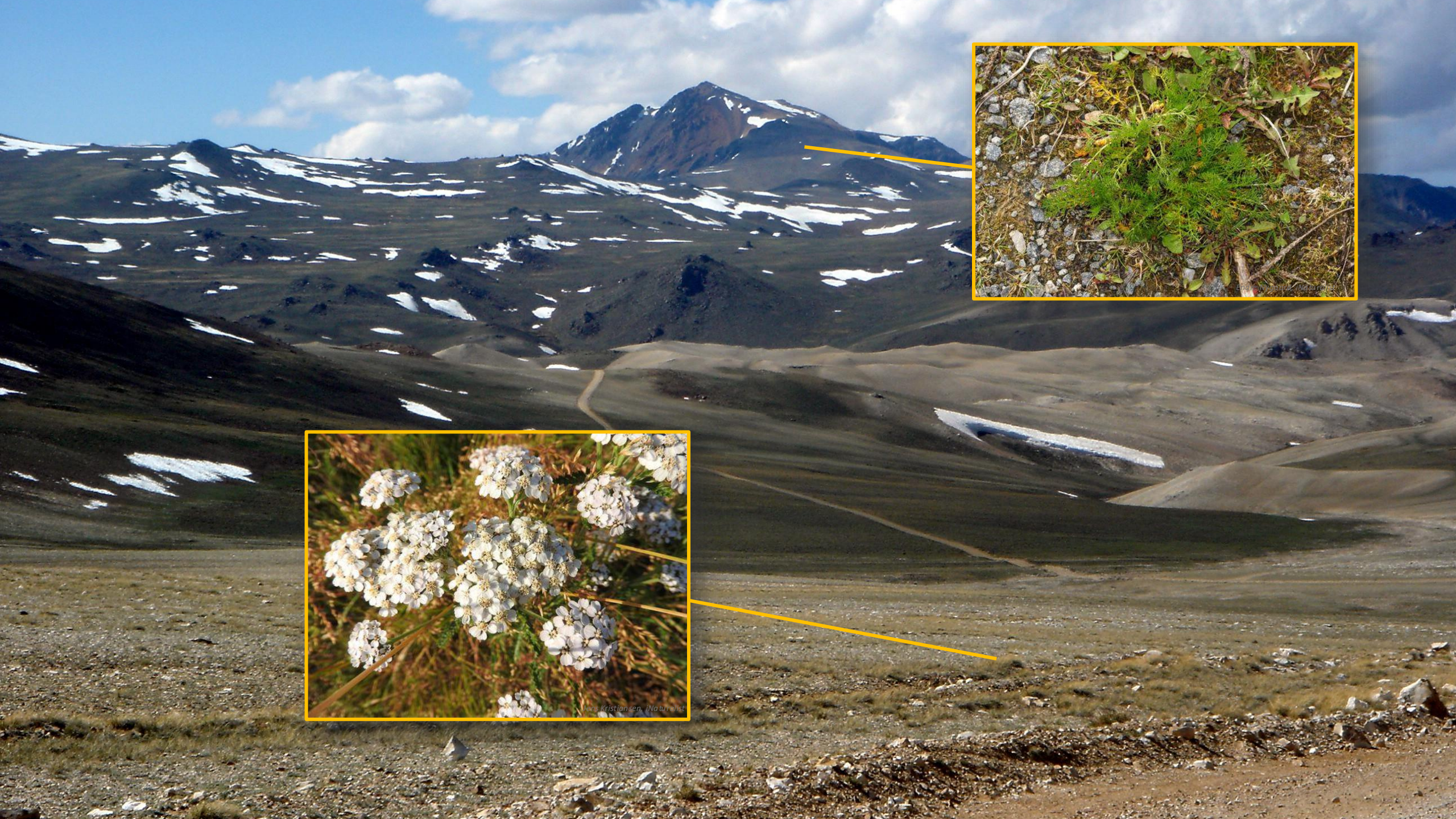


Sticky cinquefoil (*Drymocallis glandulosa*)

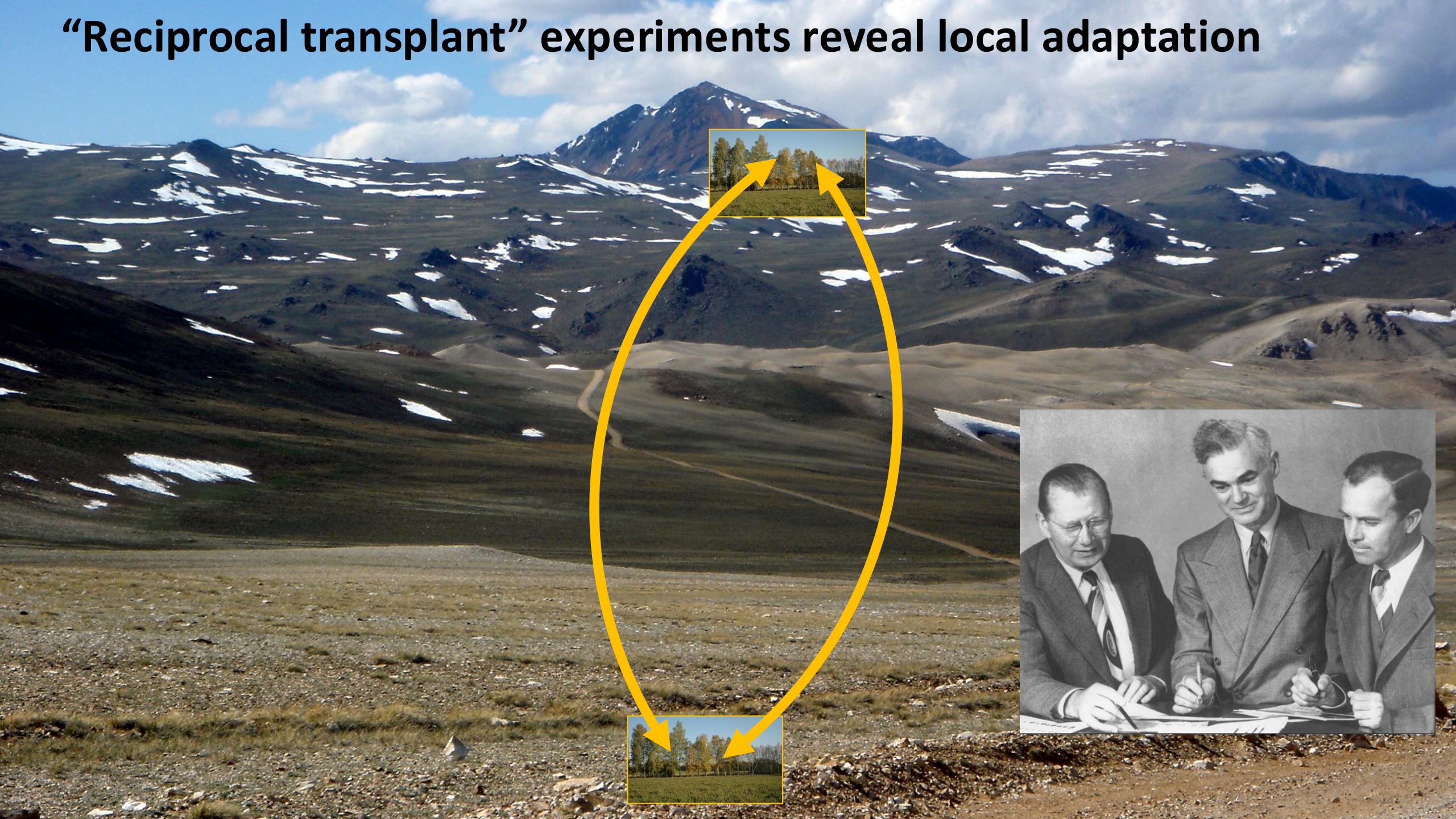


Adaptation?

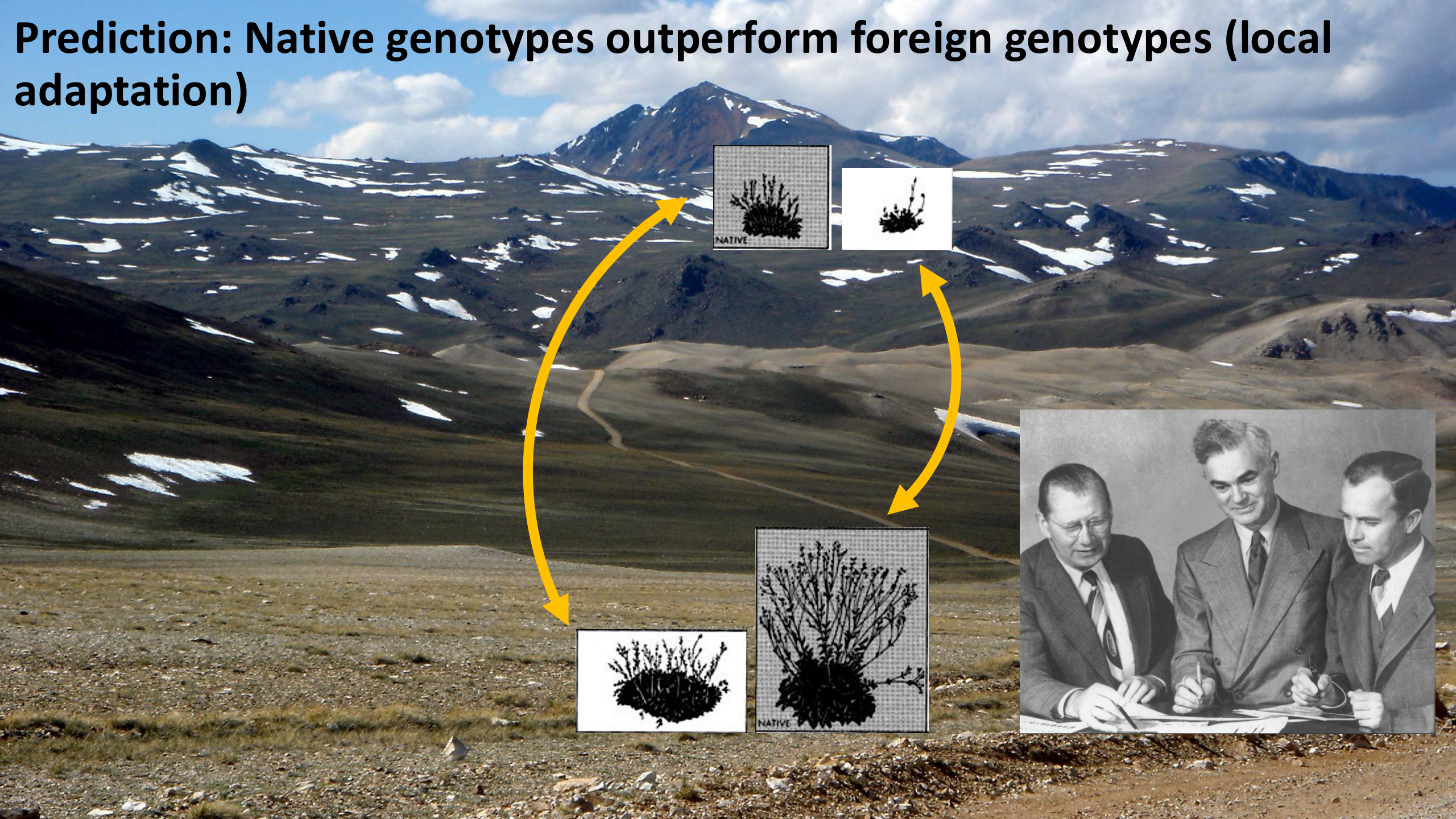




“Reciprocal transplant” experiments reveal local adaptation



Prediction: Native genotypes outperform foreign genotypes (local adaptation)



“Reciprocal transplant” experiments

- **Hypothesis:** genetic differences between populations are locally adaptive



“Reciprocal transplant” experiments

- **Hypothesis:** genetic differences between populations are locally adaptive
- **Prediction:** native genotypes will perform better than foreign genotypes when grown in home environments



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- Collect seeds or clones from different populations



“Reciprocal transplant” experiments

- **Hypothesis:** genetic differences between populations are locally adaptive
- **Prediction:** native genotypes will perform better than foreign genotypes when grown in home environments
- Collect seeds or clones from different populations
- Grow them in home sites of different populations (common gardens)



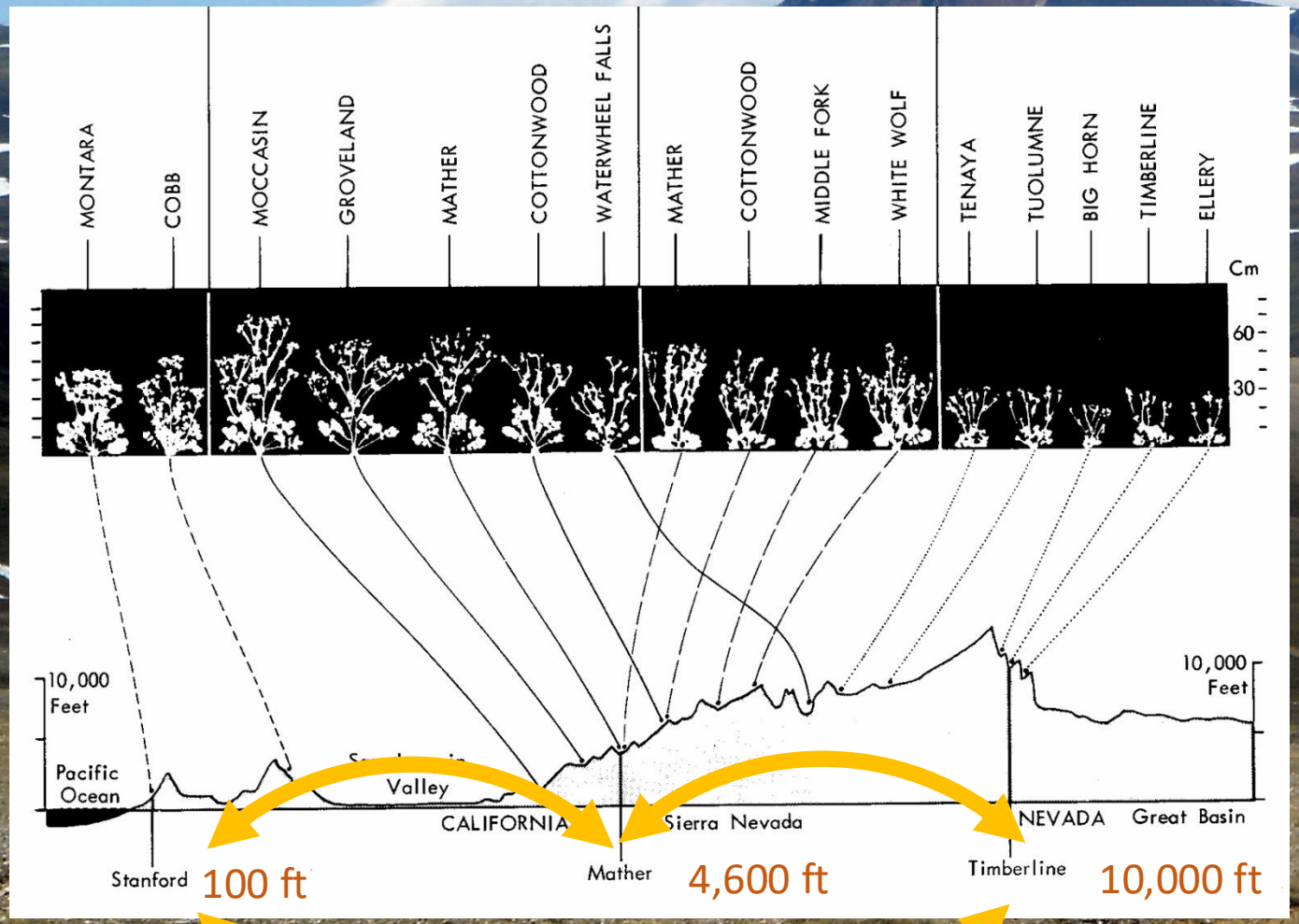
“Reciprocal transplant” experiments

- **Hypothesis:** genetic differences between populations are locally adaptive
- **Prediction:** native genotypes will perform better than foreign genotypes when grown in home environments
- Collect seeds or clones from different populations
- Grow them in home sites of different populations (common gardens)
- Observe fitness between genotypes in each environment



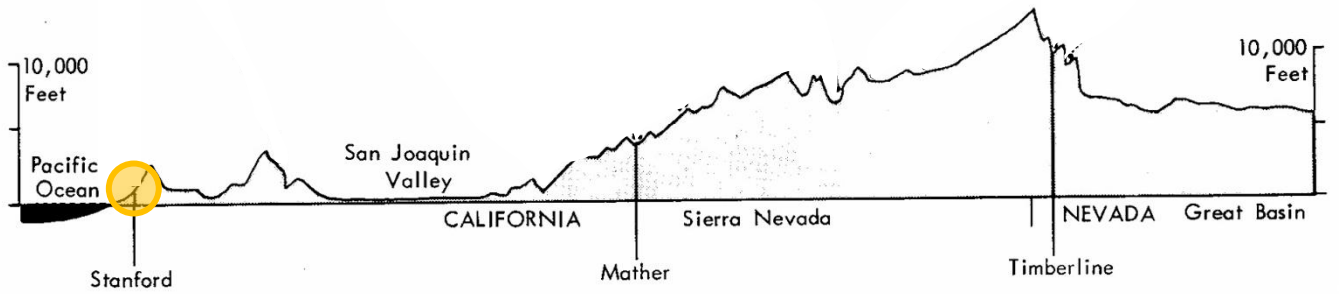
Factorial reciprocal transplant experiments

Sticky cinquefoil (*Drymocallis glandulosa*)





Local adaptation: Native genotypes outperform foreign genotypes



100 ft

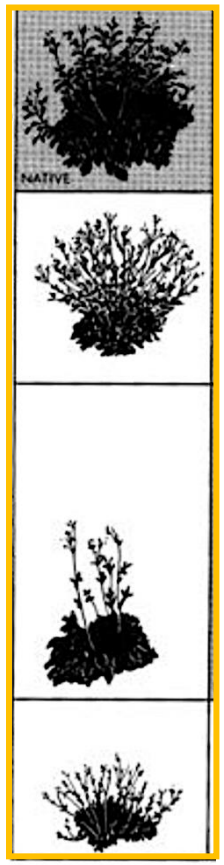
Native to:

900 ft

2,500 ft

4,600 ft

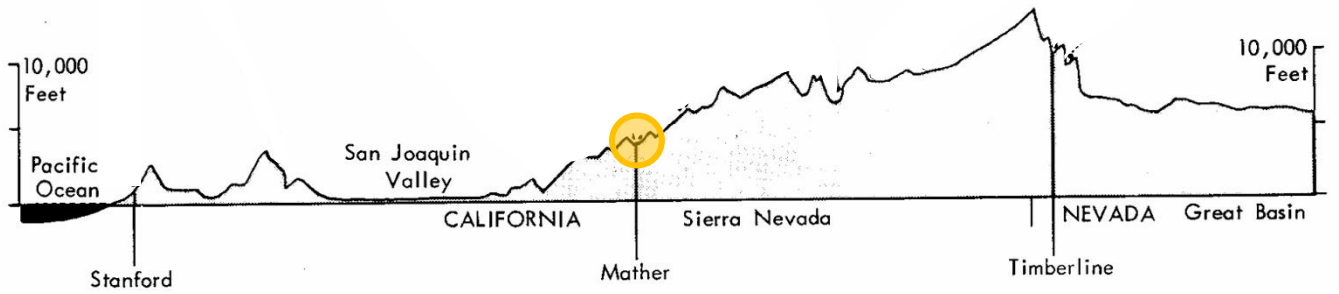
10,000 ft



Adaptation!



Local adaptation: Native genotypes outperform foreign genotypes



100 ft

4600 ft

Native to:

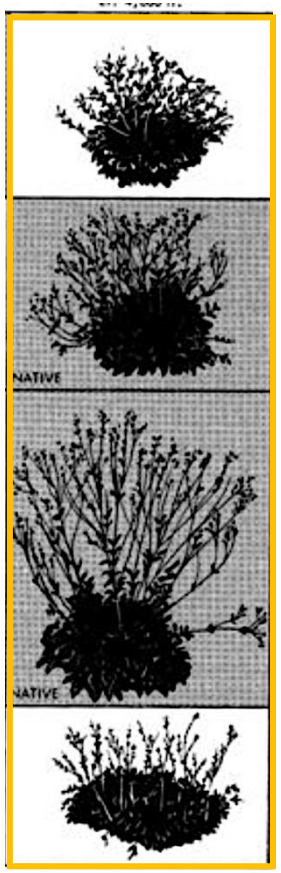
900 ft

2,500 ft

4,600 ft

10,000 ft

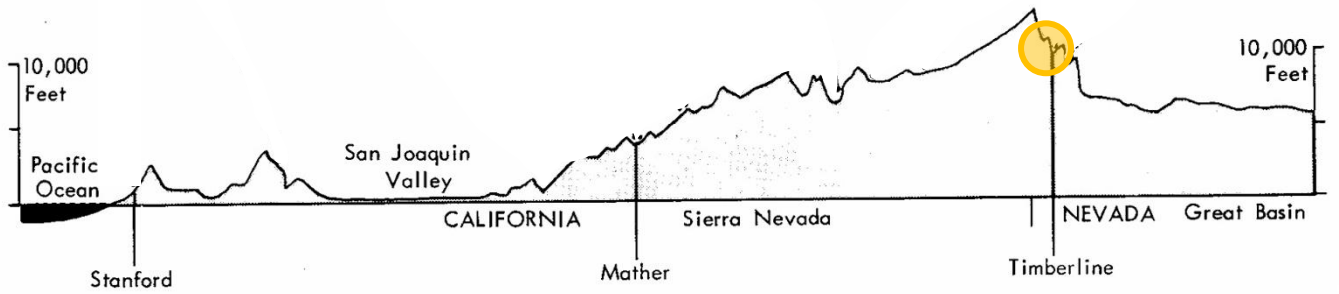




Adaptation!



Local adaptation: Native genotypes outperform foreign genotypes



Native to:

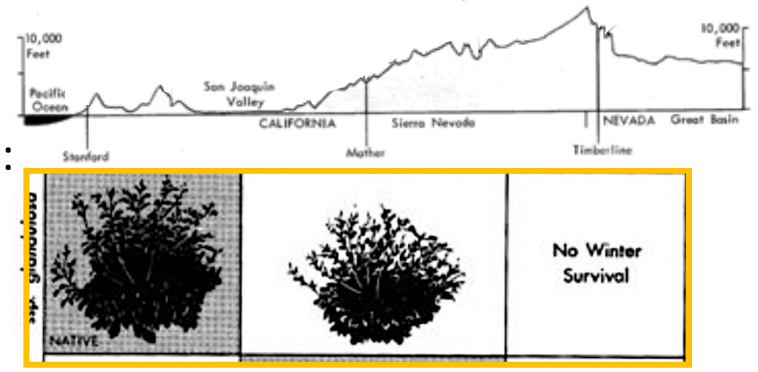
	100 ft	4600 ft	10000 ft
900 ft	NATIVE	NATIVE	No Winter Survival
2,500 ft			No Winter Survival
4,600 ft		NATIVE	
10,000 ft			NATIVE

Adaptation!

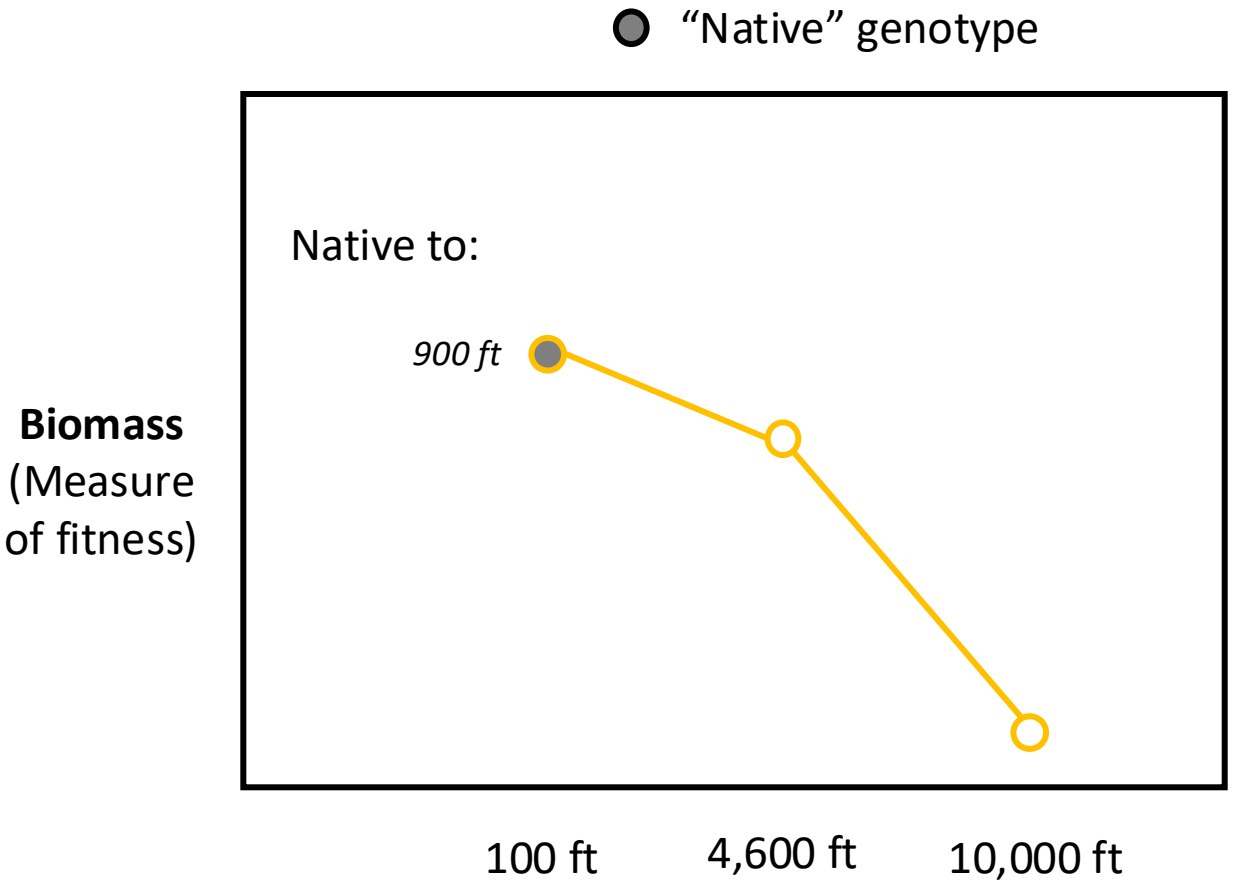


Local adaptation: Native genotypes outperform foreign genotypes

Native to:

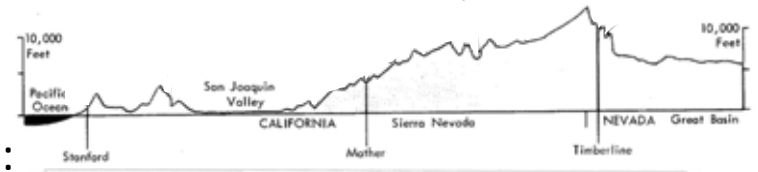


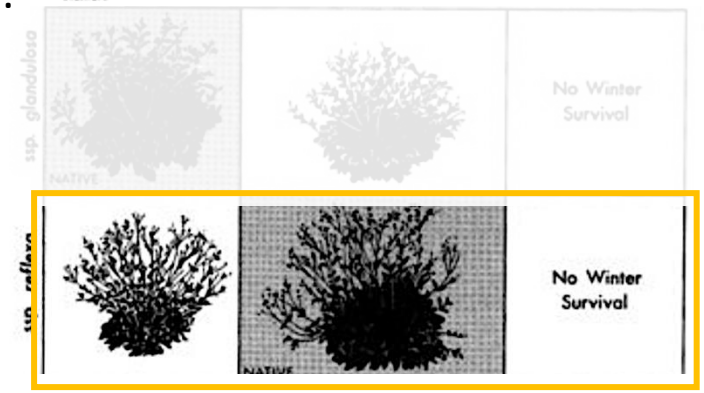
Common garden elevation



Local adaptation: Native genotypes outperform foreign genotypes

Native to:

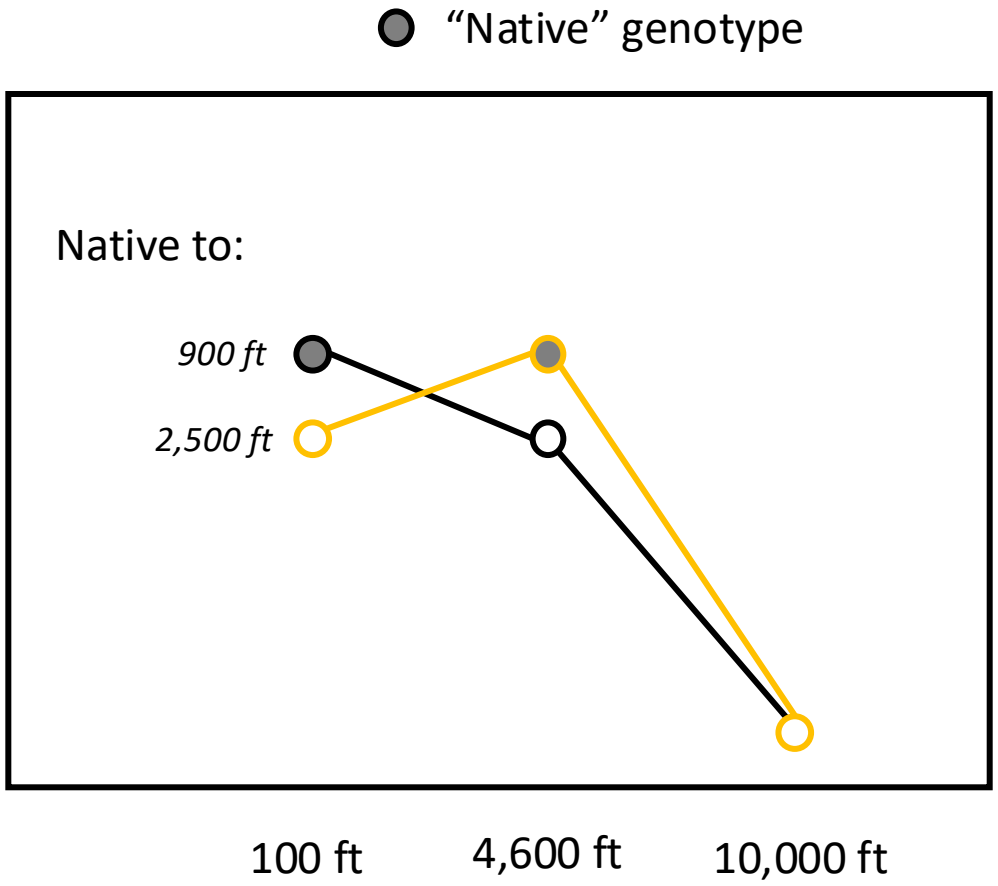




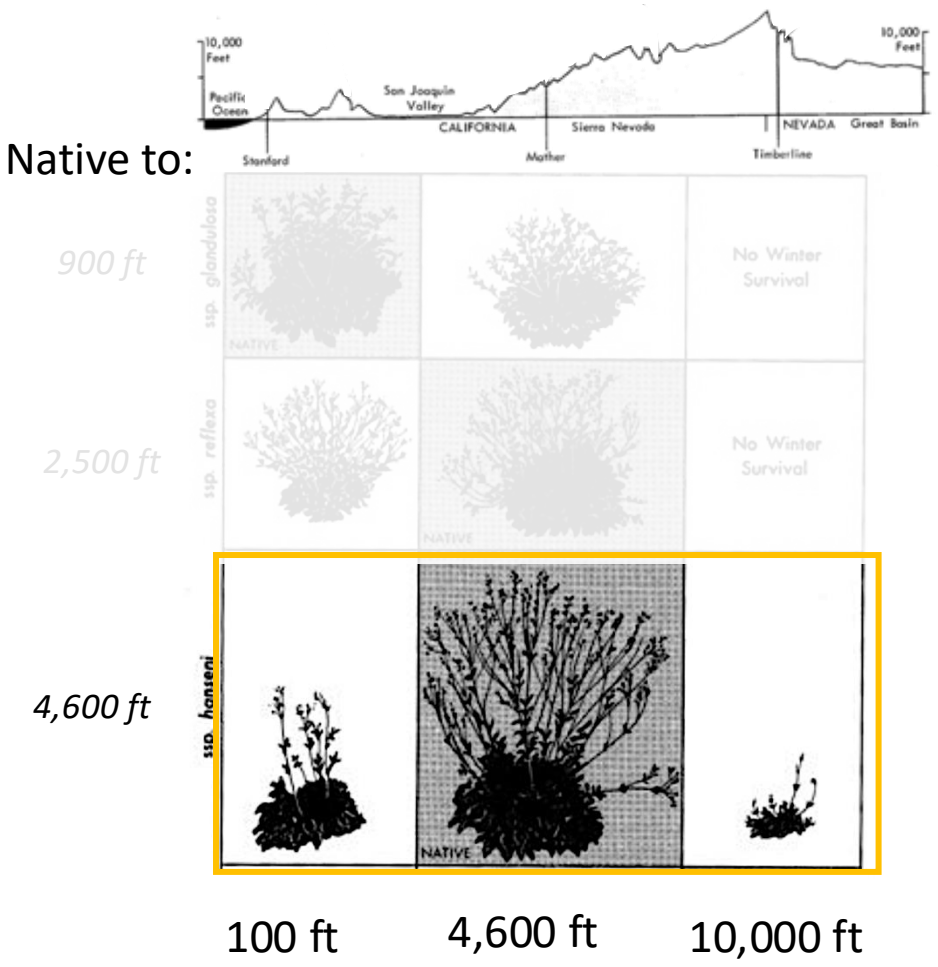
100 ft 4,600 ft 10,000 ft

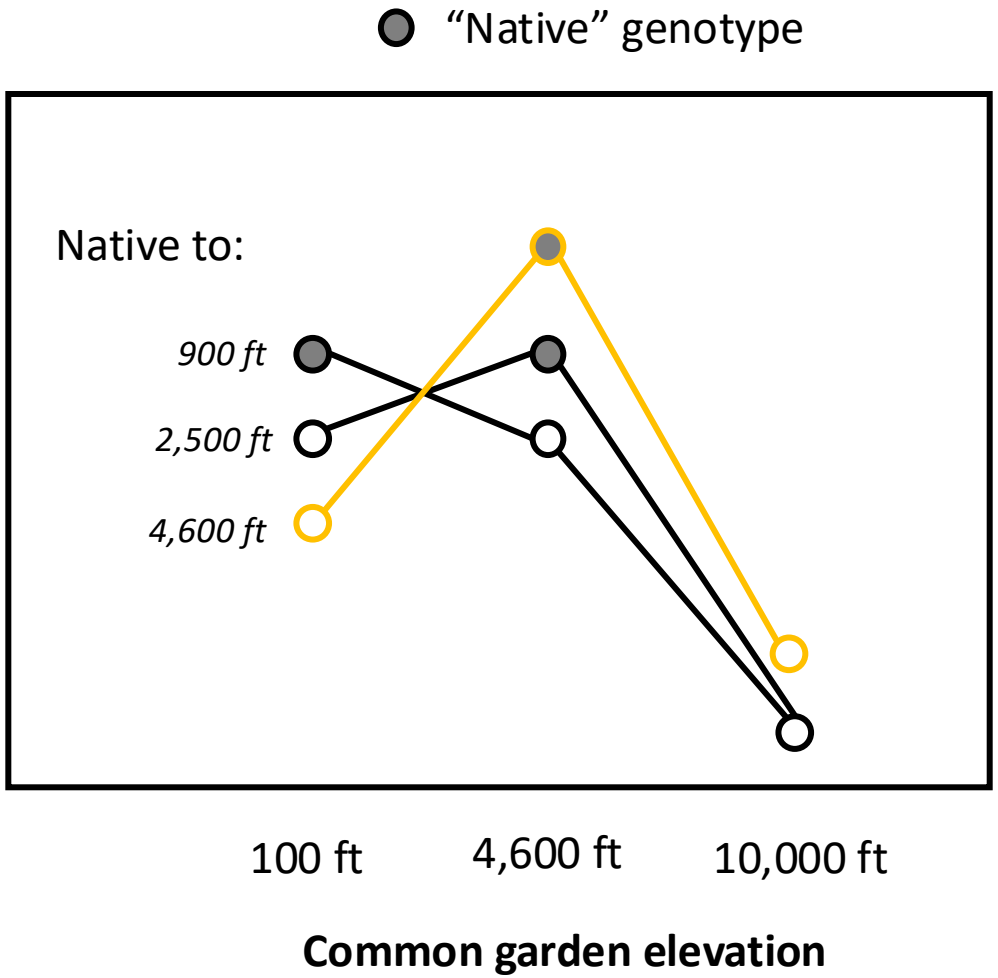
Common garden elevation

Biomass (Measure of fitness)

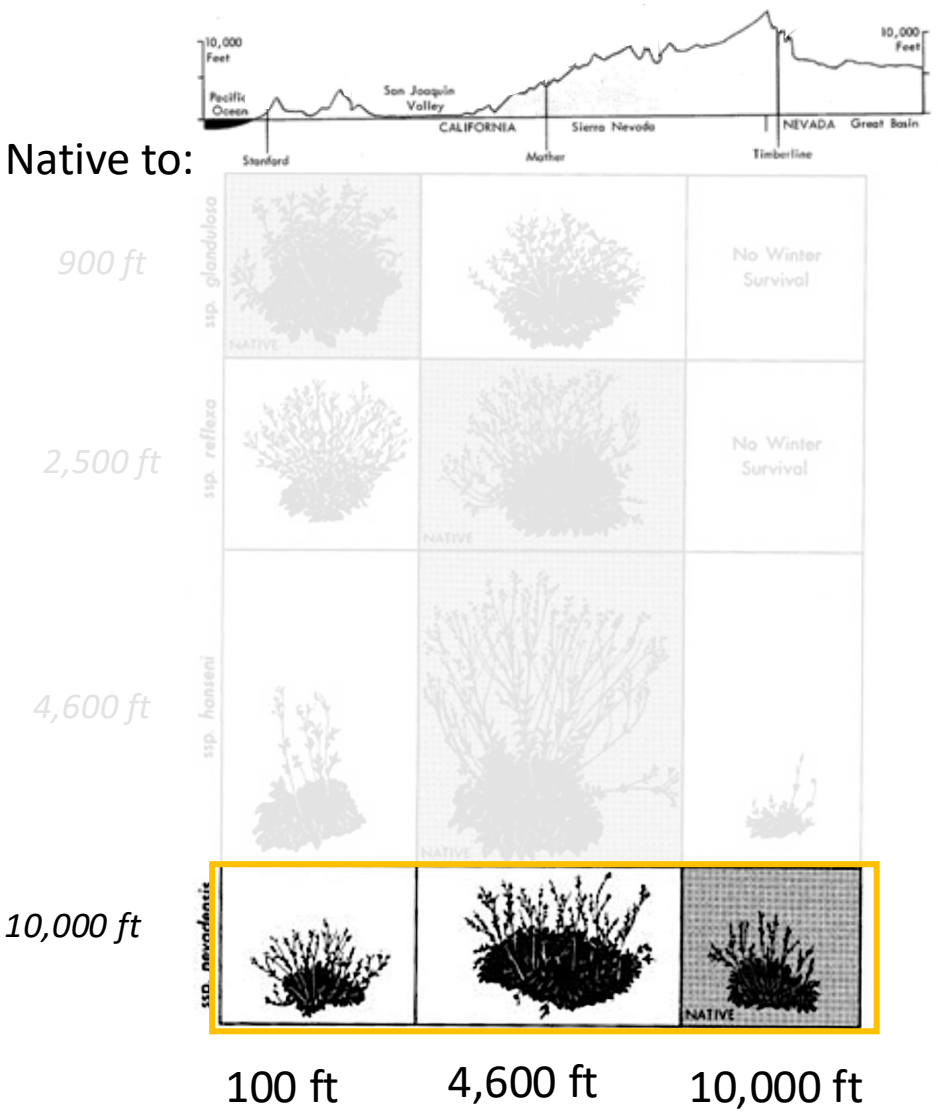


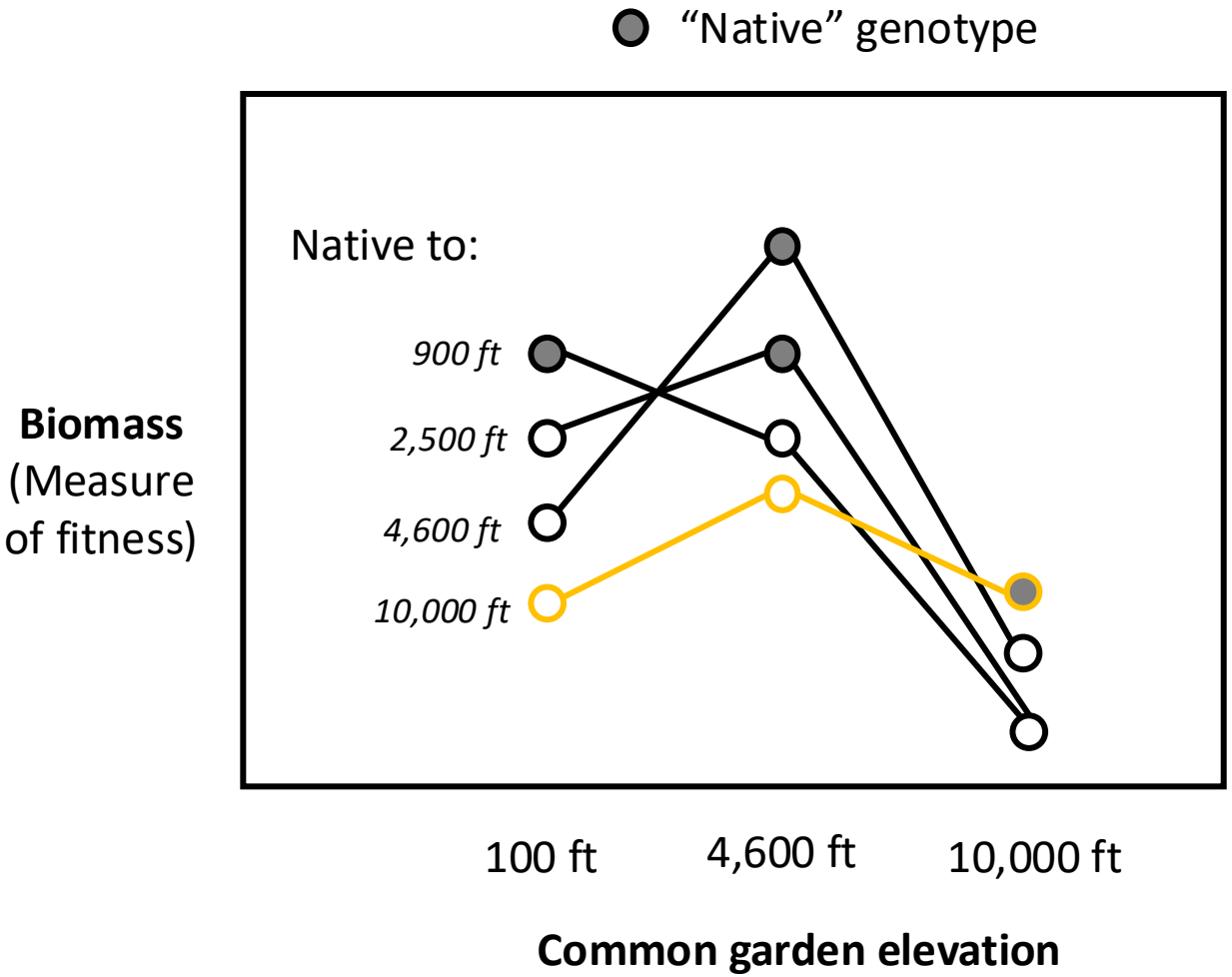
Local adaptation: Native genotypes outperform foreign genotypes



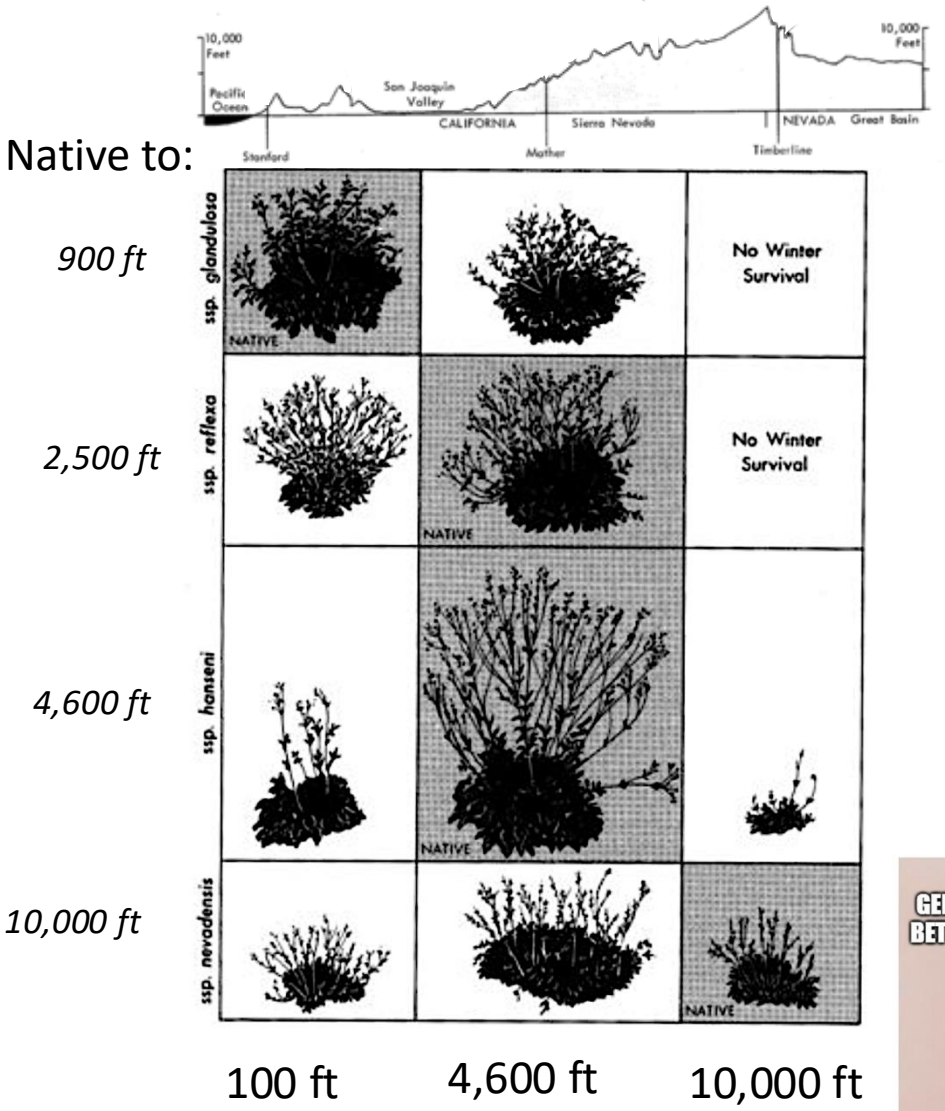


Local adaptation: Native genotypes outperform foreign genotypes

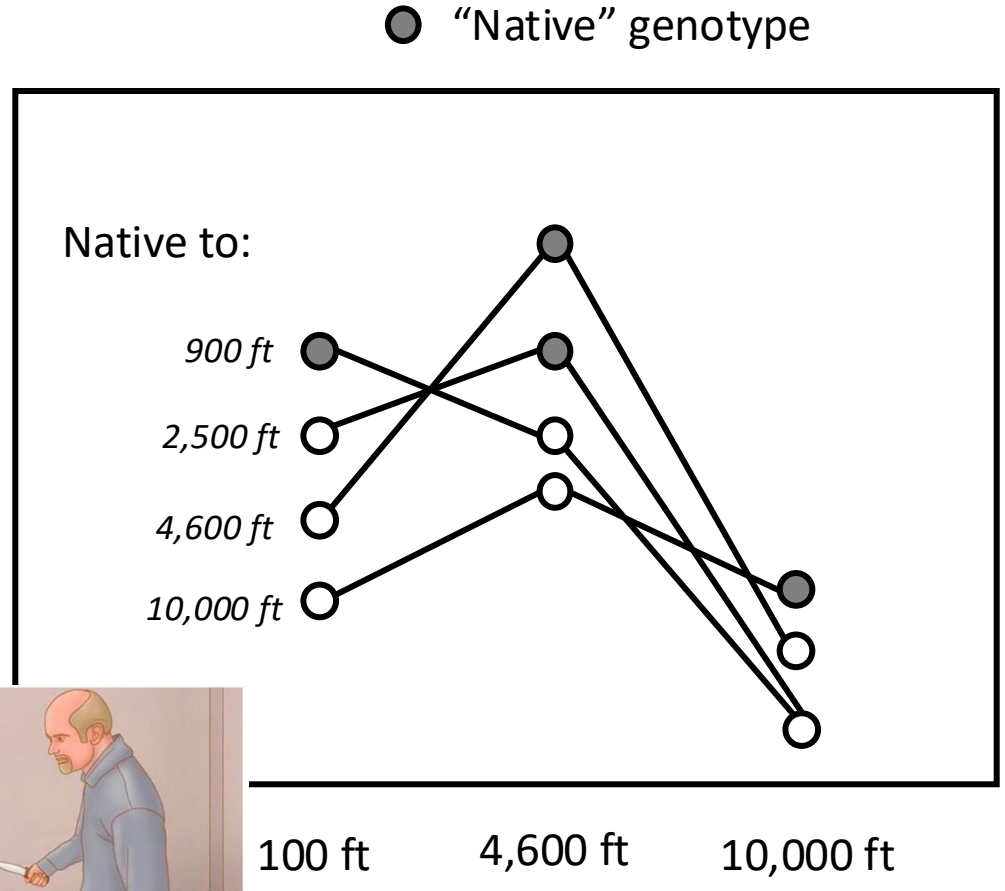




Local adaptation: Native genotypes outperform foreign genotypes







Read the garden

Kahoot-style game

4 garden results
Which pattern is it?
Fast and correct wins.

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-ca-garden

host screen



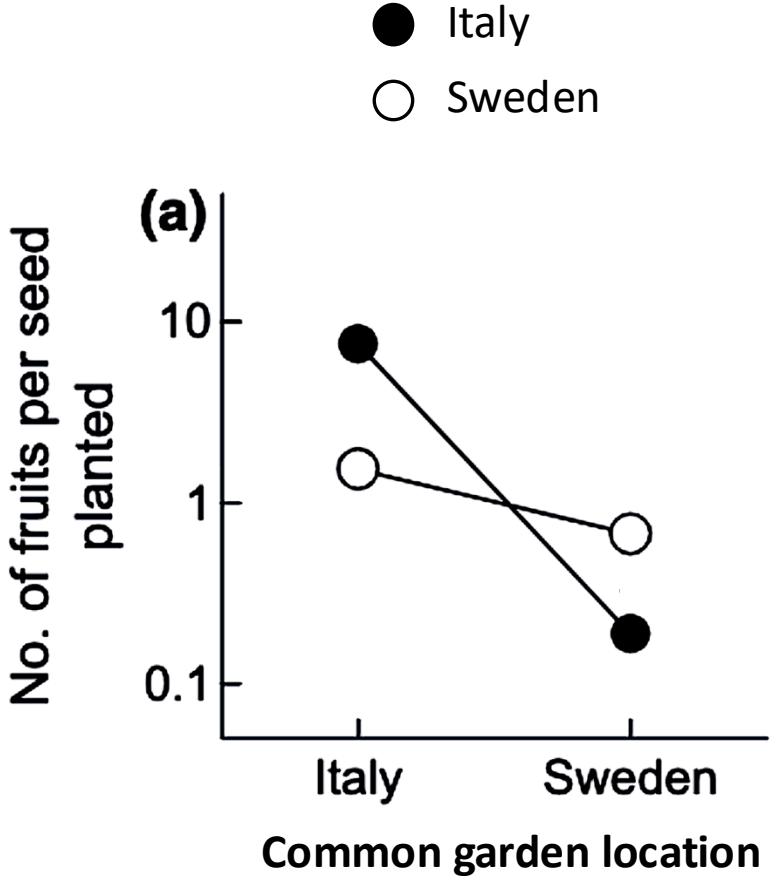
Case study:

Arabidopsis thaliana local adaptation in Europe



Ågren & Schemske 2012







Ågren & Schemske 2012

Live poll

Multiple choice

These data indicate that the Italian and Swedish genotypes are locally adapted.

Scan with your phone

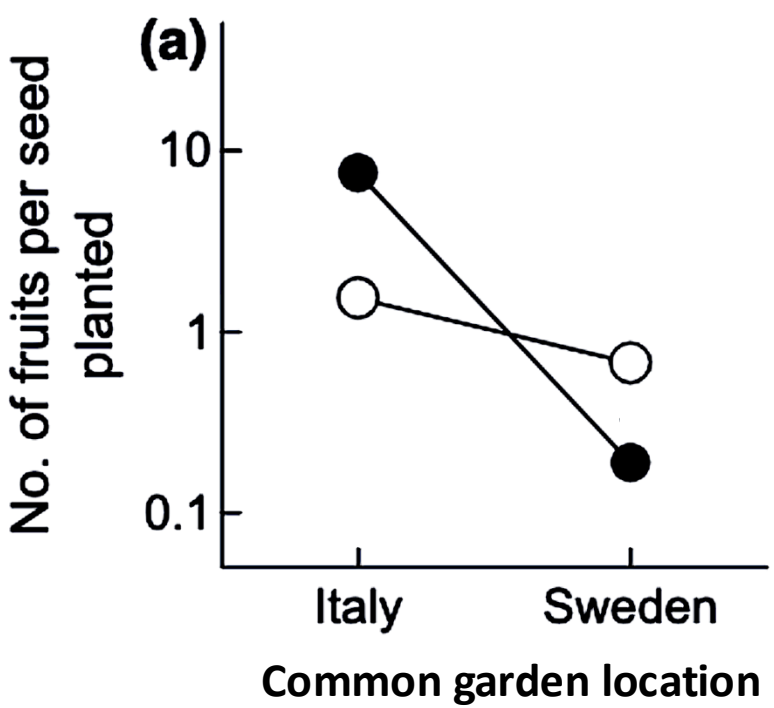
Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ca-agren

host screen

These data indicate that the Italian and Swedish genotypes are locally adapted.

A. **True**. In each environment, the native genotype has the highest fitness.

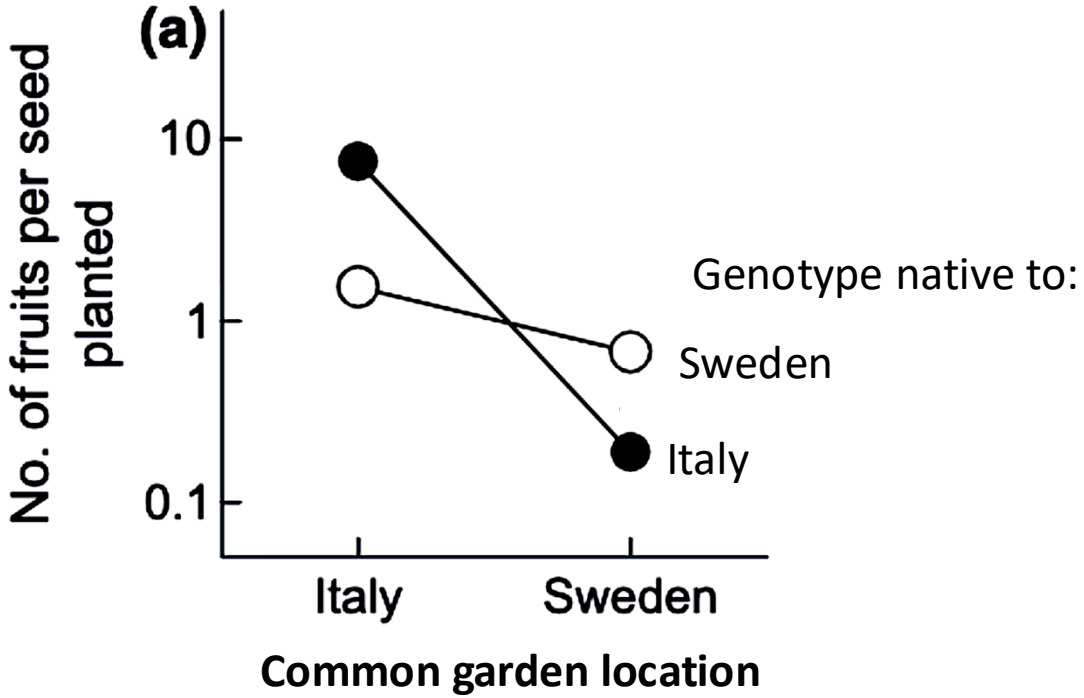
B. **False.** The Swedish genotype has higher fitness when grown in the foreign environment, Italy, than in it's home environment, Sweden.





Ågren & Schemske 2012

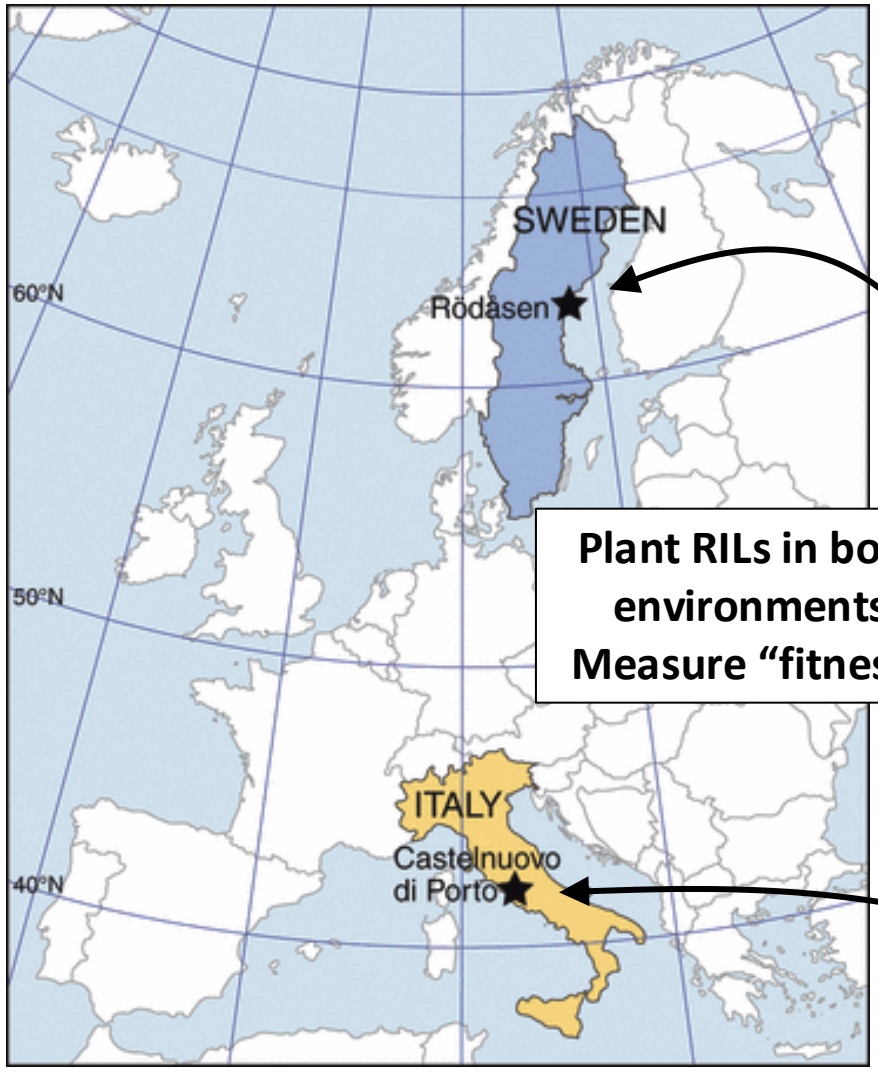
What is the genetic basis of local adaptation?



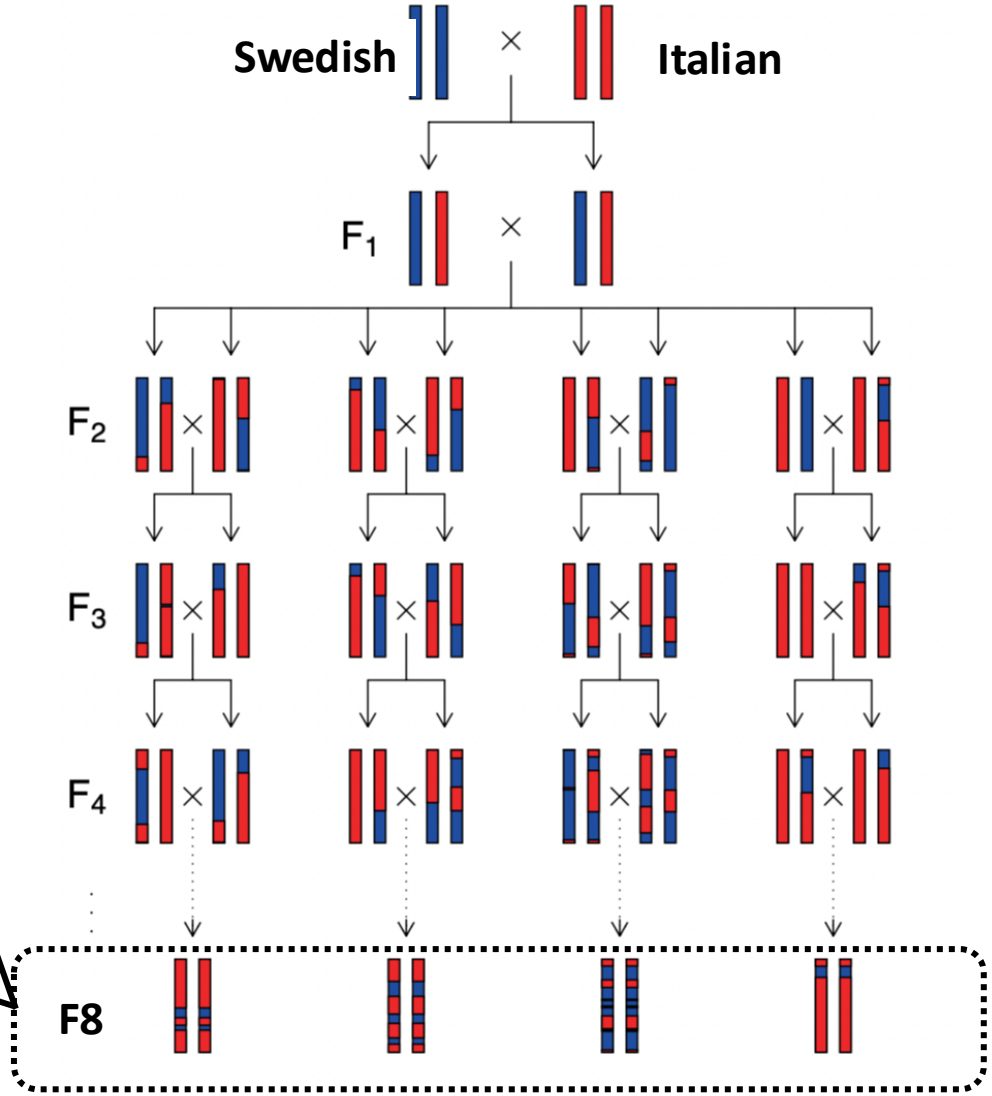


Ågren & Schemske 2012

Recombinant Inbred Line (RIL) population for QTL mapping



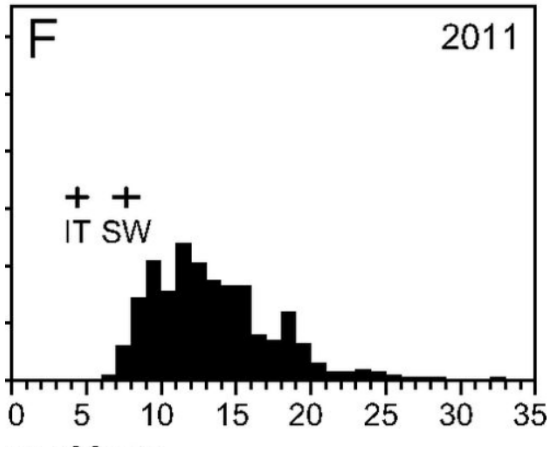
Plant RILs in both environments
Measure "fitness"

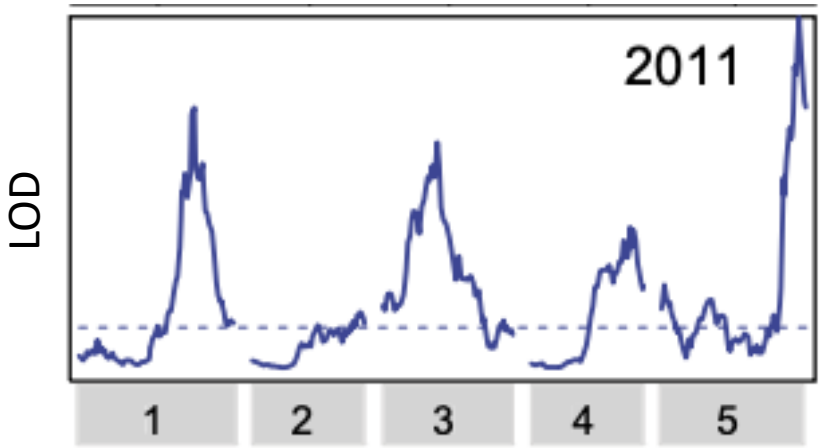


Discovering genome regions explaining fitness

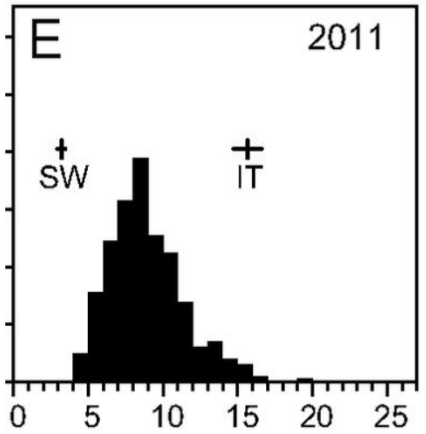


In Sweden

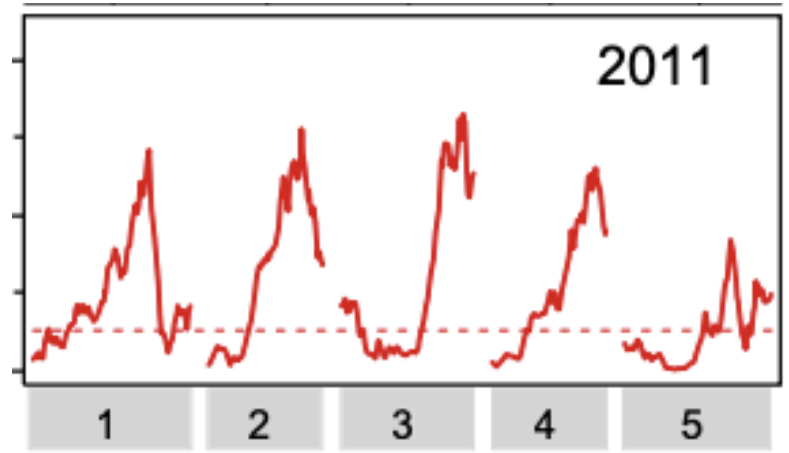




In Italy

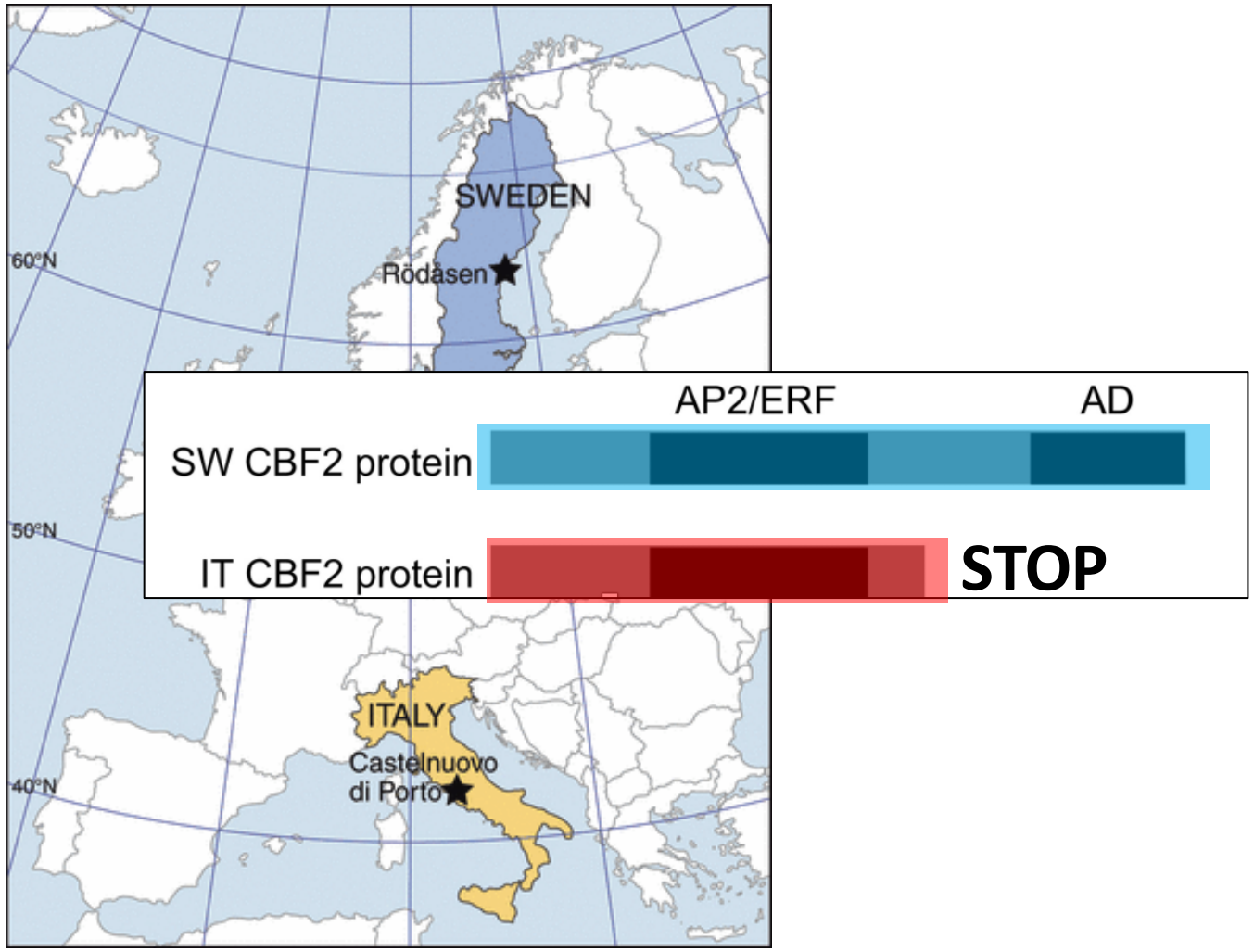


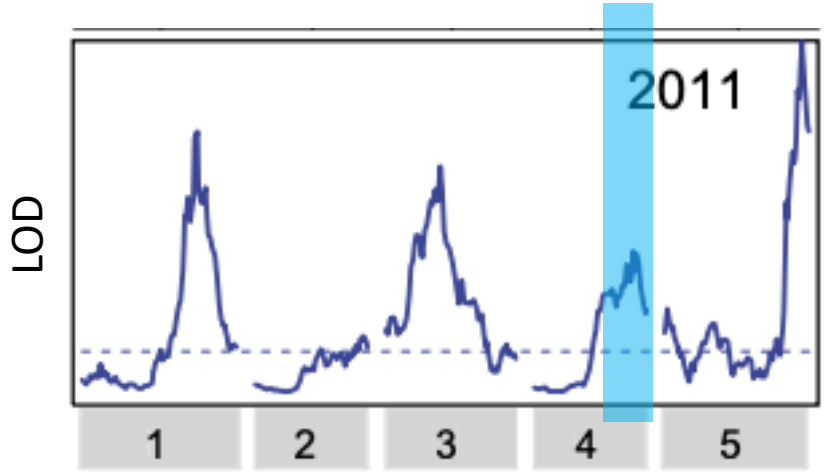
Fruits

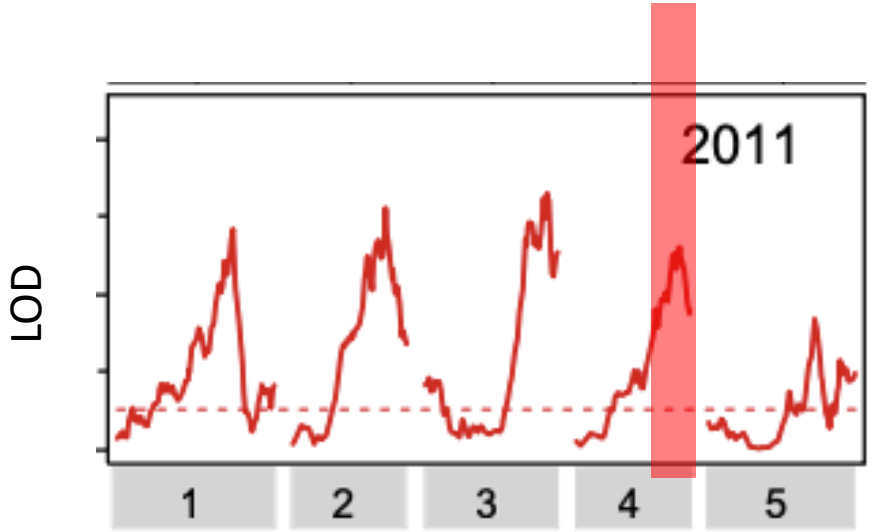


Chromosome

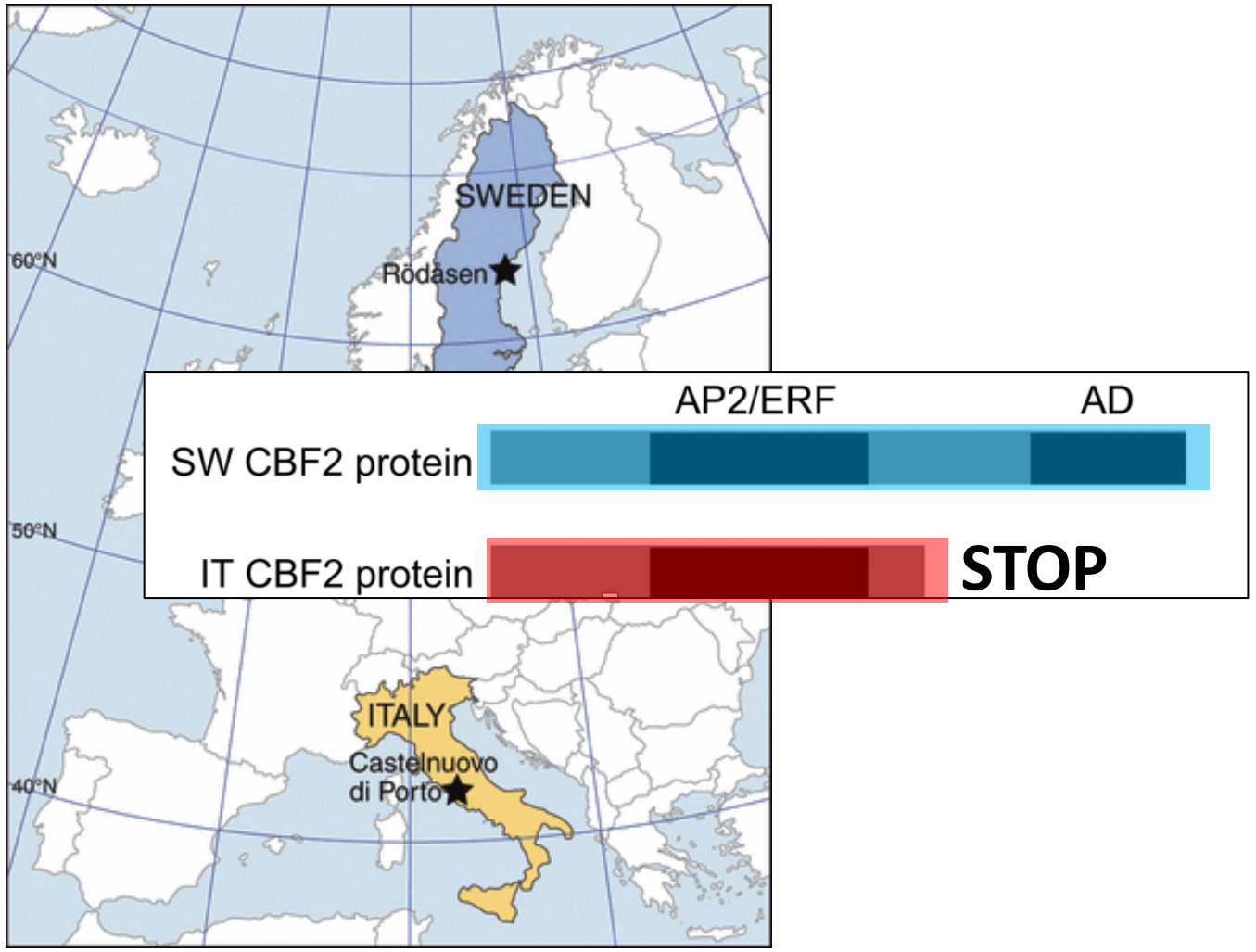
Italian line has a stop codon mutation in CBF2, a freezing tolerance gene

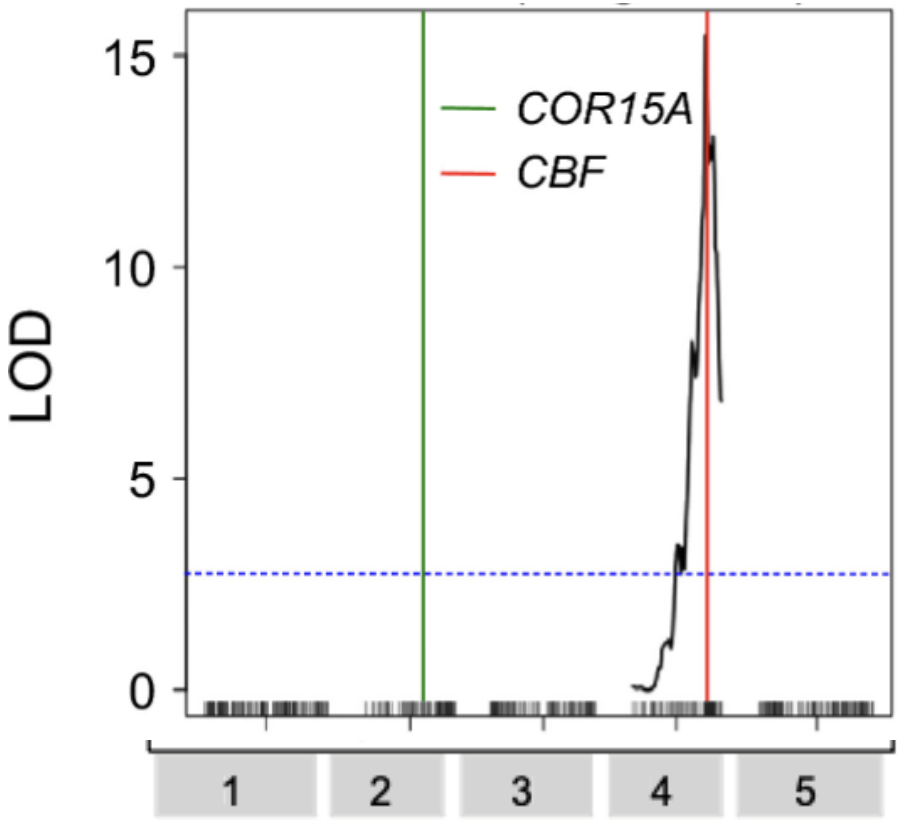




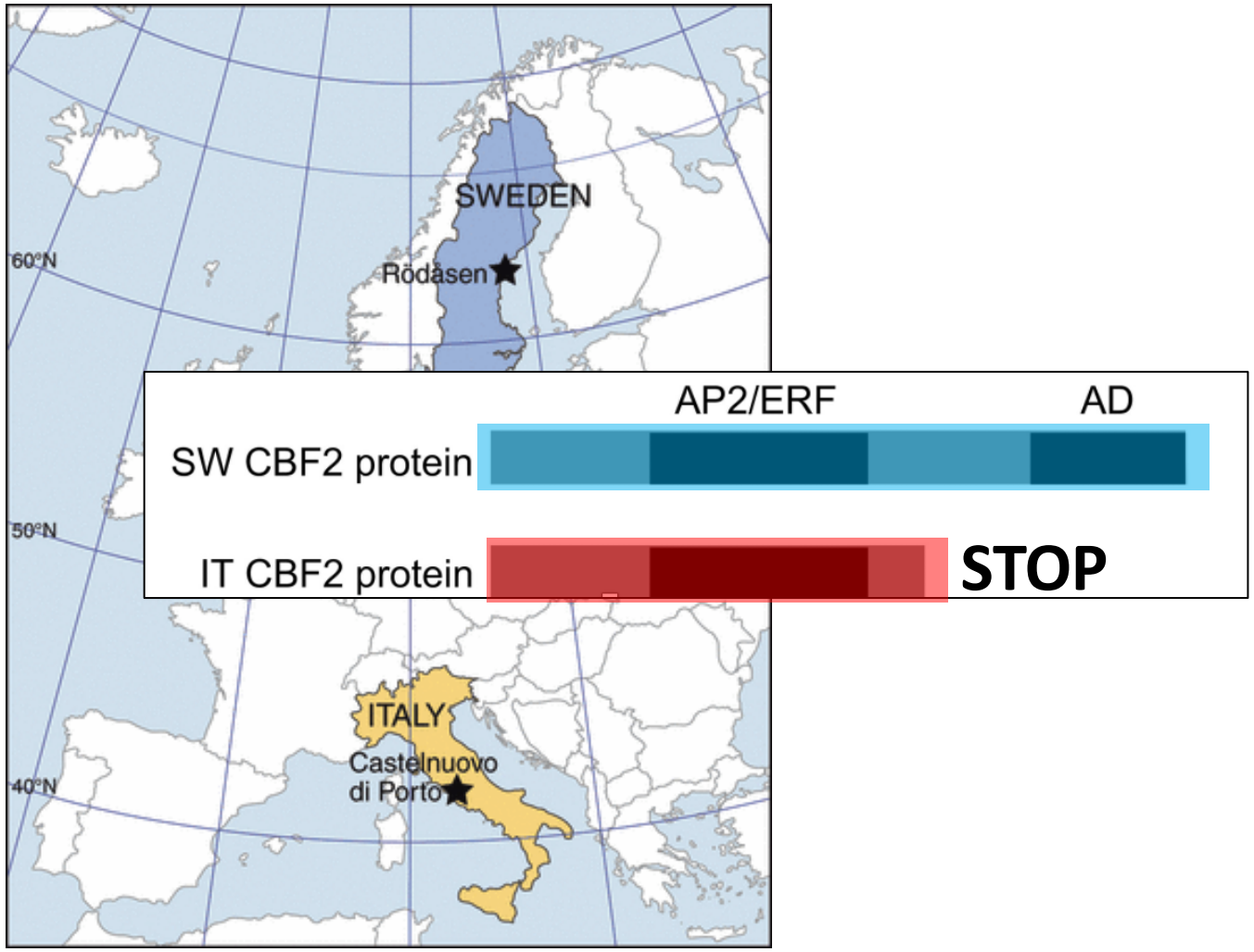


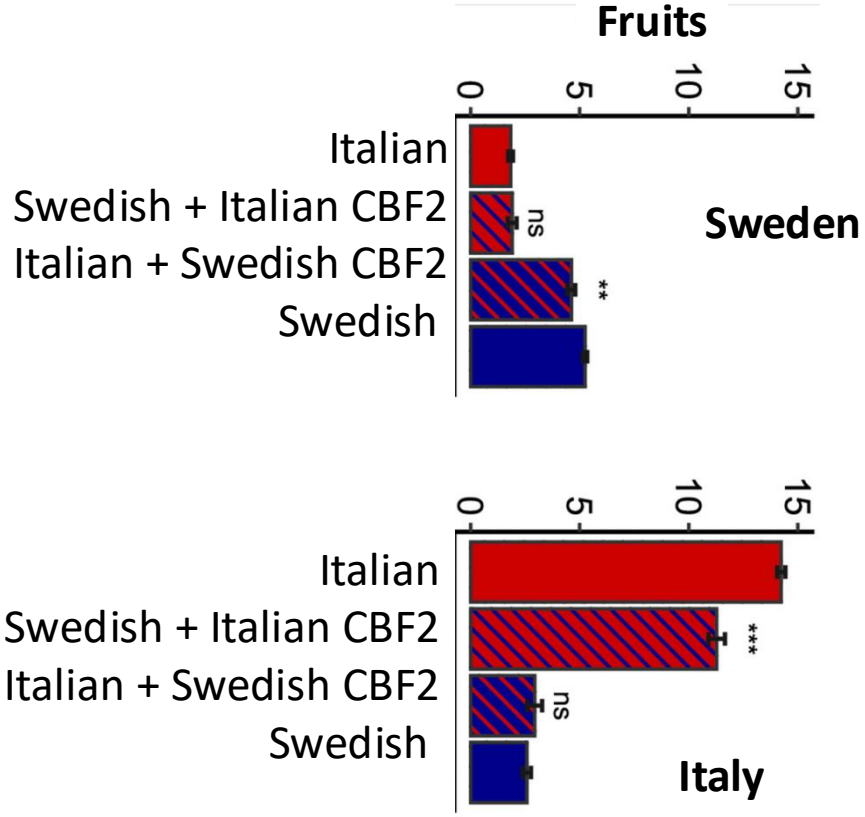
Italian CBF2 causes reduced freezing tolerance…





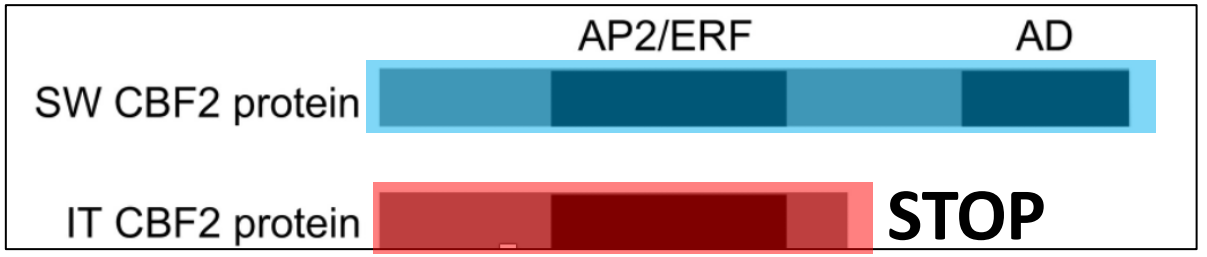
Functional CBF2 (Swedish allele) reduces fitness in Italy

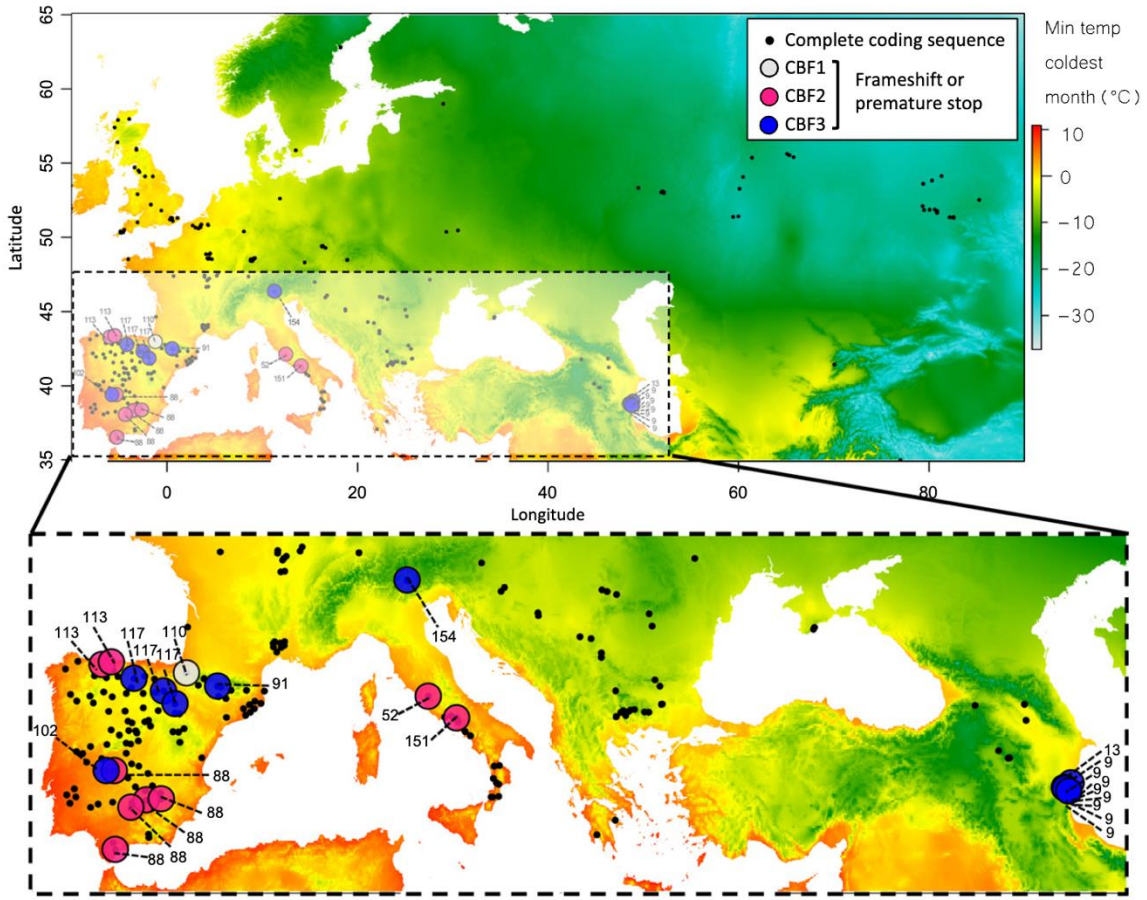




Functional CBF2 (Swedish allele) reduces fitness in Italy





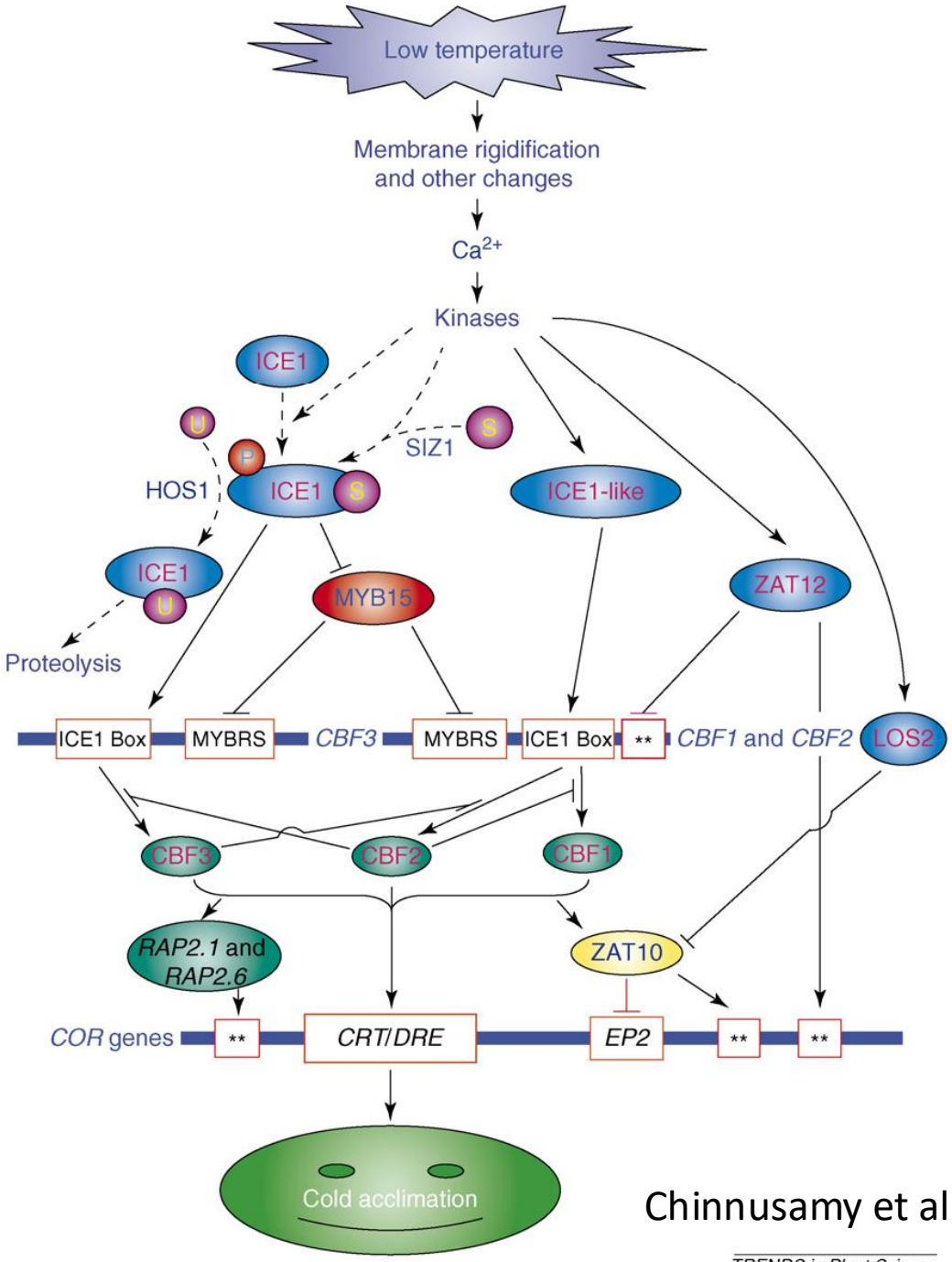


Example of “gene regulatory network” for freezing tolerance



Example of “gene regulatory network” for freezing tolerance

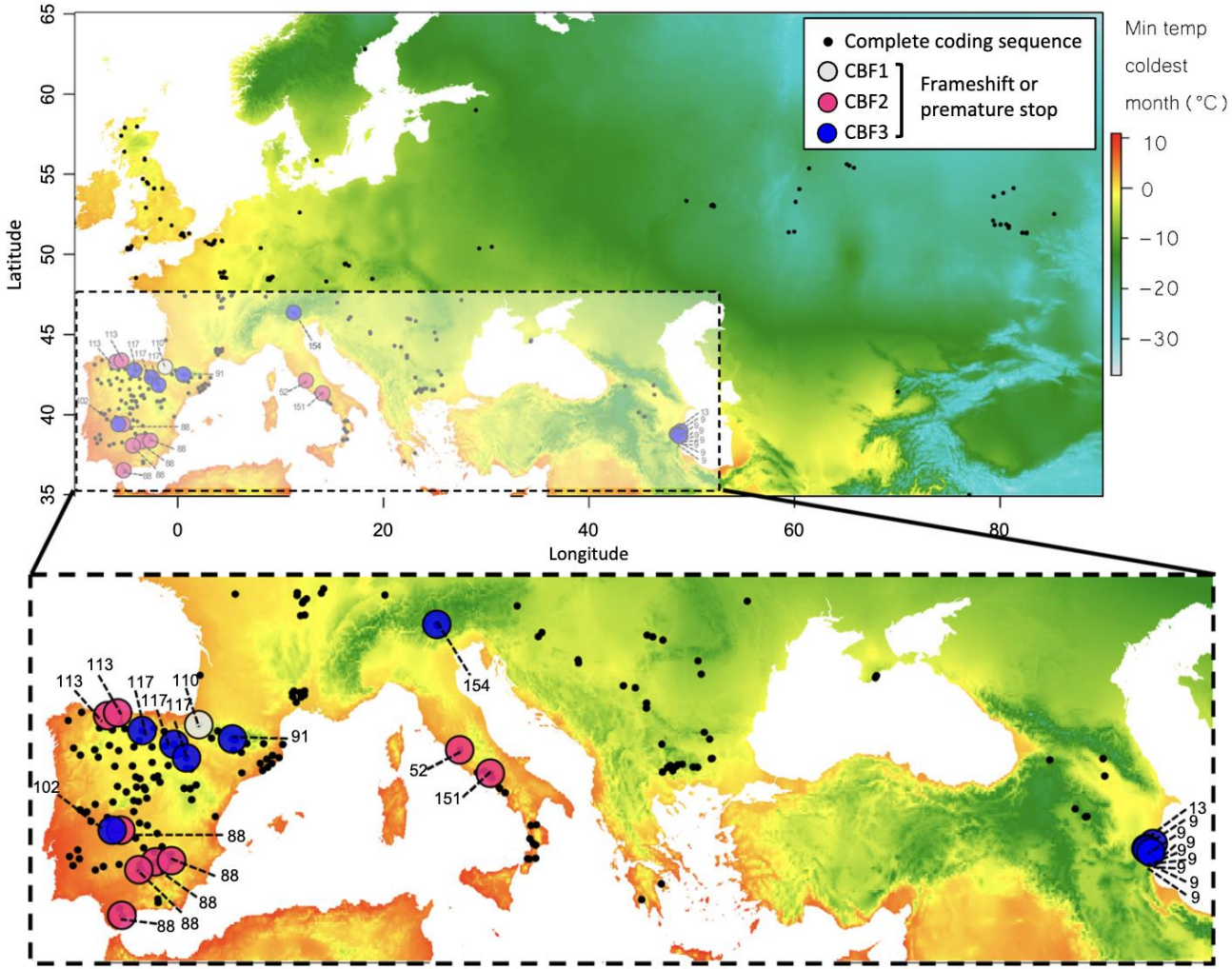


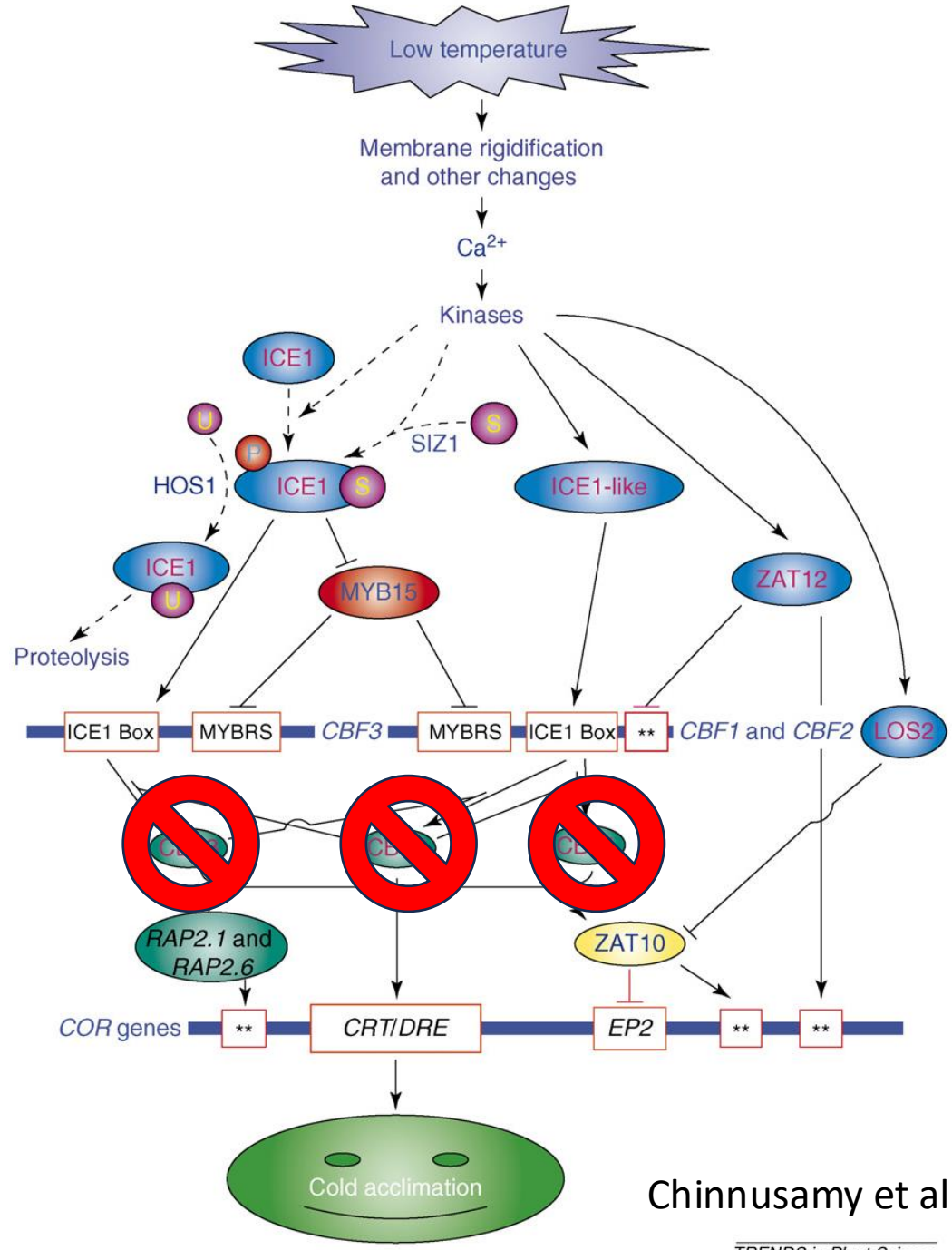


Chinnusamy et al. 2007

TRENDS in Plant Science

CBF genes lost in warm climates





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JOURNAL ARTICLE

InDel variation and contraction of the C-repeat binding factor family contribute to cold sensitivity in *Betula fujianensis*

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Wheat

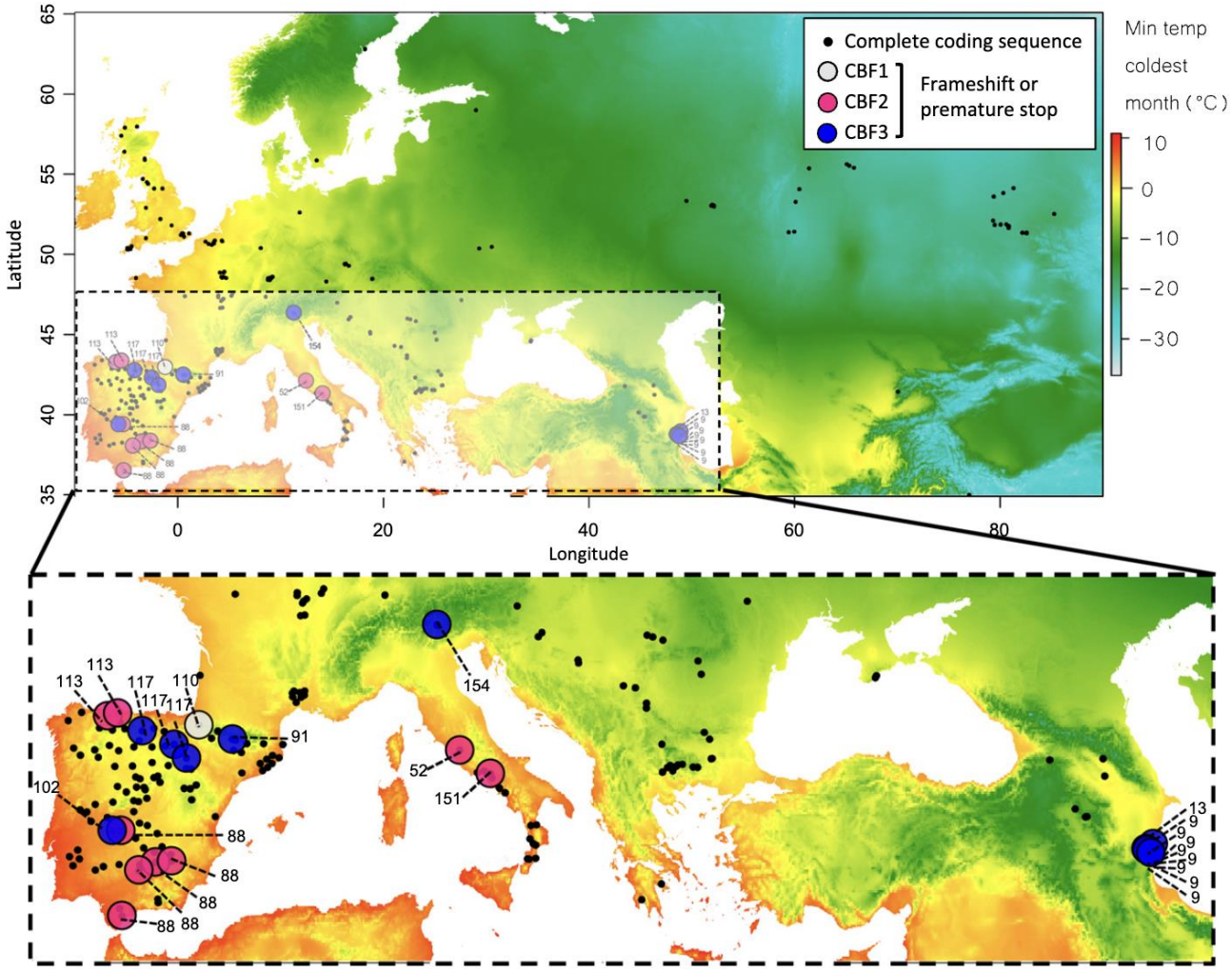
Lettuce



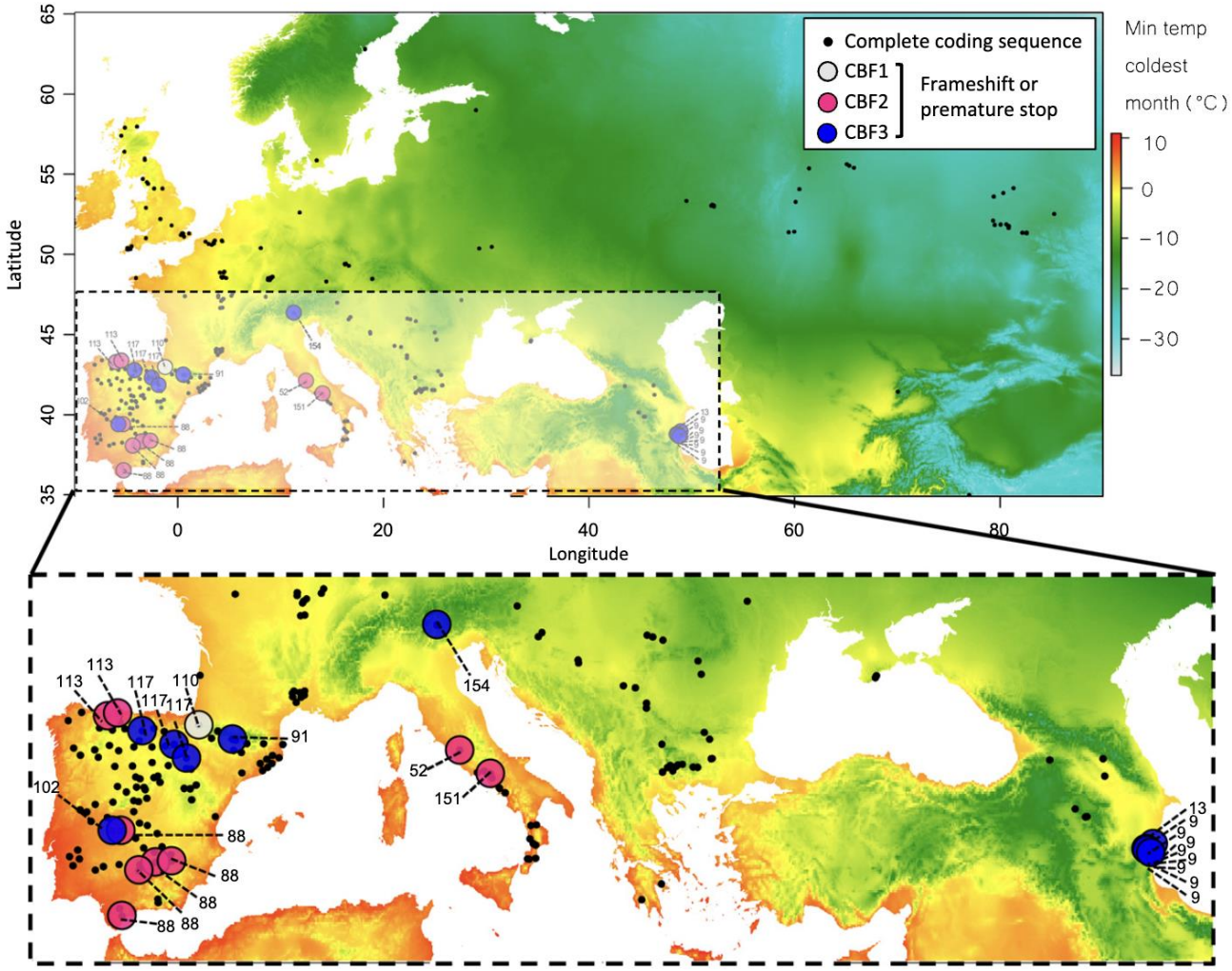
Low frequency of the wild-type freezing-tolerance *LsCBF7* allele among lettuce population suggests a negative selection during domestication and breeding

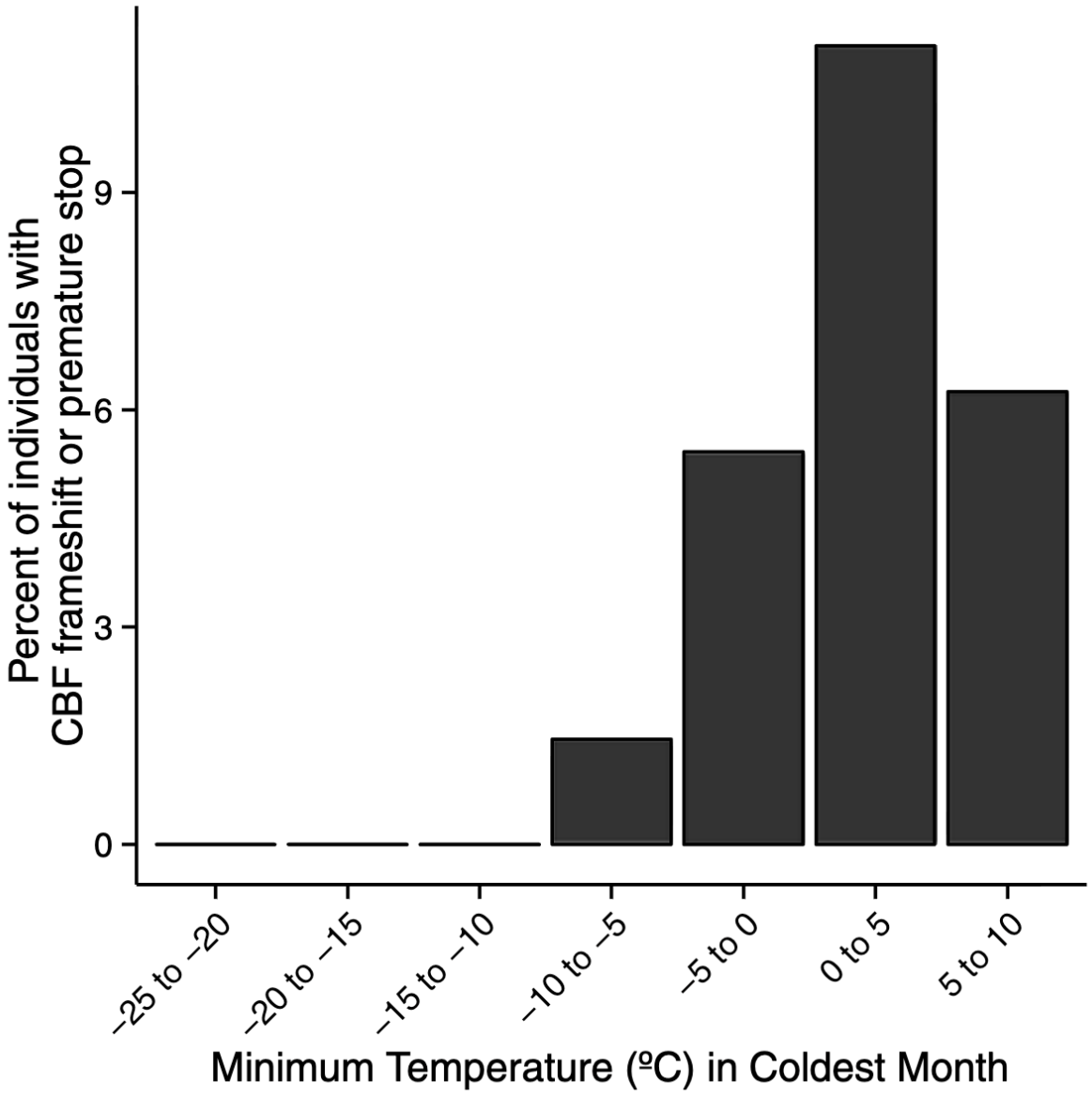
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CBF genes lost in warm climates



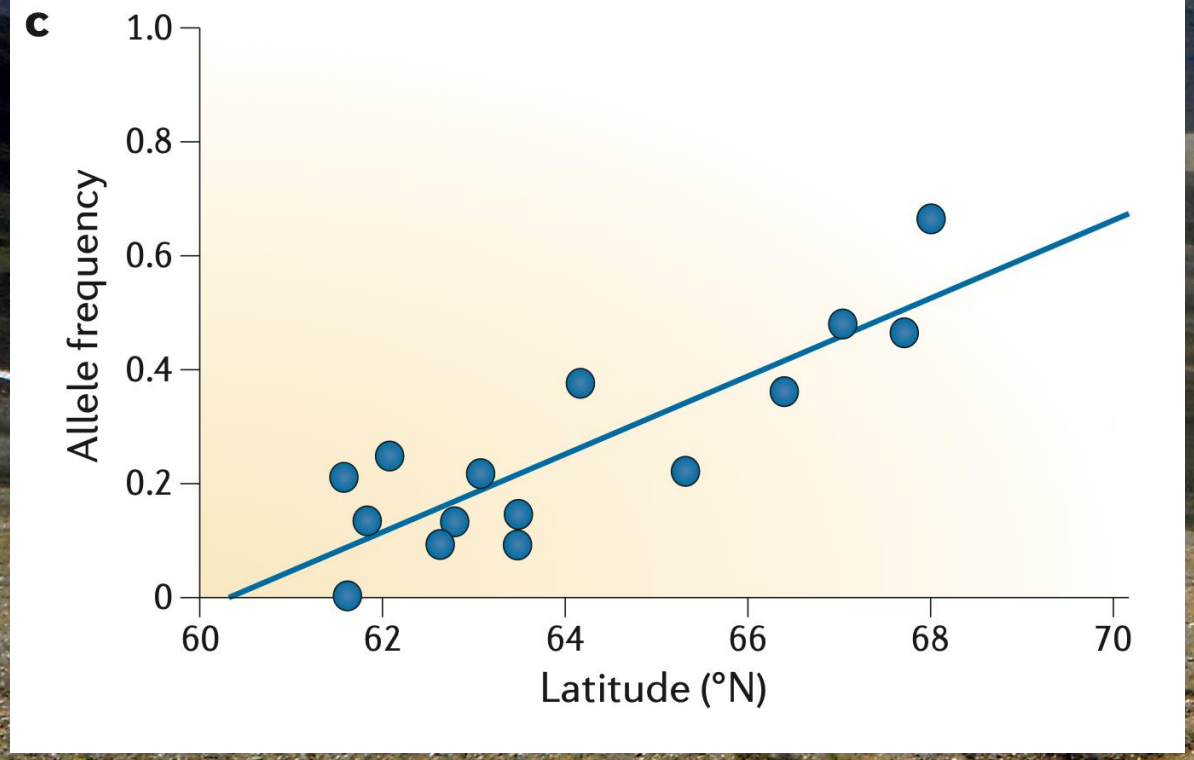
CBF genes lost in warm climates





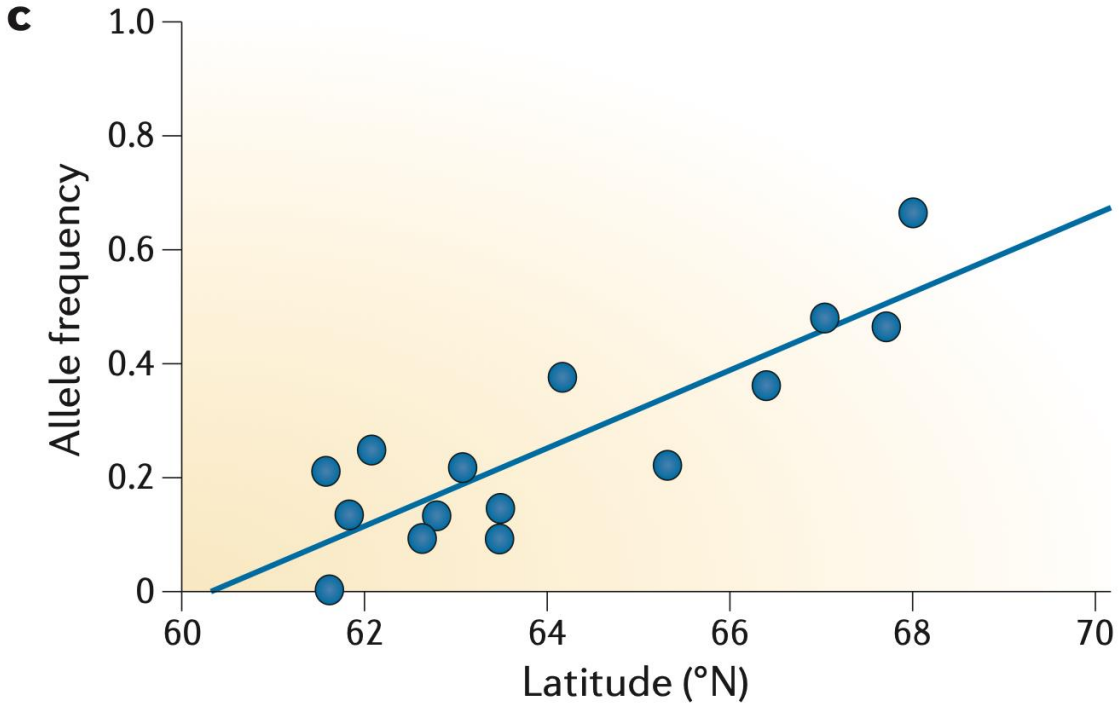
Evidence of local adaptation: alleles

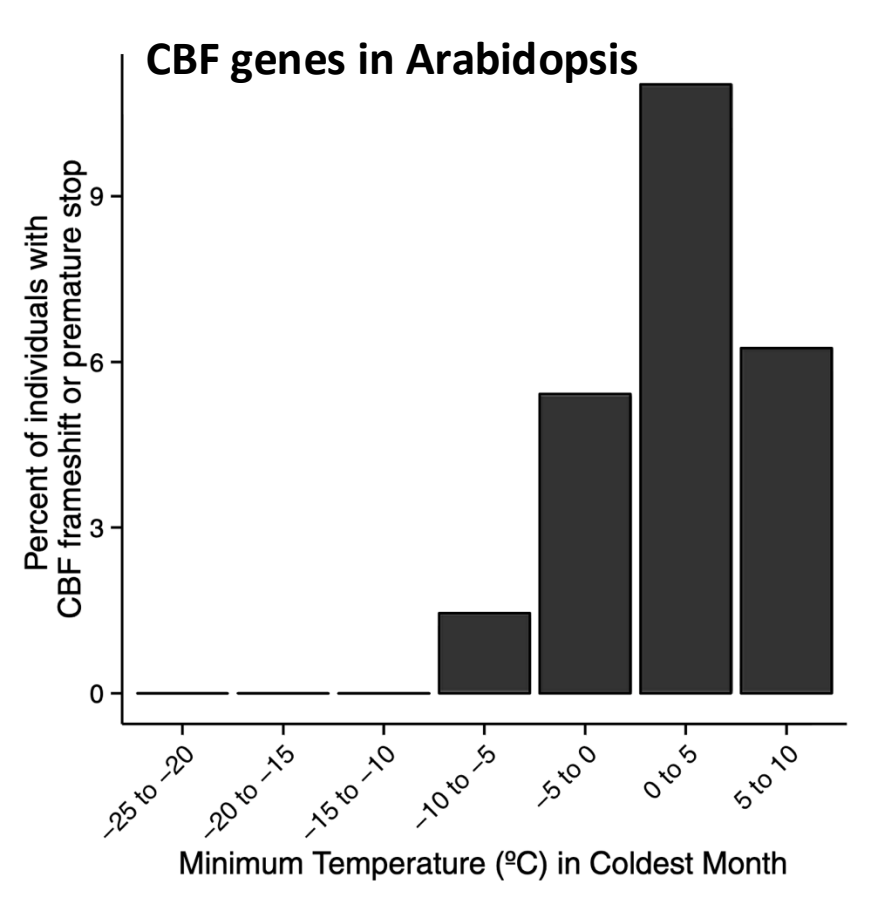
Climate-allele associations



Evidence of local adaptation: alleles

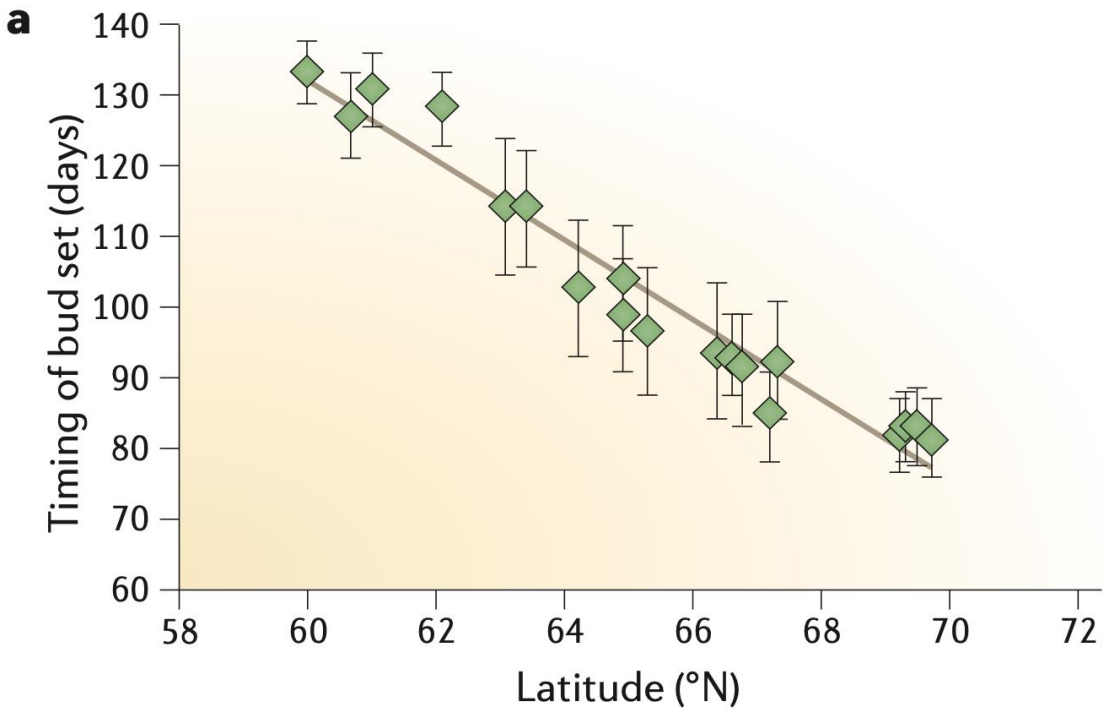
Climate-allele associations

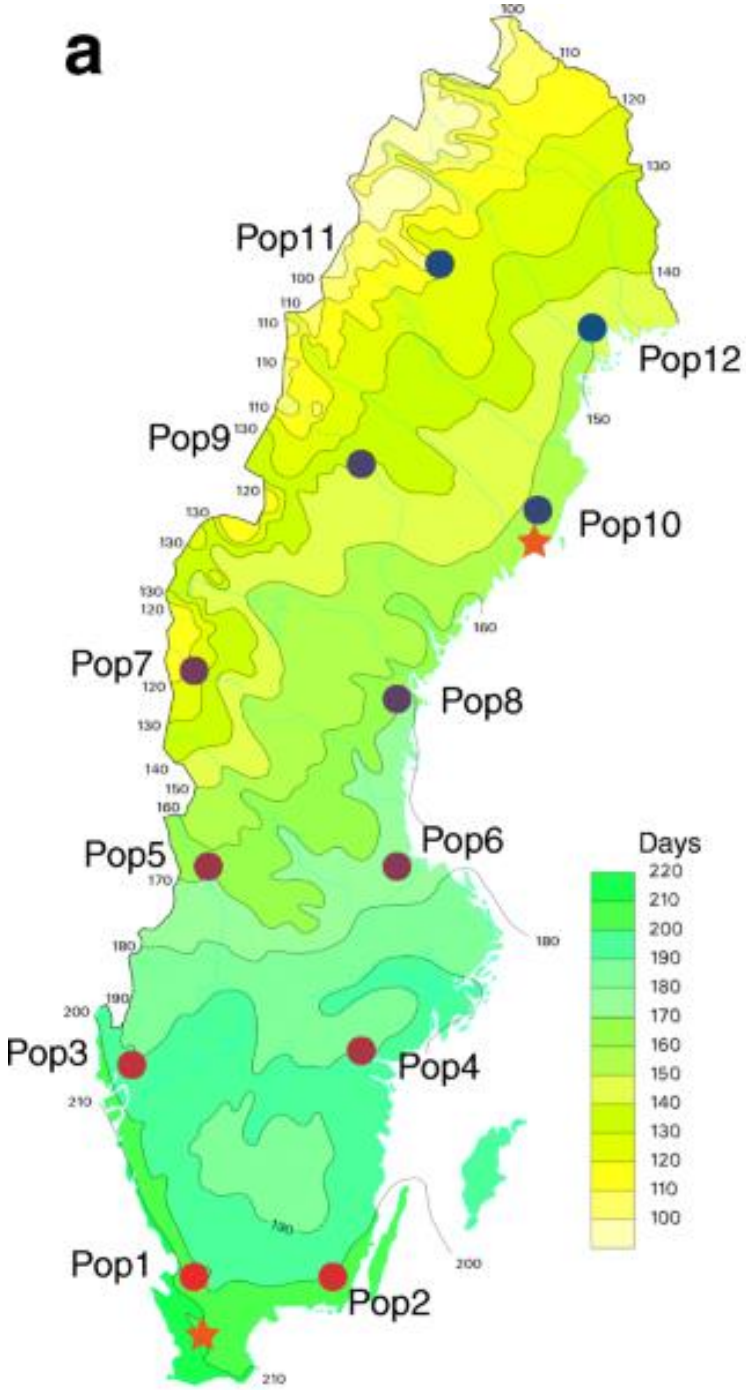




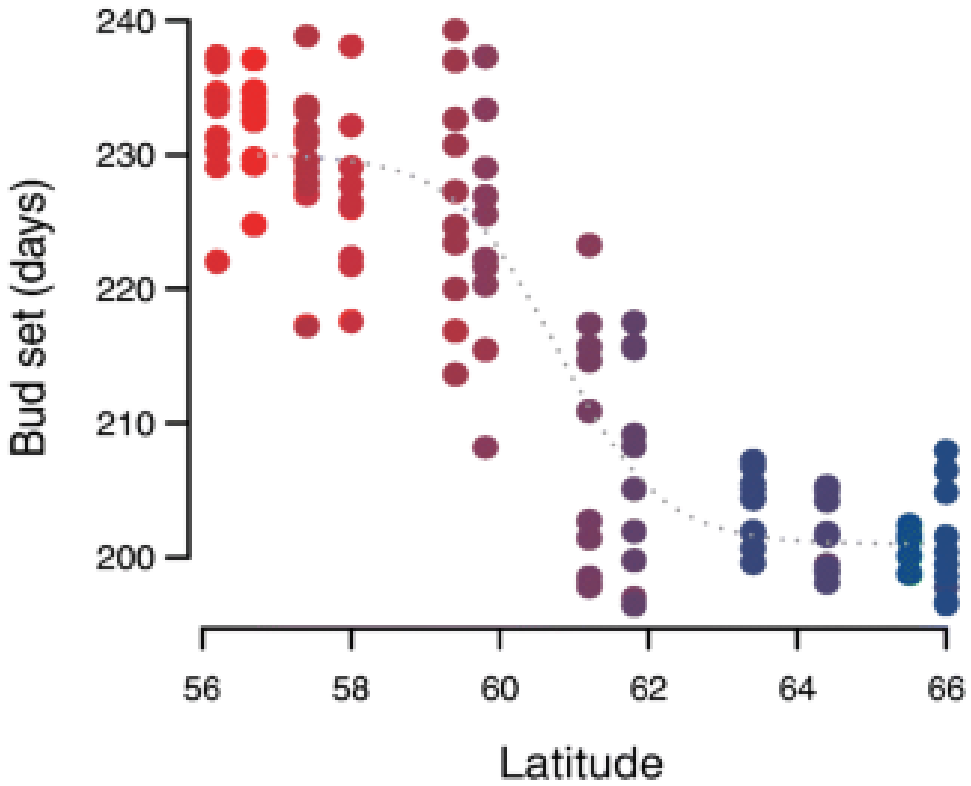
Evidence of local adaptation: traits

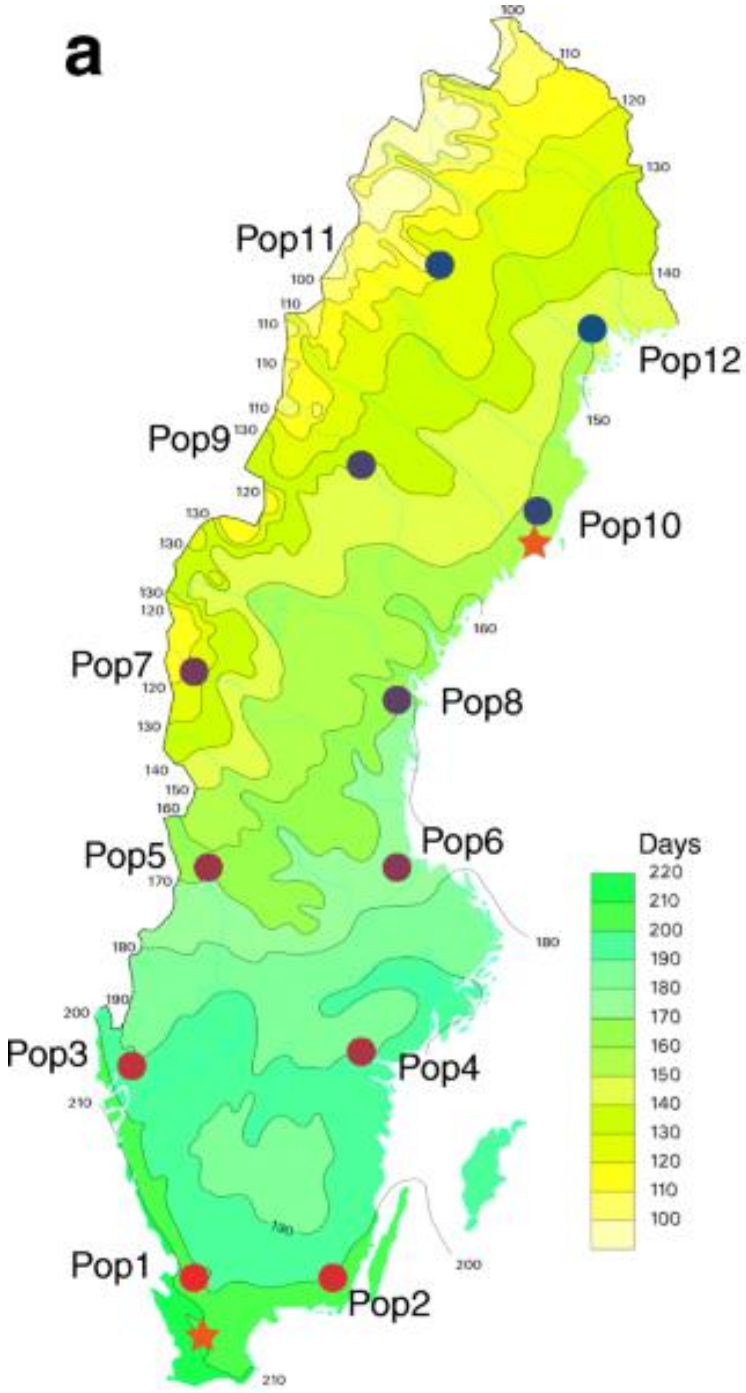
Climate-trait associations

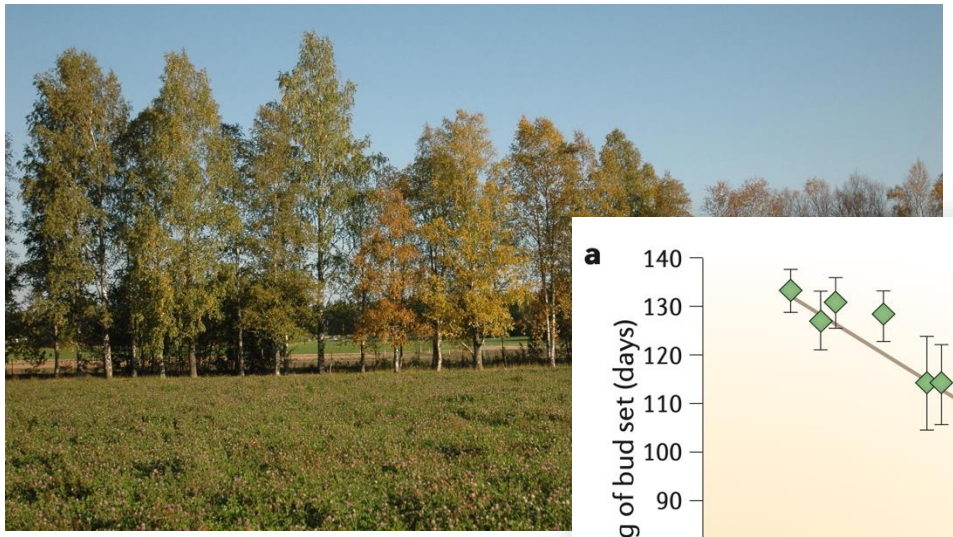


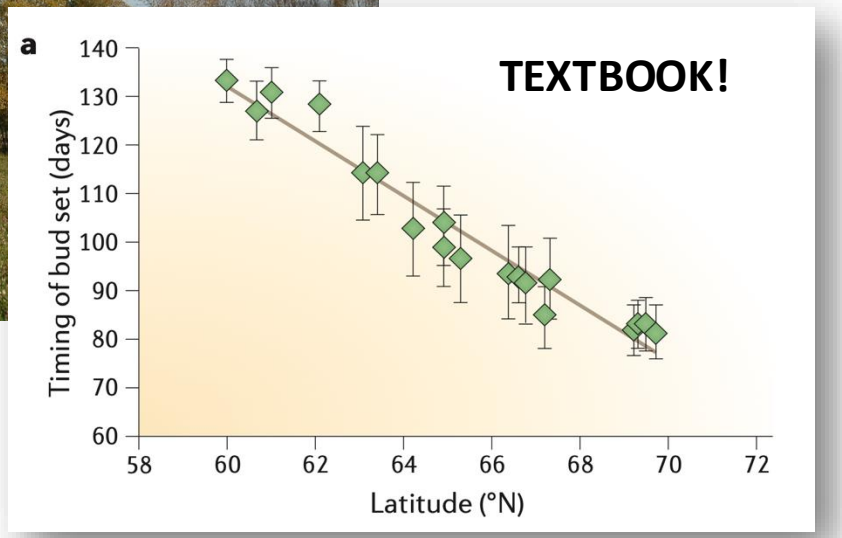

a
Pop11
Pop12
Pop9
Pop10
Pop7
Pop8
Pop5
Pop6
Pop3
Pop4
Pop1
Pop2
Days
220
210
200
190
180
170
160
150
140
130
120
110
100

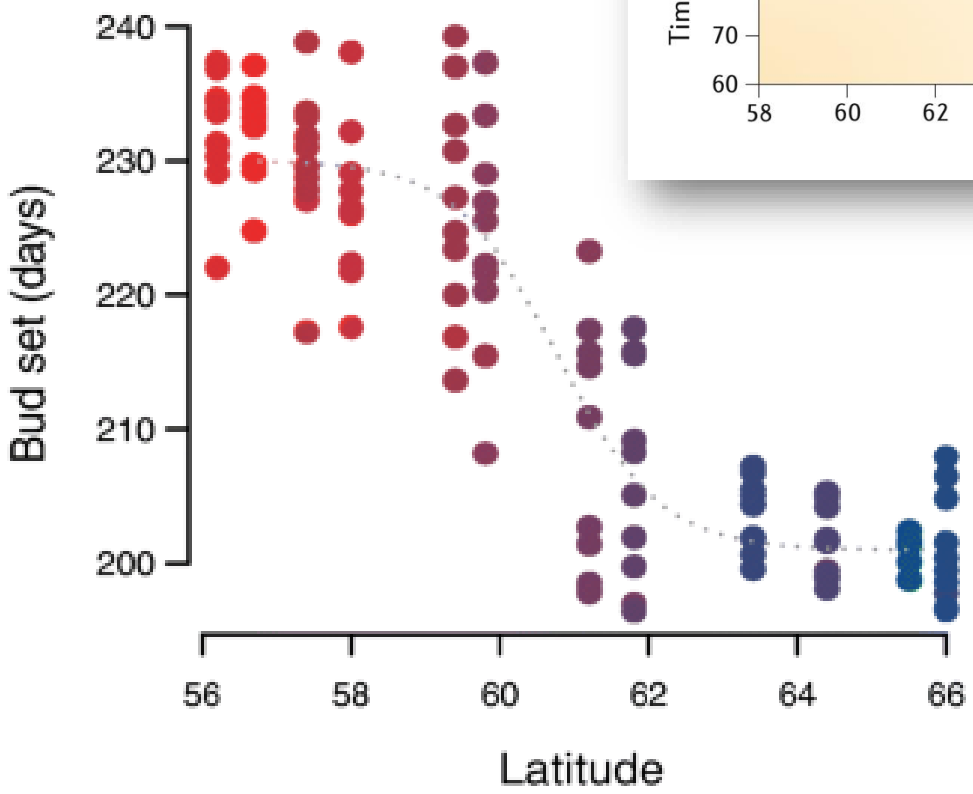



Bud set (days)
240
230
220
210
200
56
58
60
62
64
66
Latitude

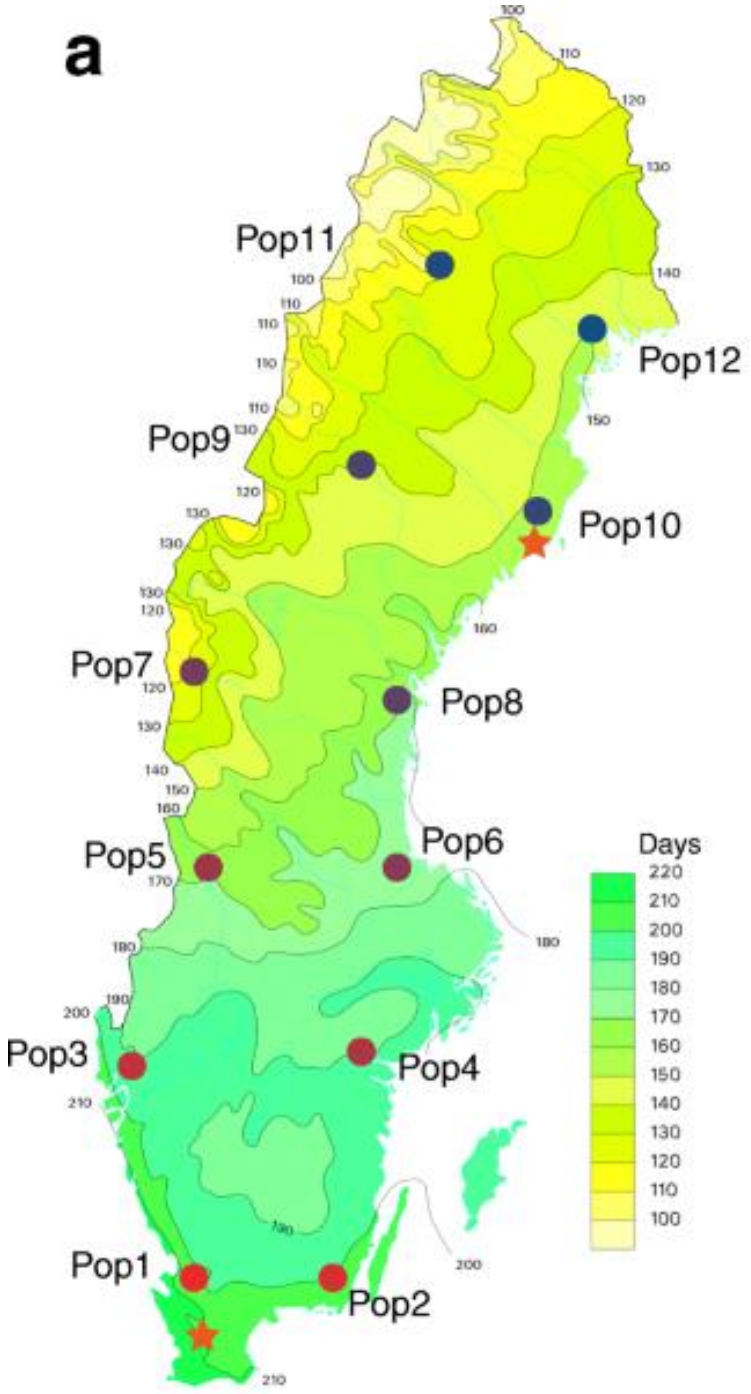

a
Pop11
Pop12
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Days
220
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190
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160
150
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120
110
100

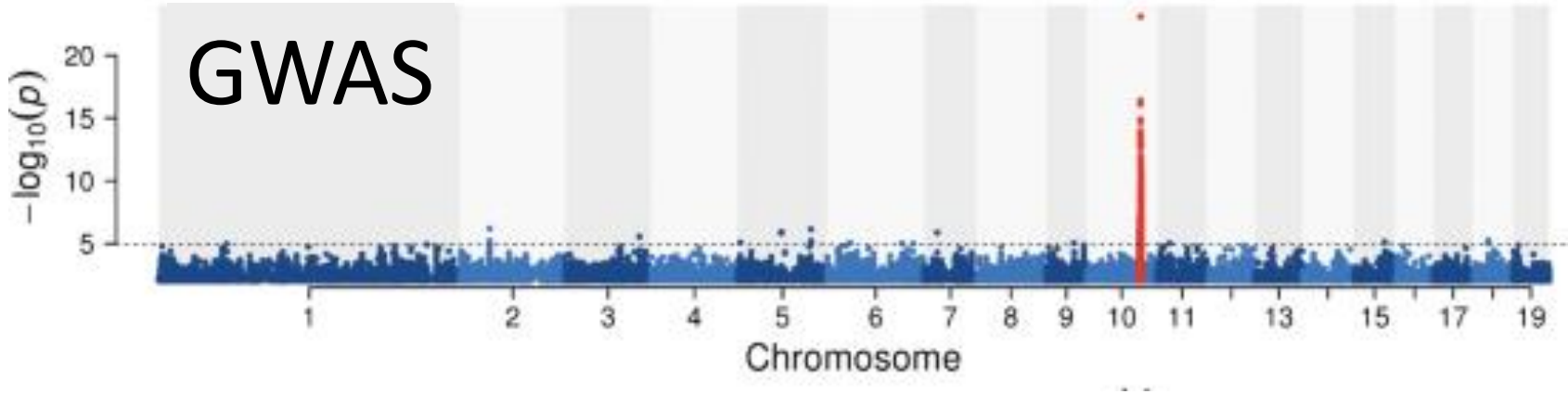



a
TEXTBOOK!
Timing of bud set (days)
140
130
120
110
100
90
80
70
60
58
60
62
64
66
68
70
72
Latitude (°N)

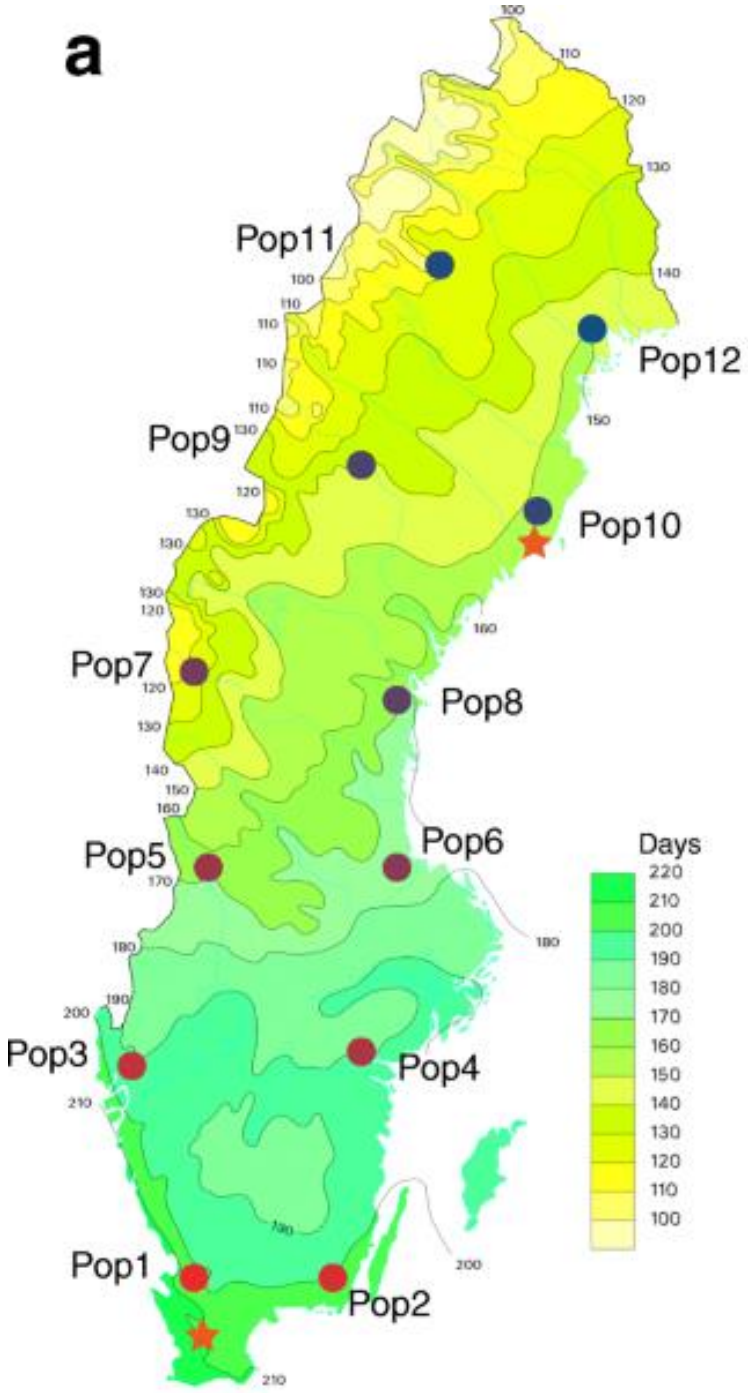

Bud set (days)
240
230
220
210
200
56
58
60
62
64
66
Latitude

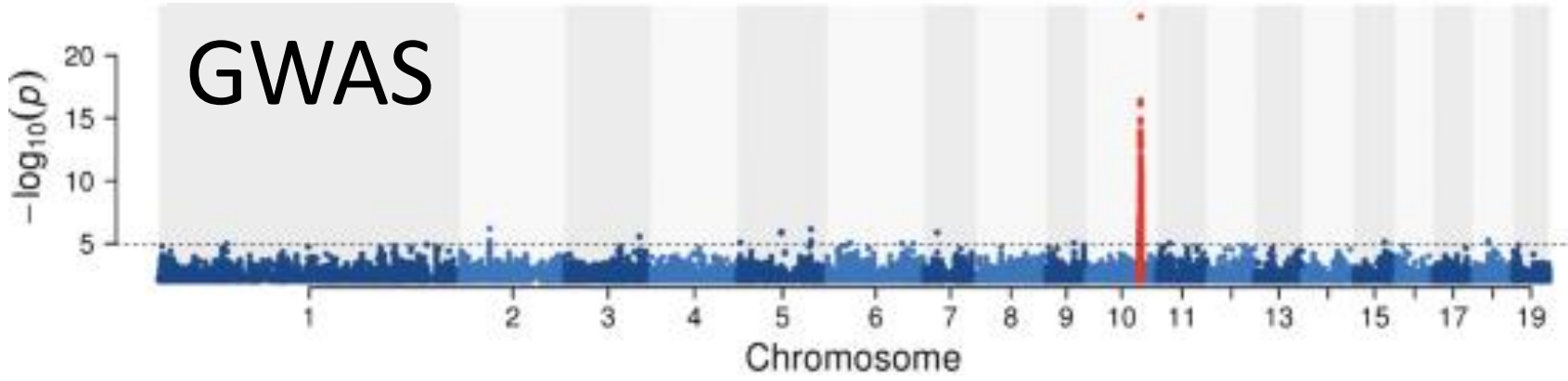
FT gene allele causes change in budset timing

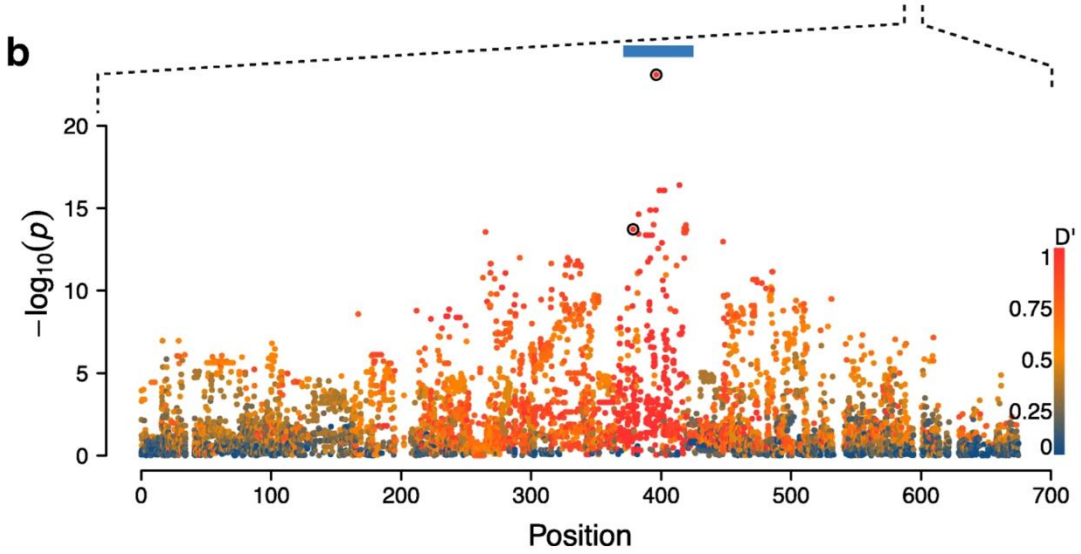




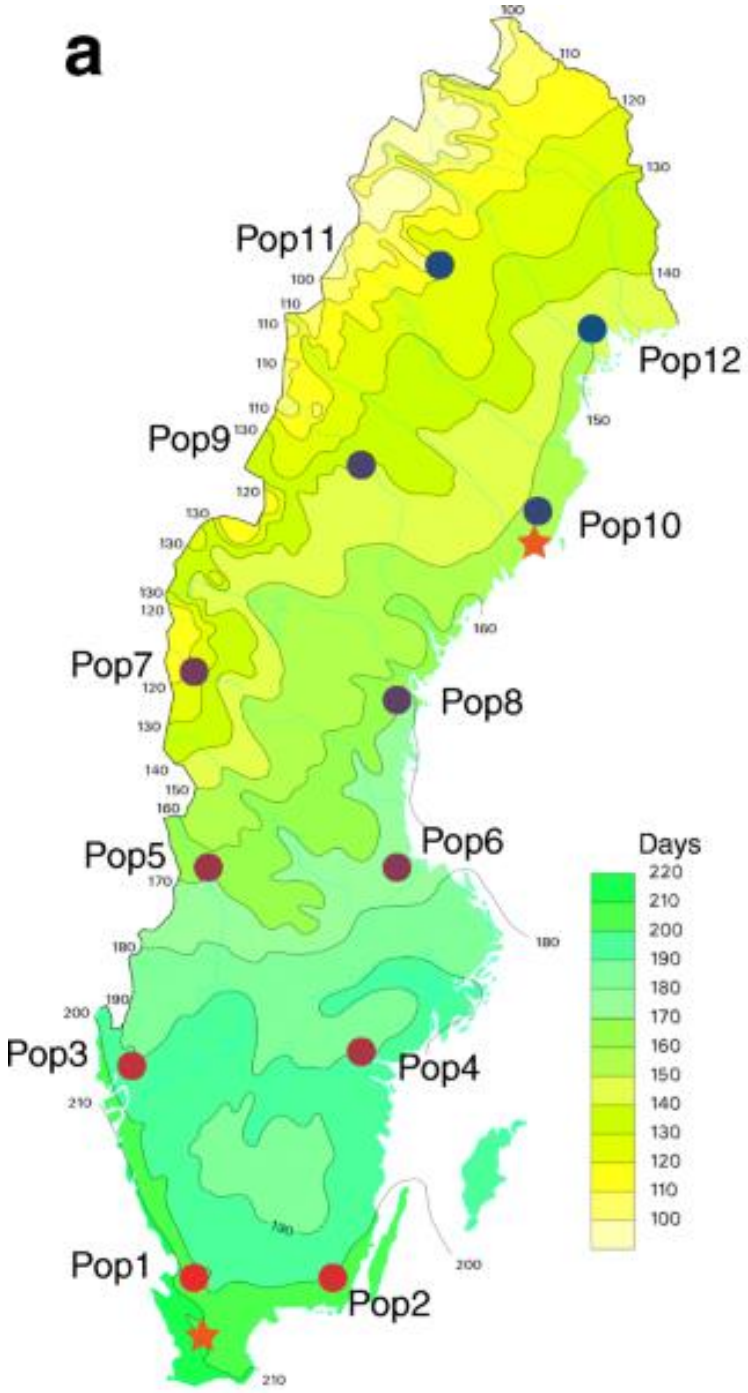
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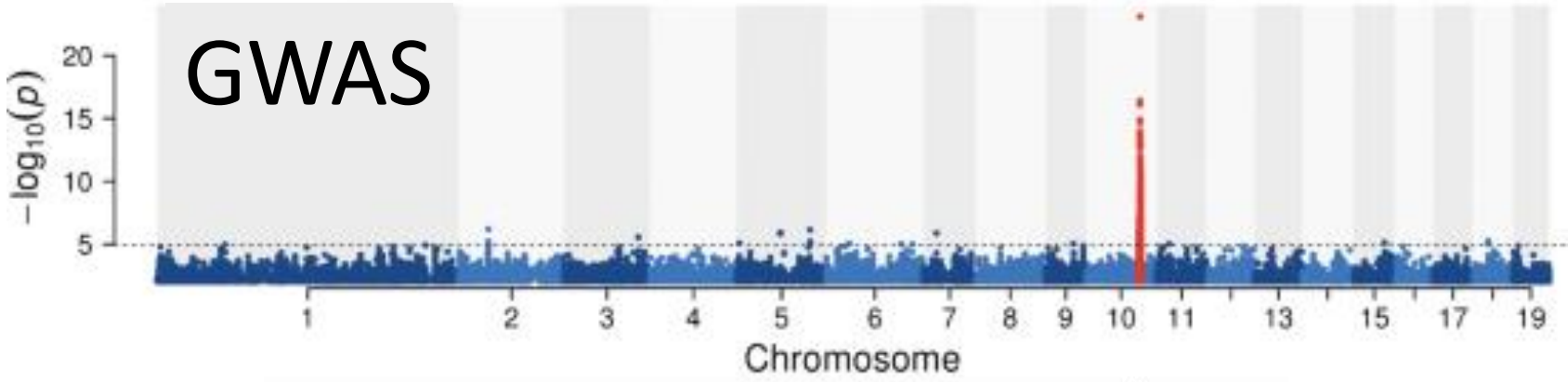


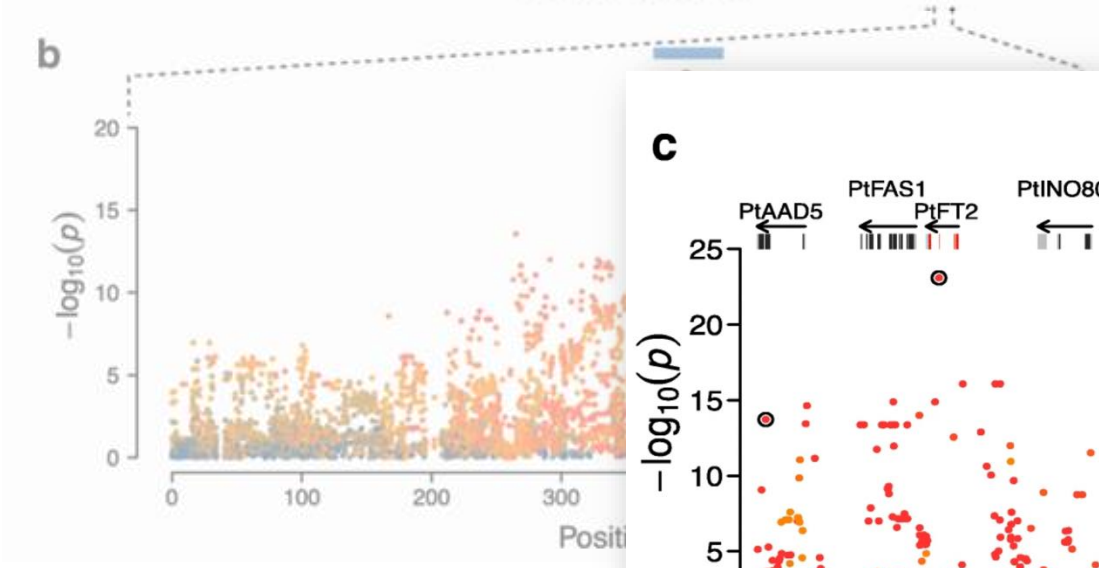


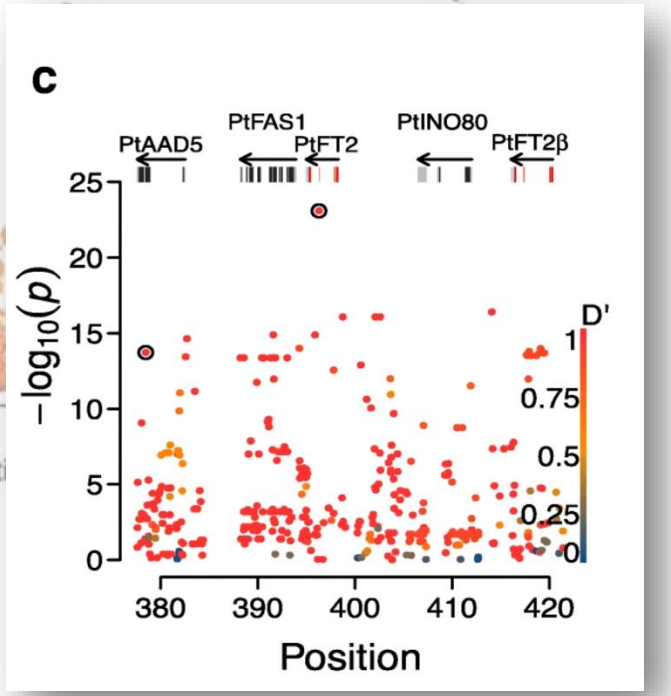
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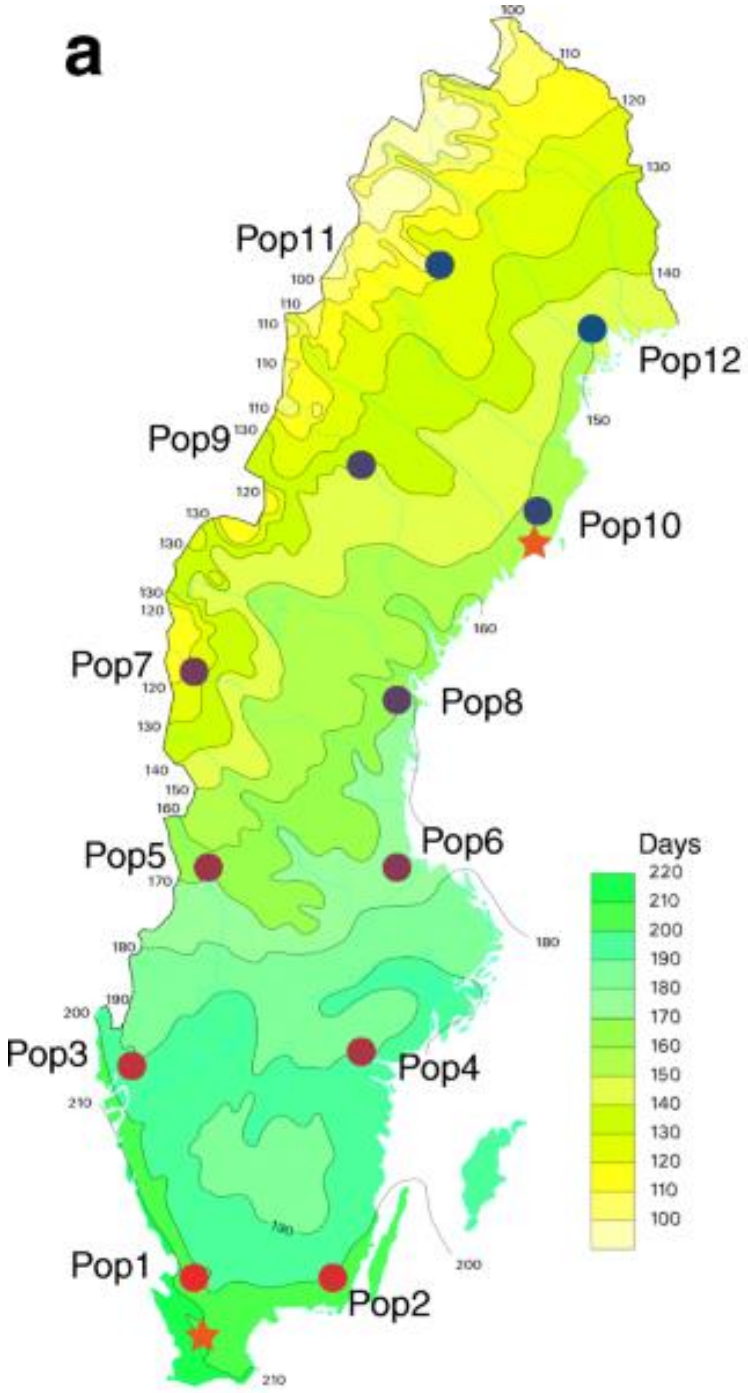
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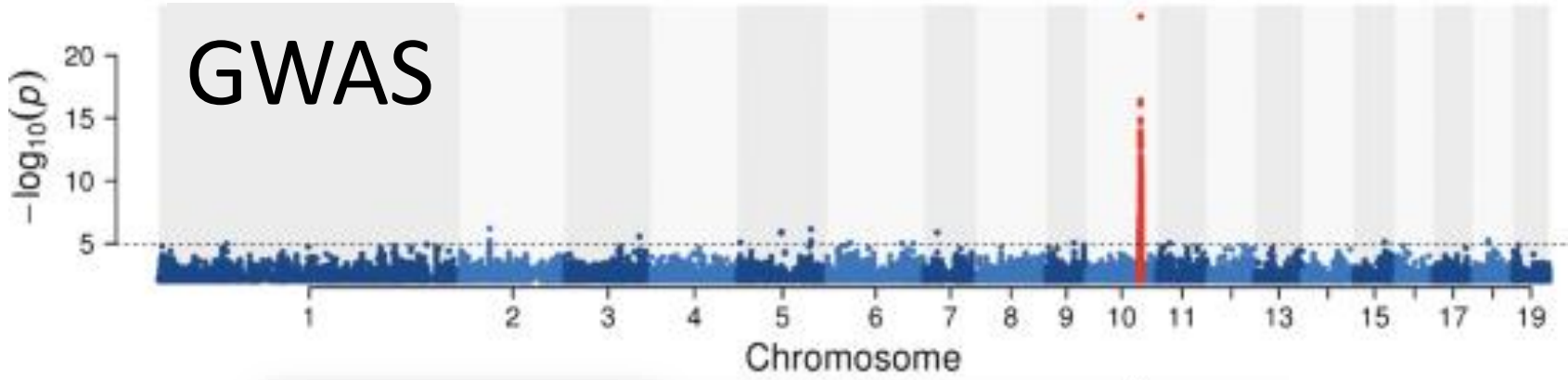


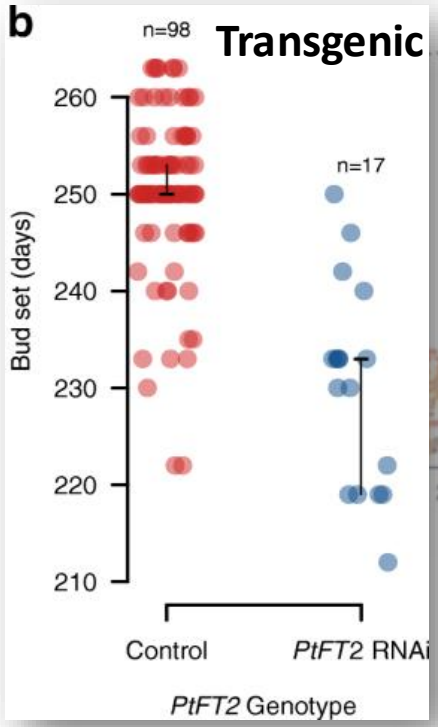


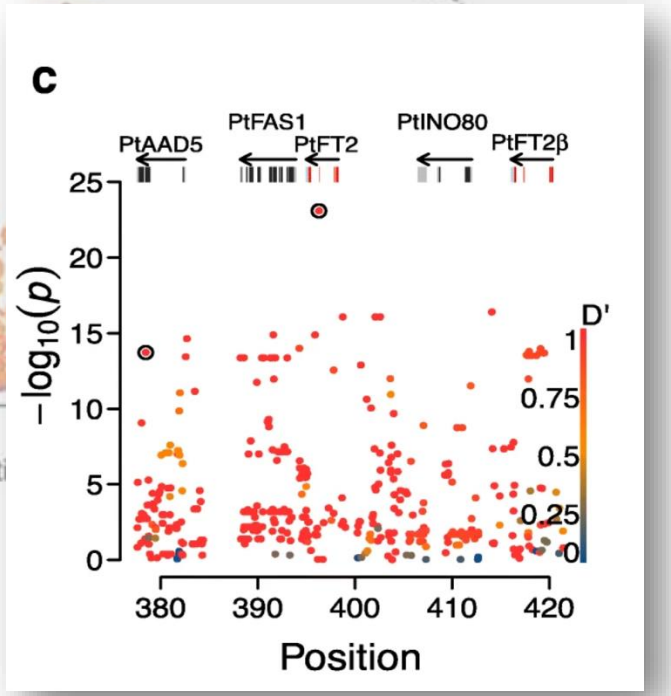


FT gene allele causes change in budset timing

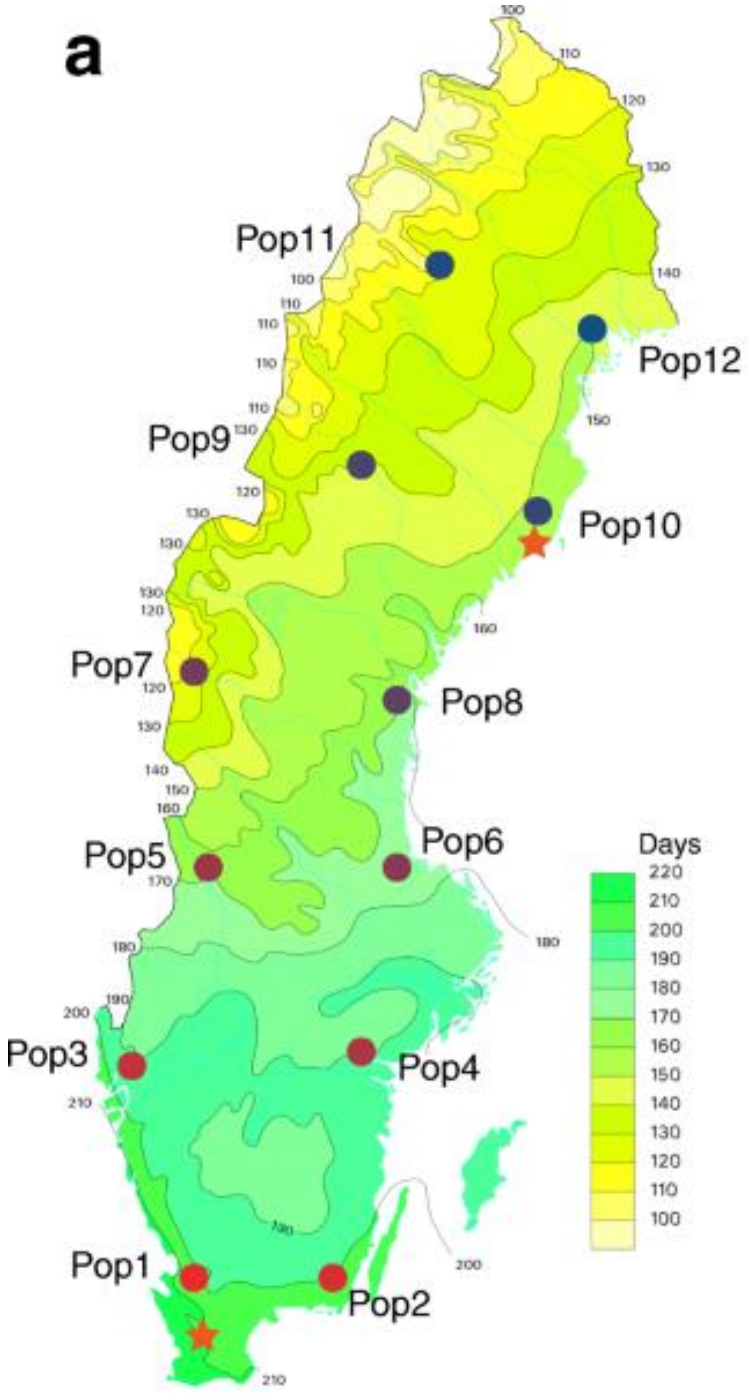




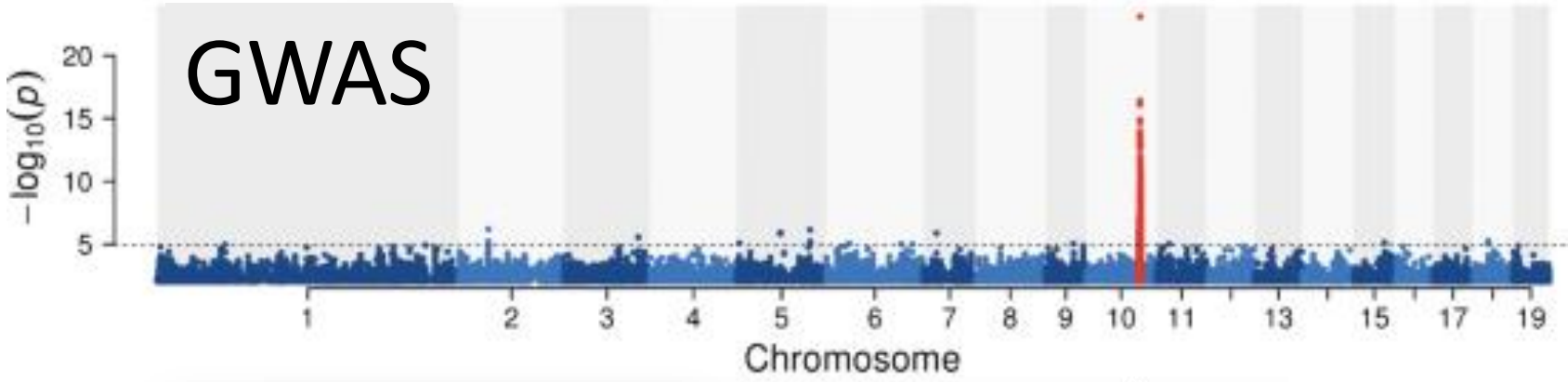


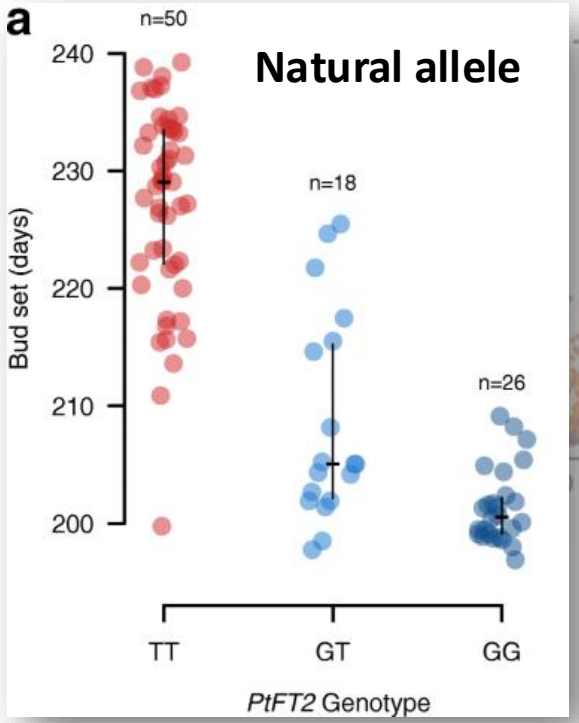


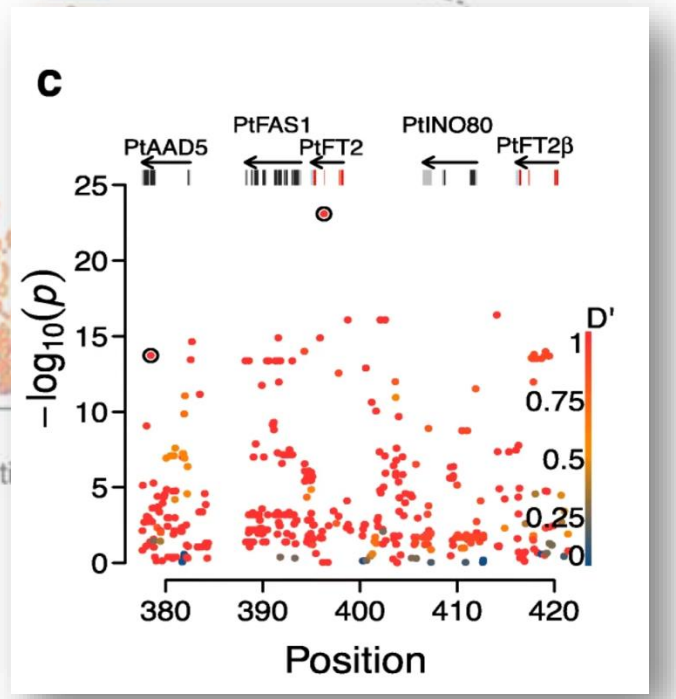
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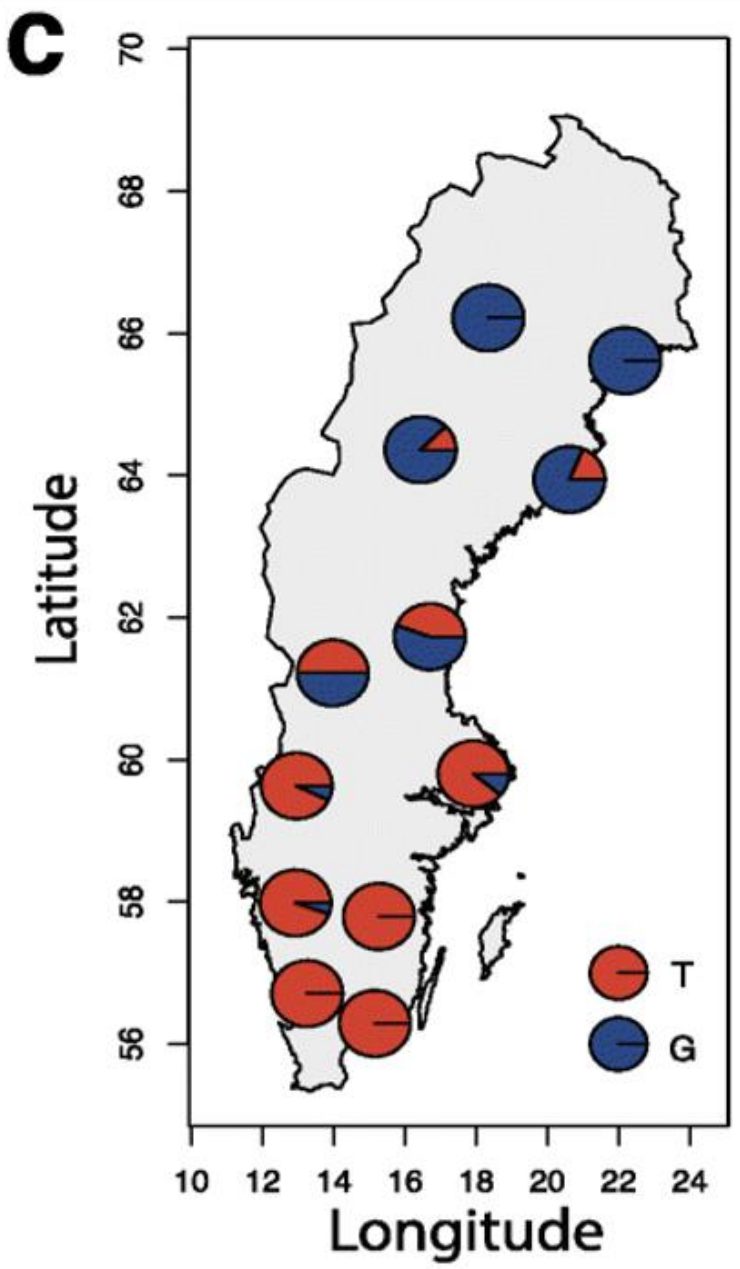
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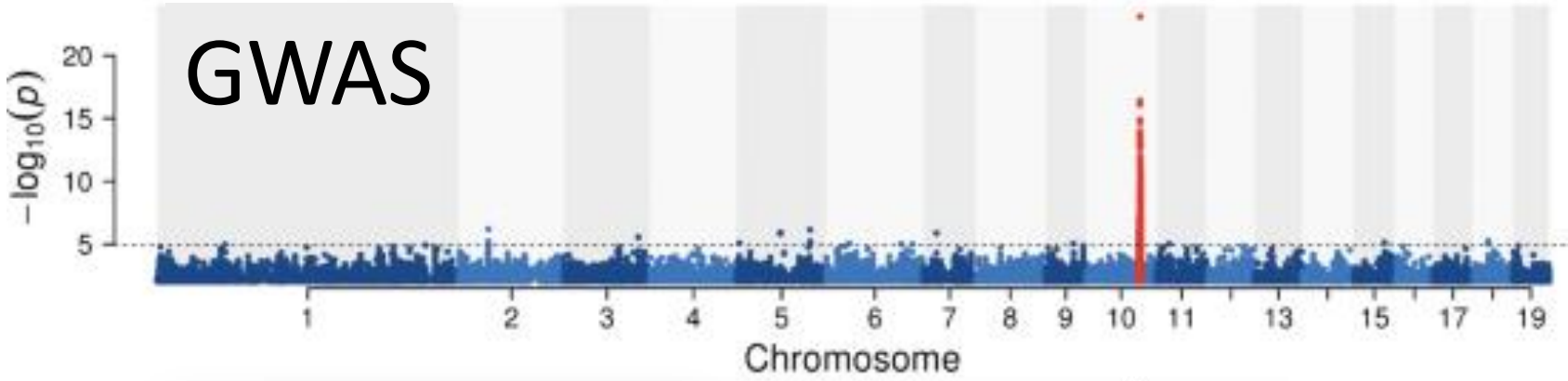


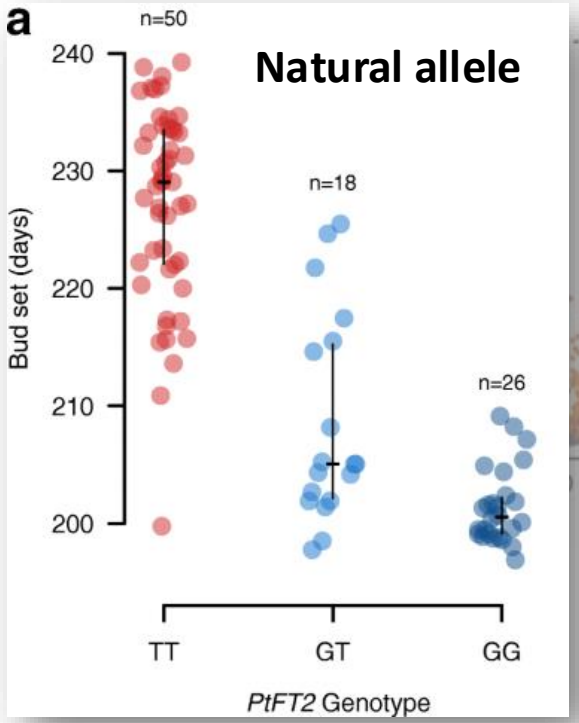


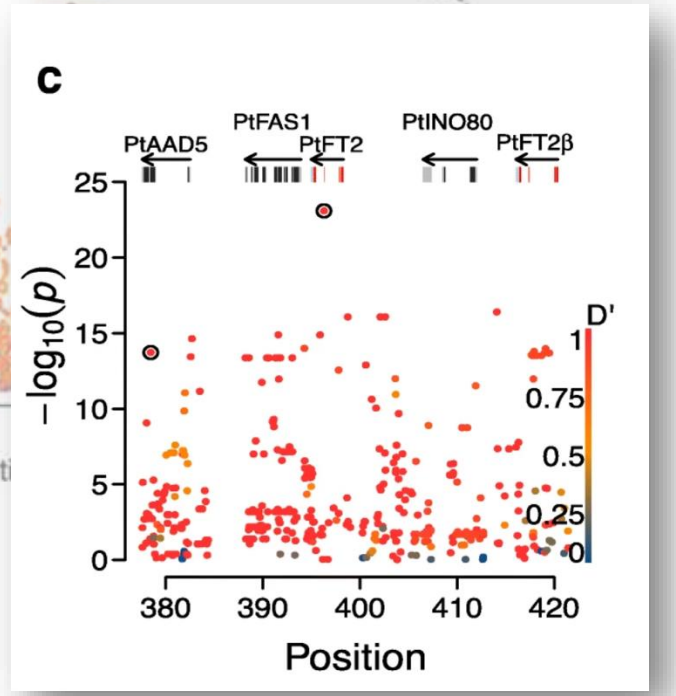
FT gene allele correlated with climate!



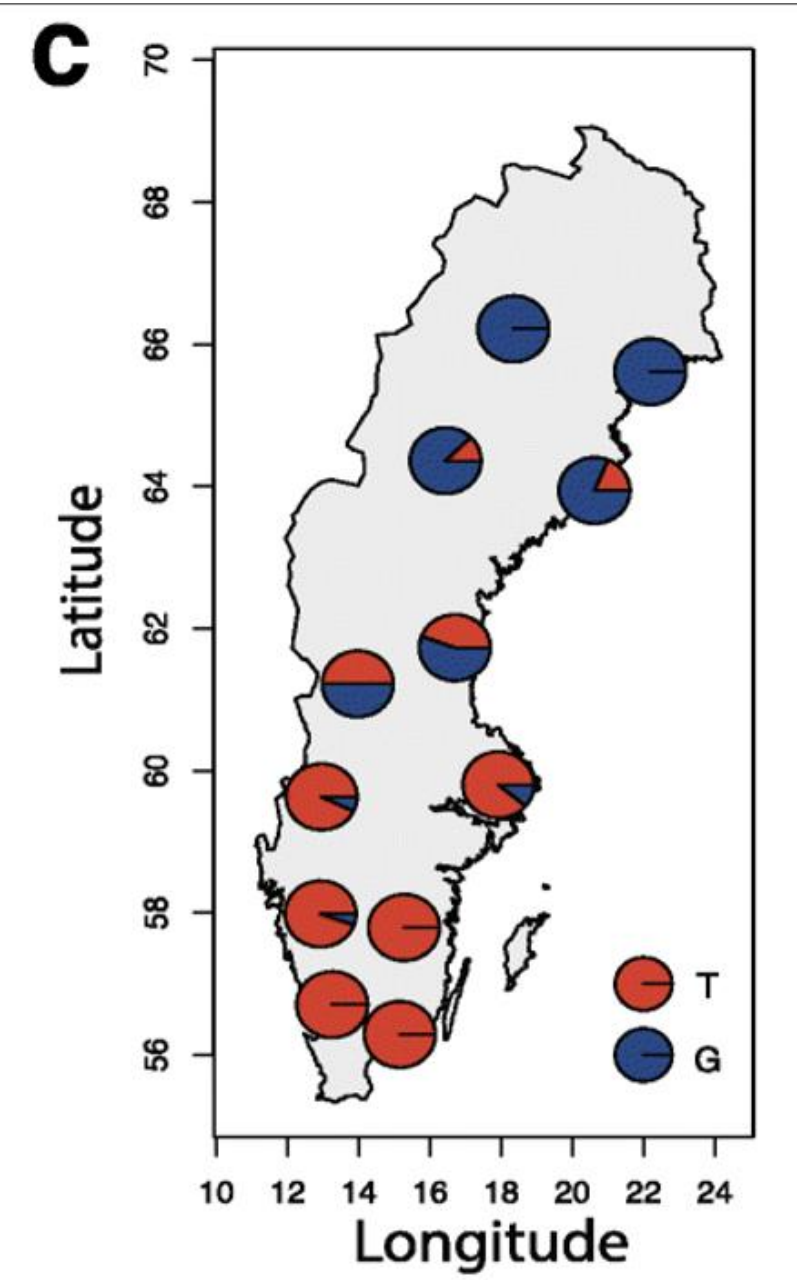
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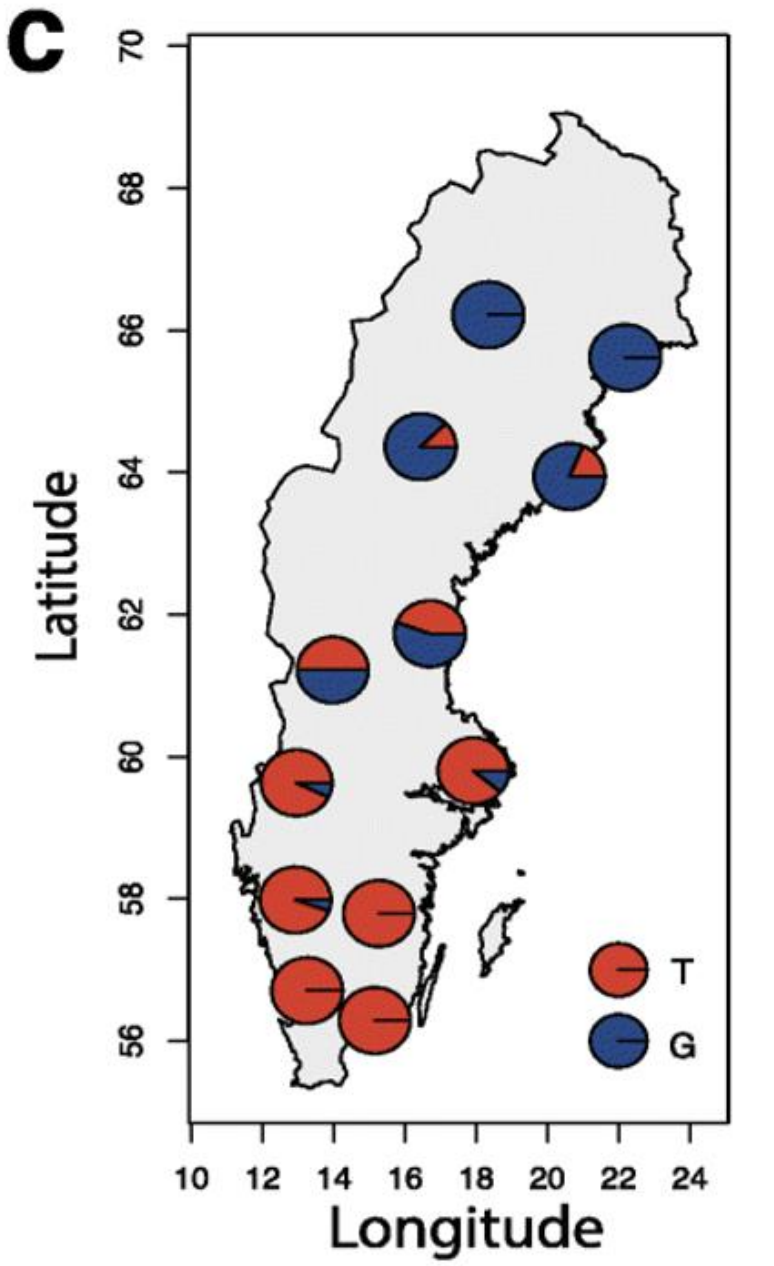
Climate-allele association

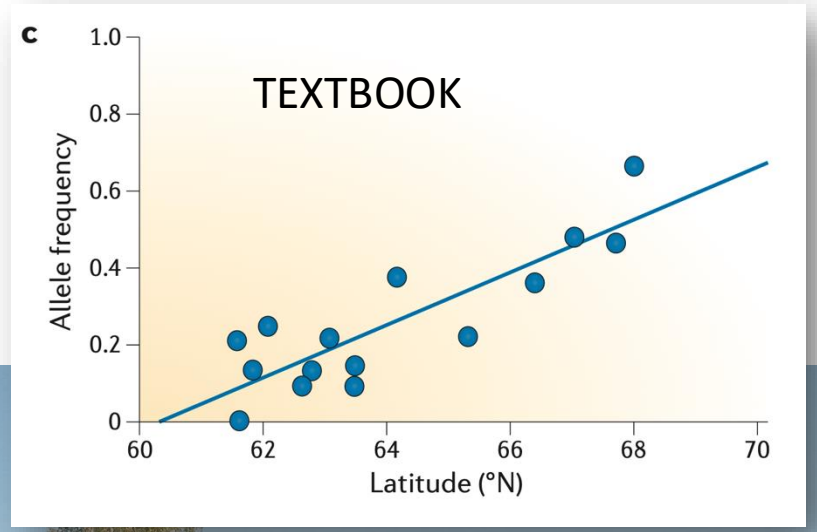


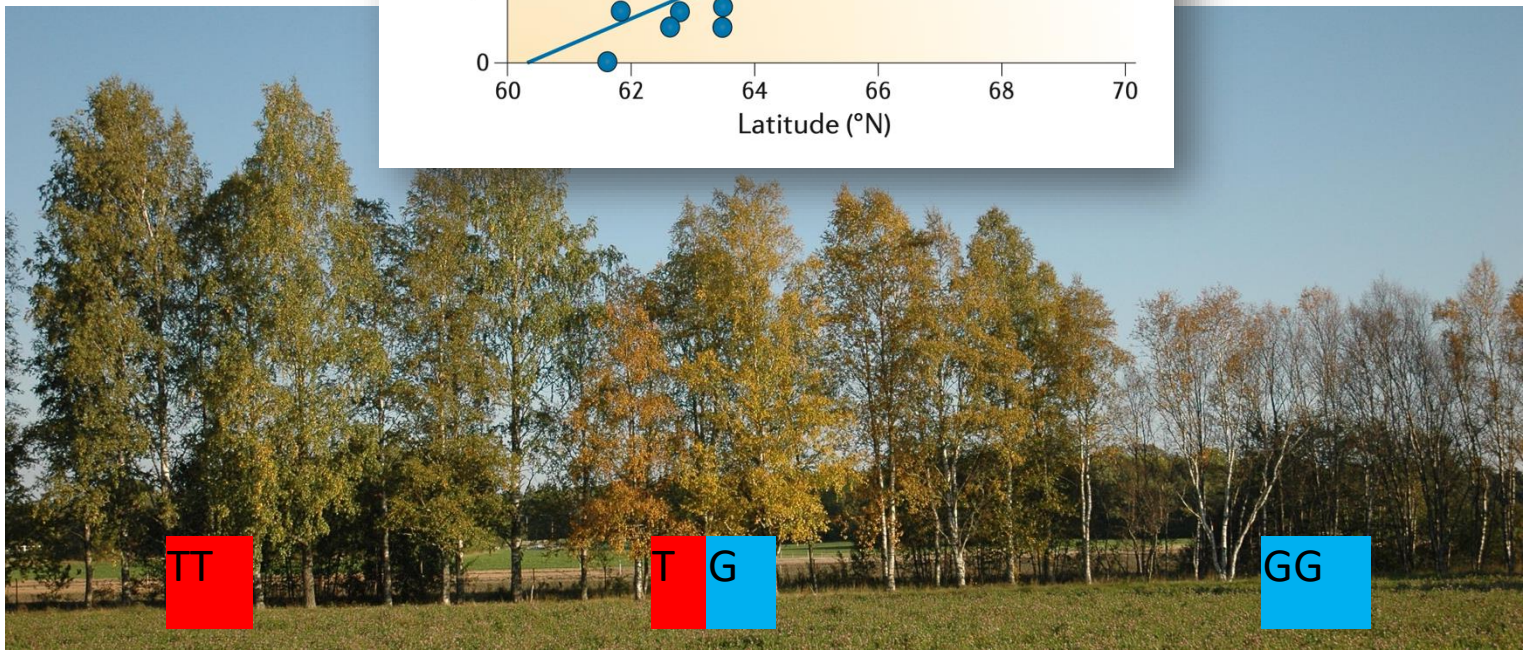
Mutation in FT2 gene



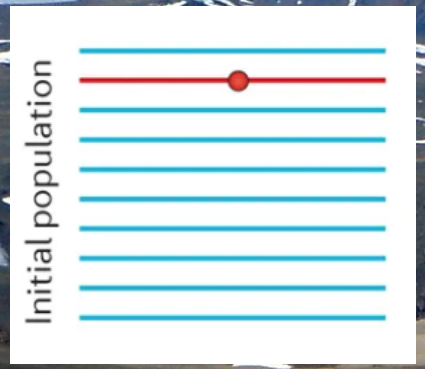
Climate-allele association



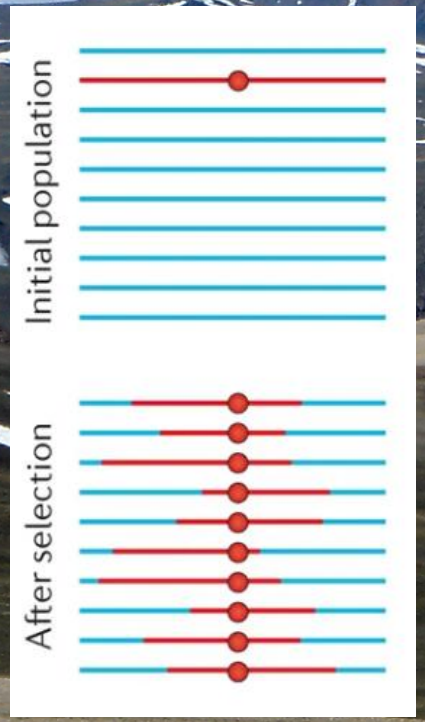




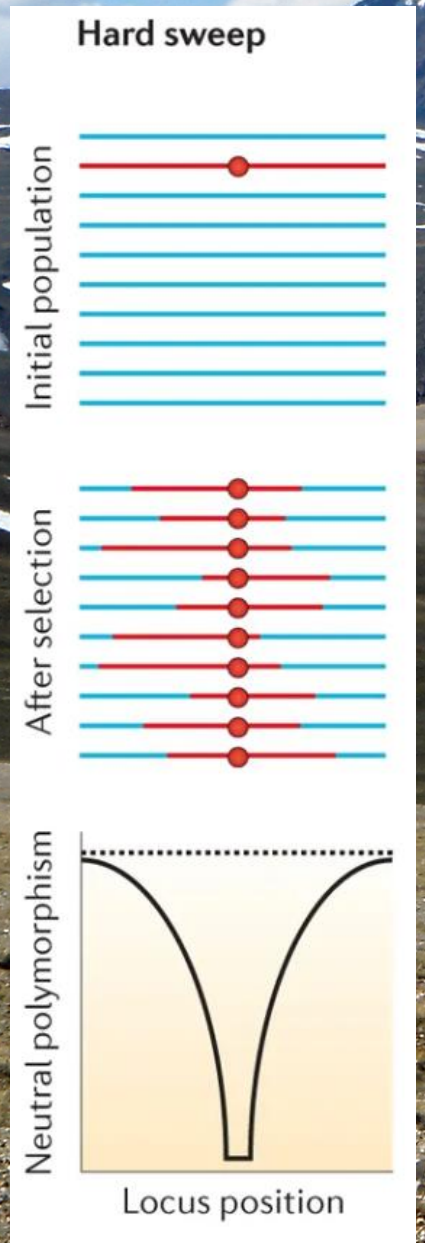
Evidence of natural selection



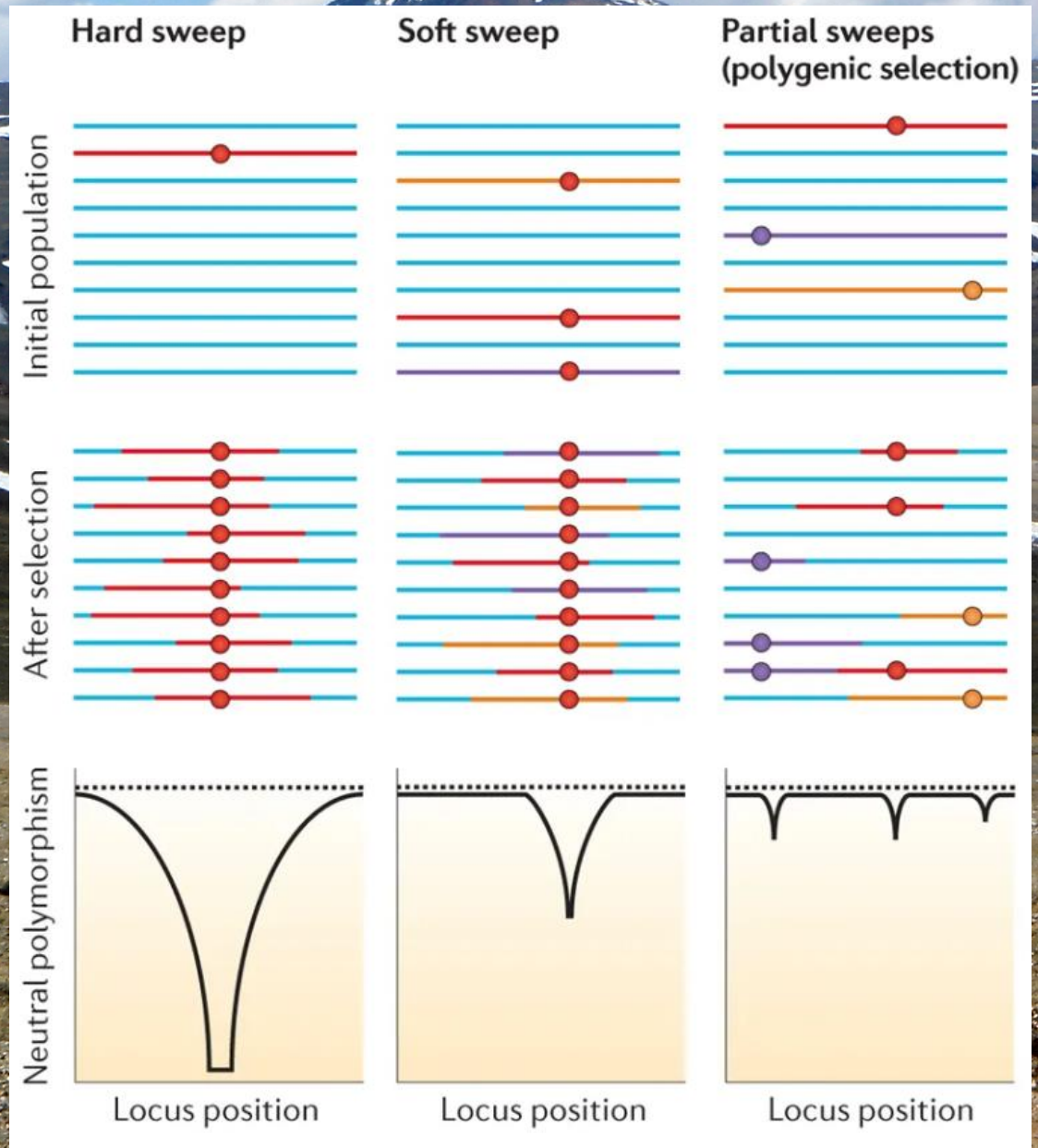
Evidence of natural selection



Evidence of natural selection



Evidence of natural selection



Population genetics

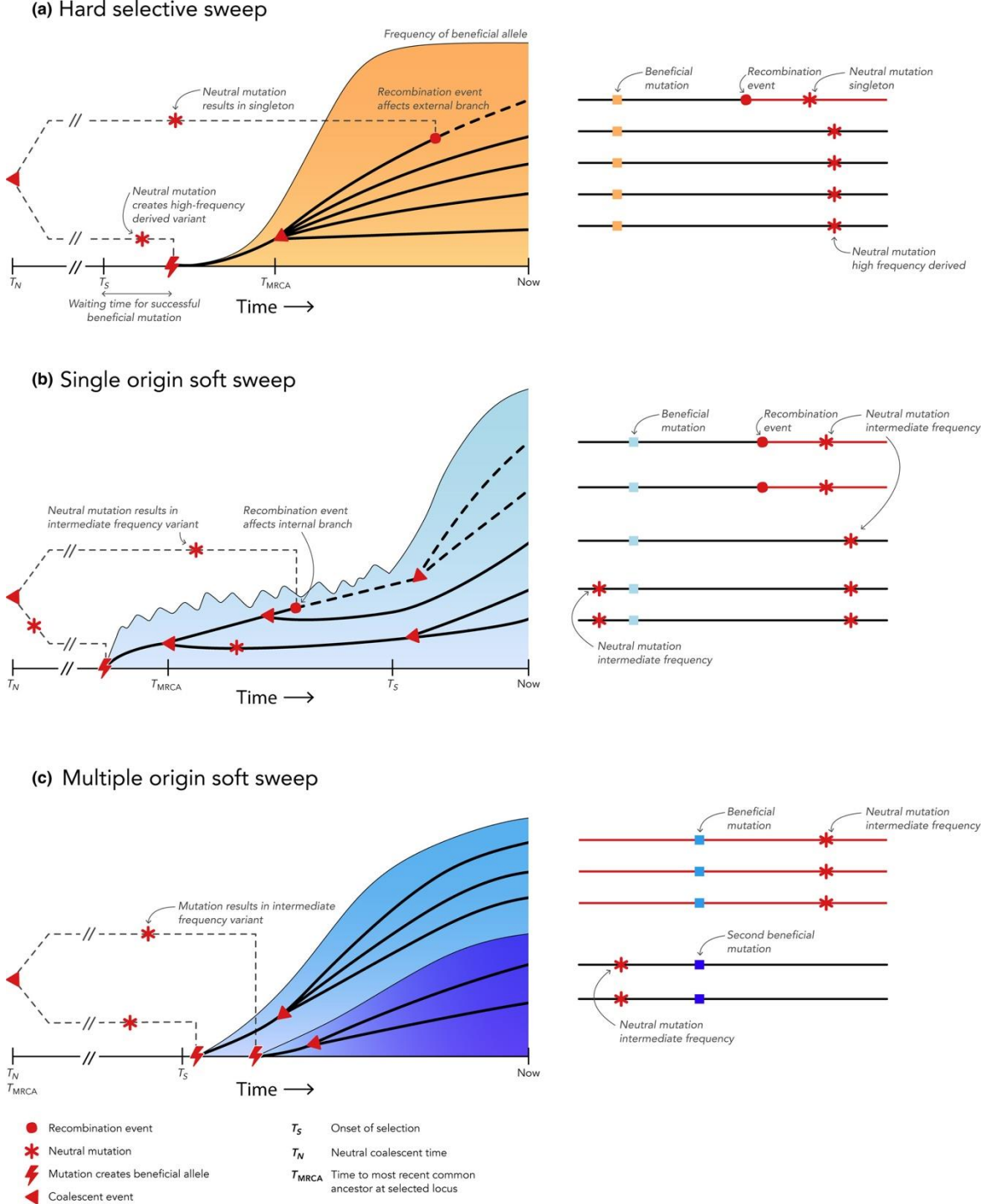
Reduced nucleotide diversity (π) — Variation is “wiped out” around the selected allele because the sweep drags along a single haplotype.

Skewed site frequency spectrum (excess of rare derived alleles) — After the sweep, new mutations start accumulating around the fixed haplotype, so you see many low-frequency variants.

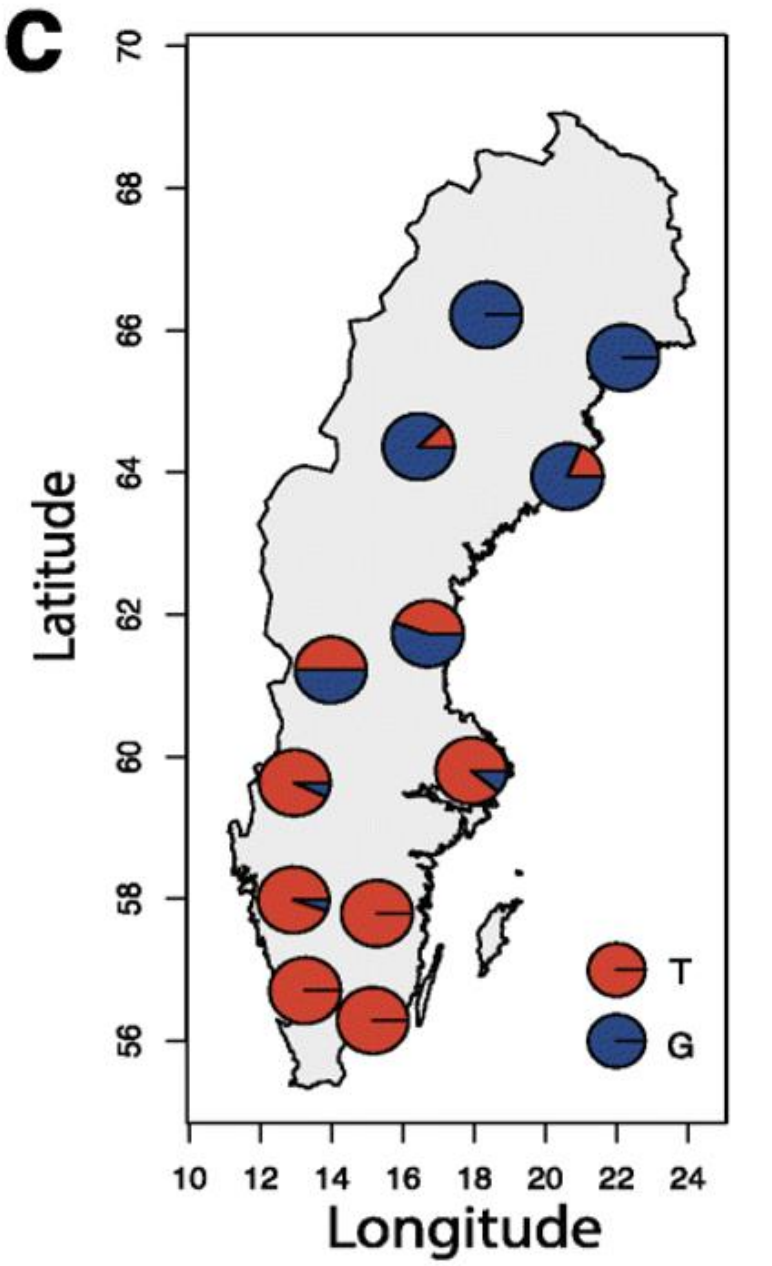
High linkage disequilibrium (LD), extended haplotype homozygosity (EHH) — Long stretches of identical haplotypes (i.e. reduced recombination) around the selected locus, because recombination hasn’t had time to break things up.

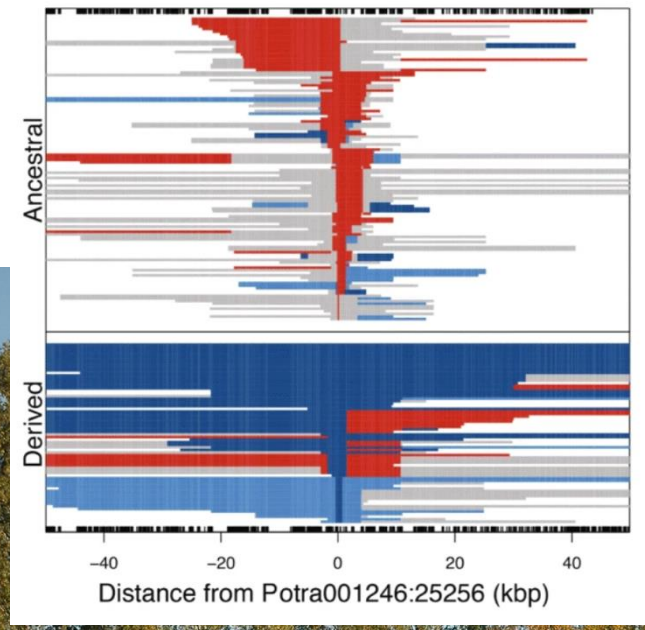
High differentiation (e.g. F_{ST}) between populations — If the selective sweep is specific to the domesticated population (or to a sub-population), that region may show unusually high differentiation compared to neutral loci.

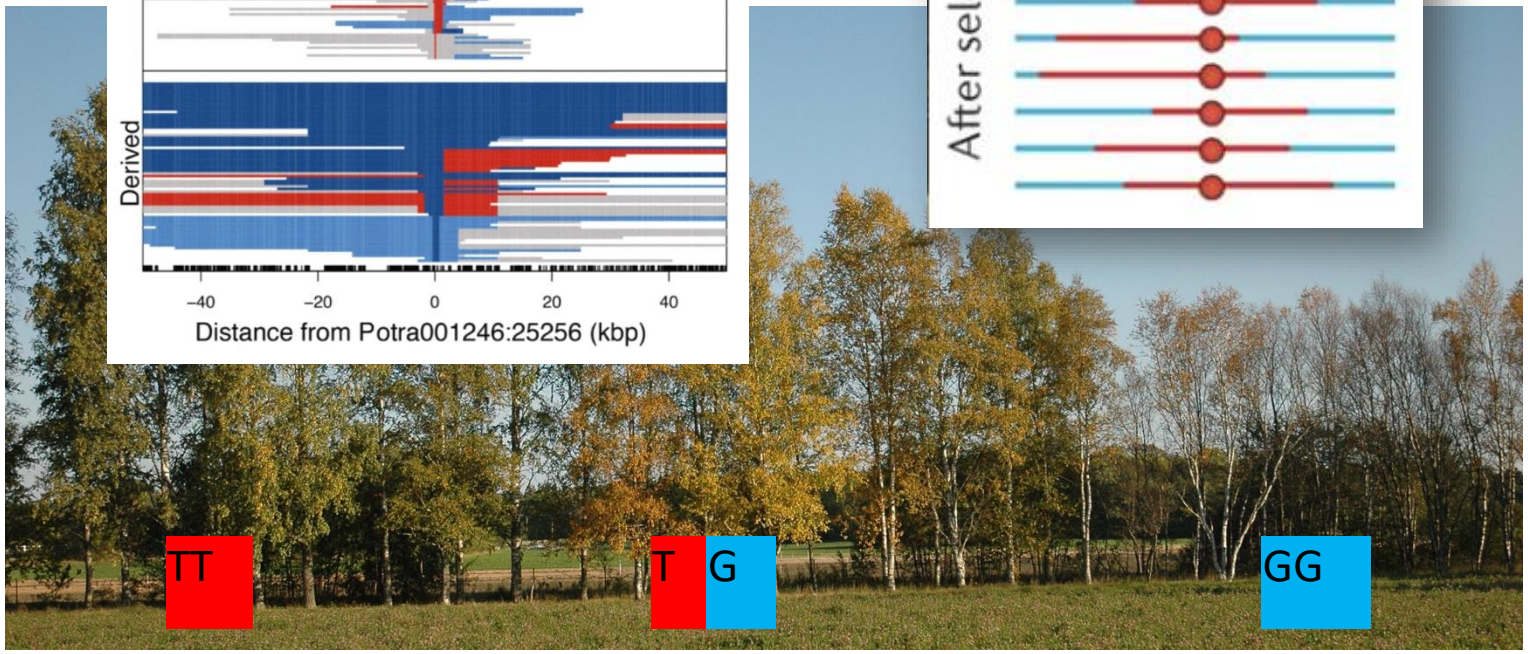
Excess of high-frequency derived alleles (e.g. Fay & Wu’s H or other tests) — Because the derived (selected) allele just rose to high frequency, there will be patterns detectable by statistics that compare derived allele frequency versus neutrality.

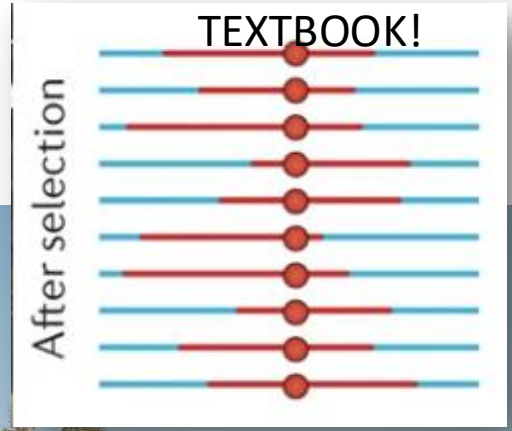


Evidence of natural selection

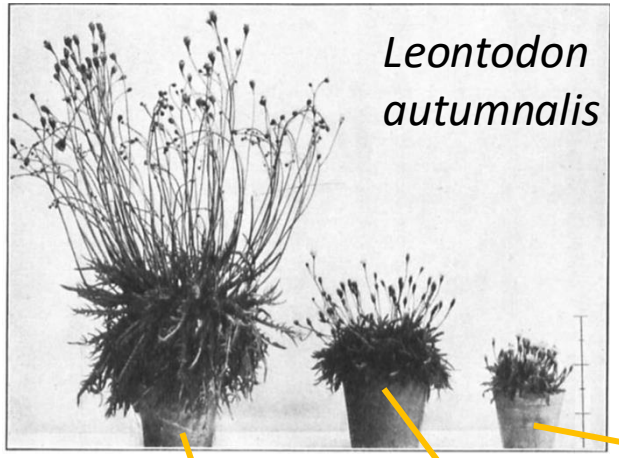


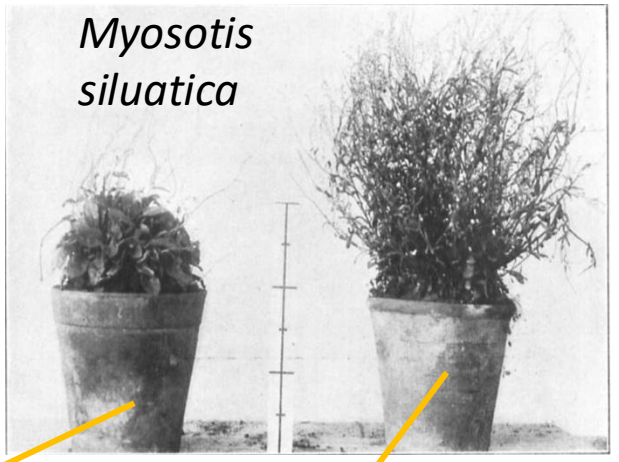


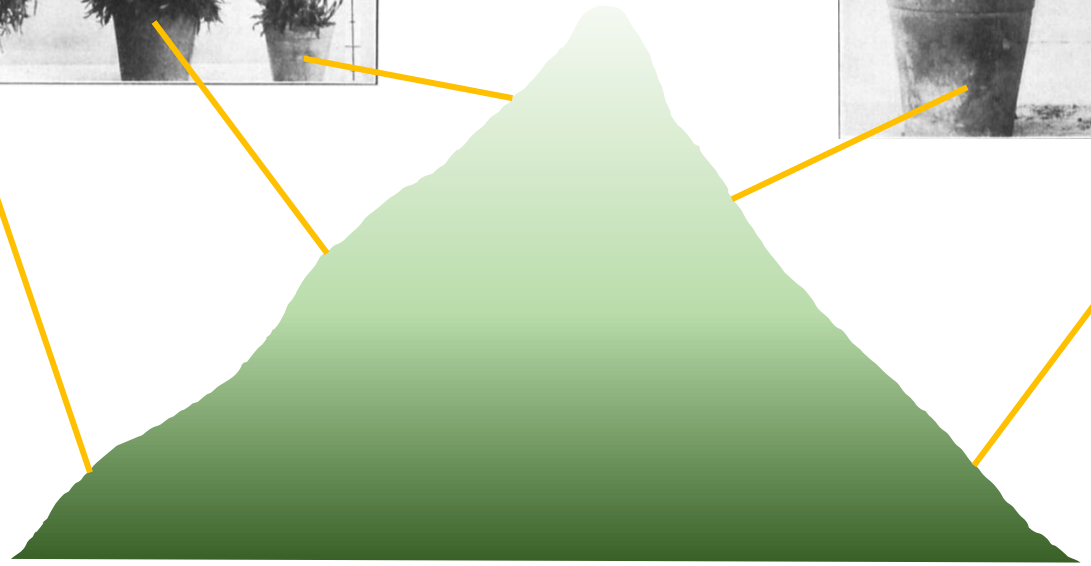




Turesson's common garden experiments reveal genetic variation in relation to climate

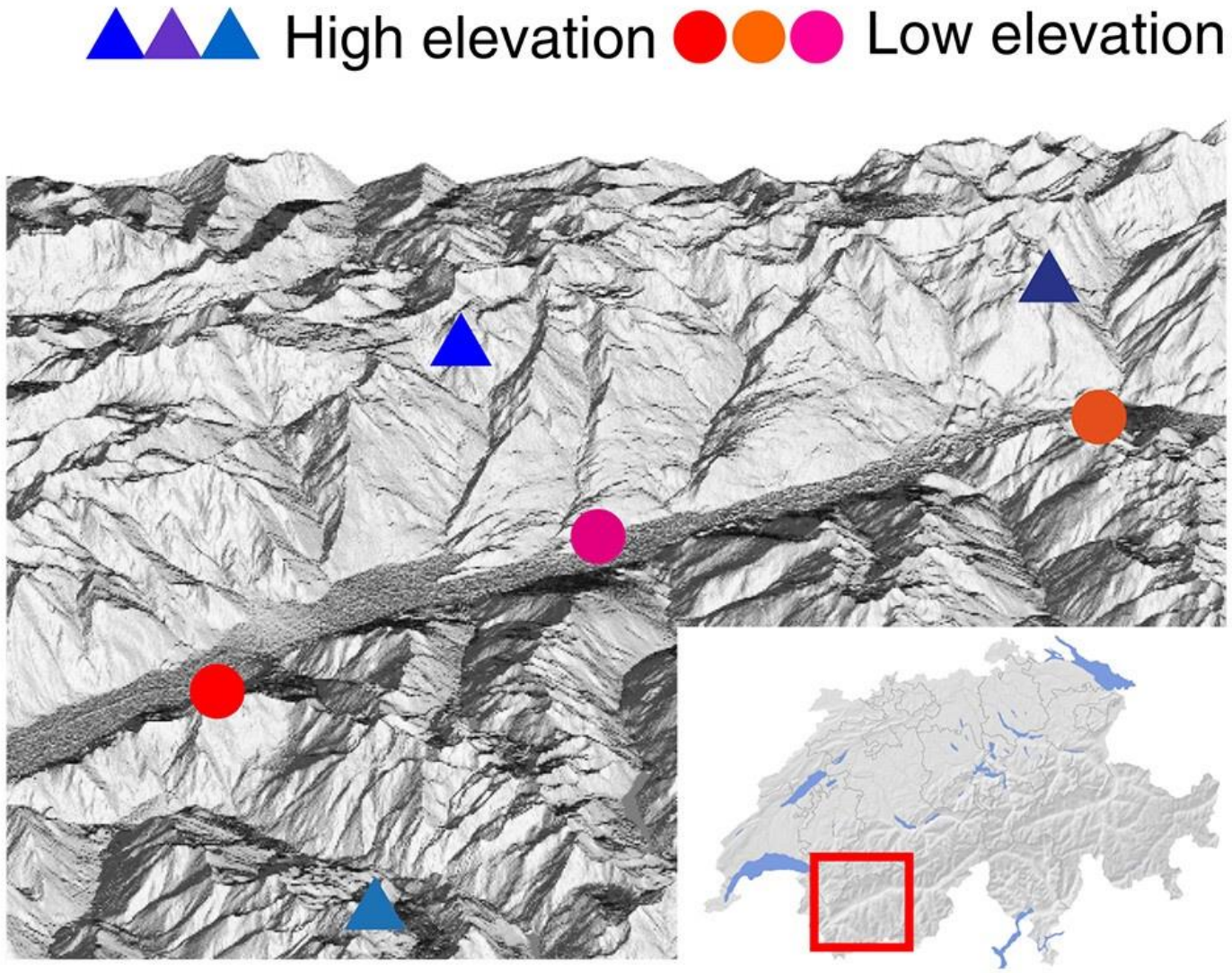






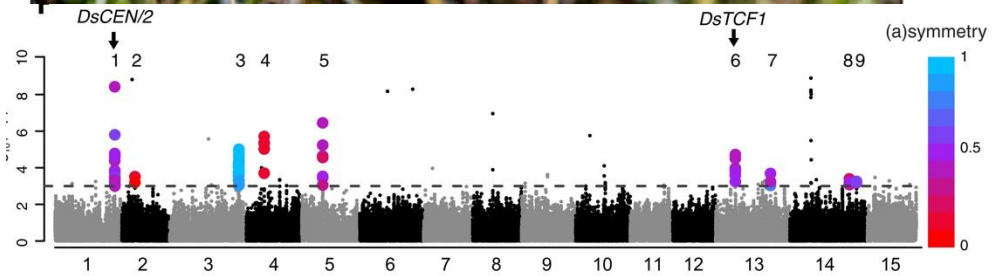
Genetic variation!





Dianthus sylvestris





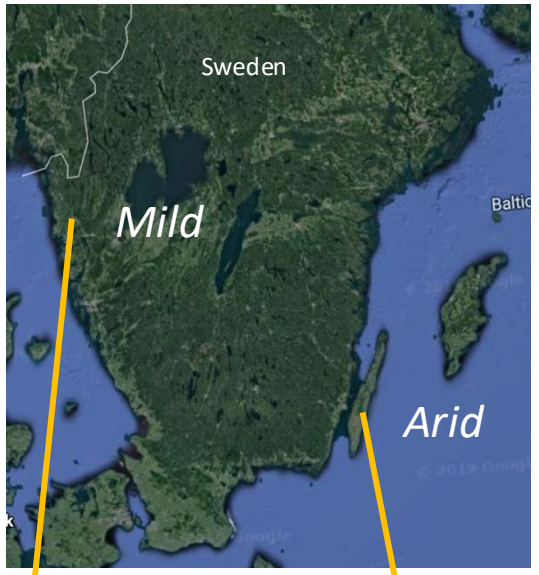
Turesson's common garden experiments reveal genetic variation in relation to climate

Silene maritima





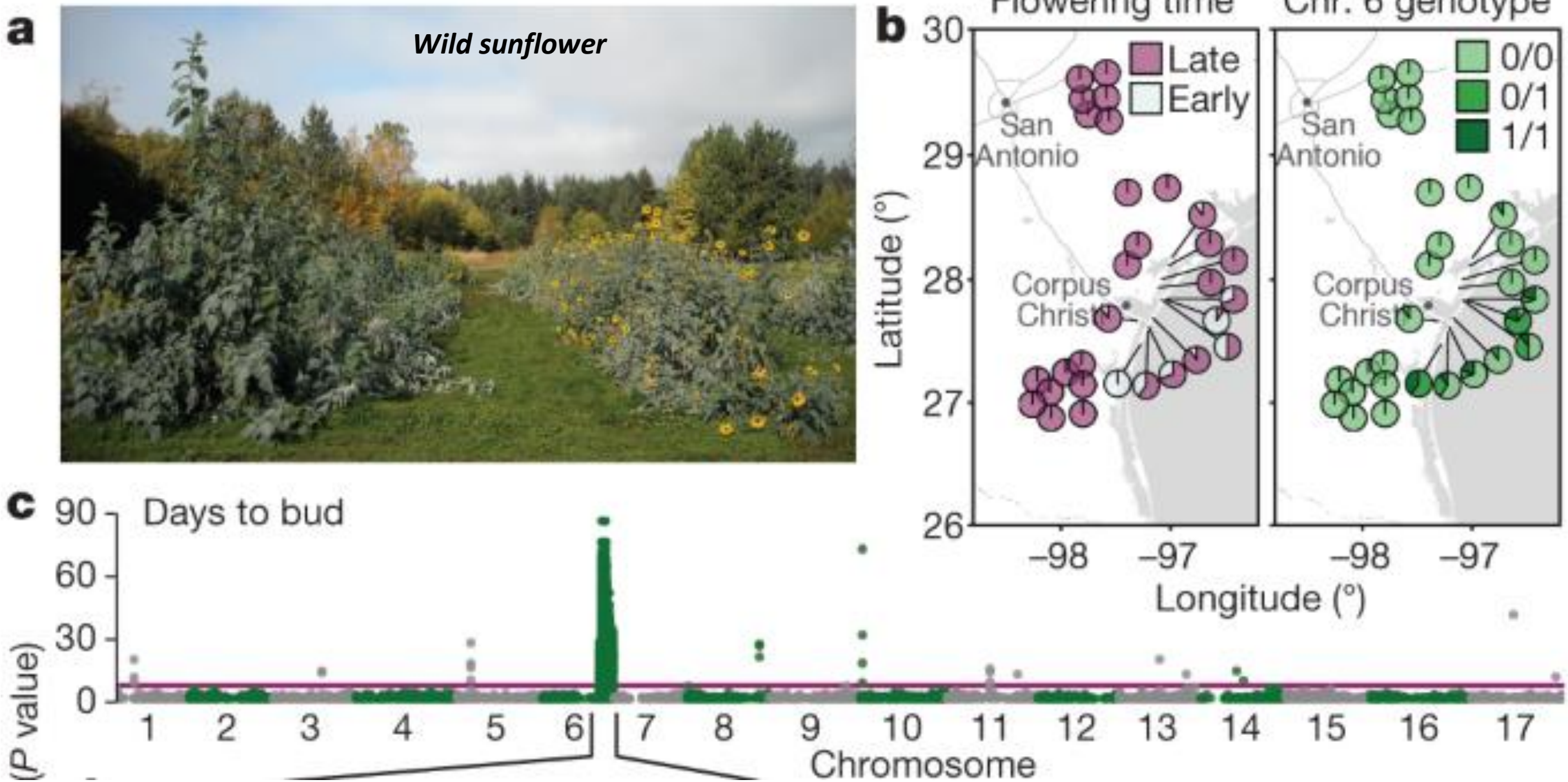
Allitum schoenoprasum





Genetic variation!




a
Wild sunflower
b
Flowering time
Chr. 6 genotype
Late
Early
0/0
0/1
1/1
San Antonio
Corpus Christi
Latitude (°)
30
29
28
27
26
–98
–97
Longitude (°)
c
Days to bud
90
60
30
0
(P value)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
Chromosome

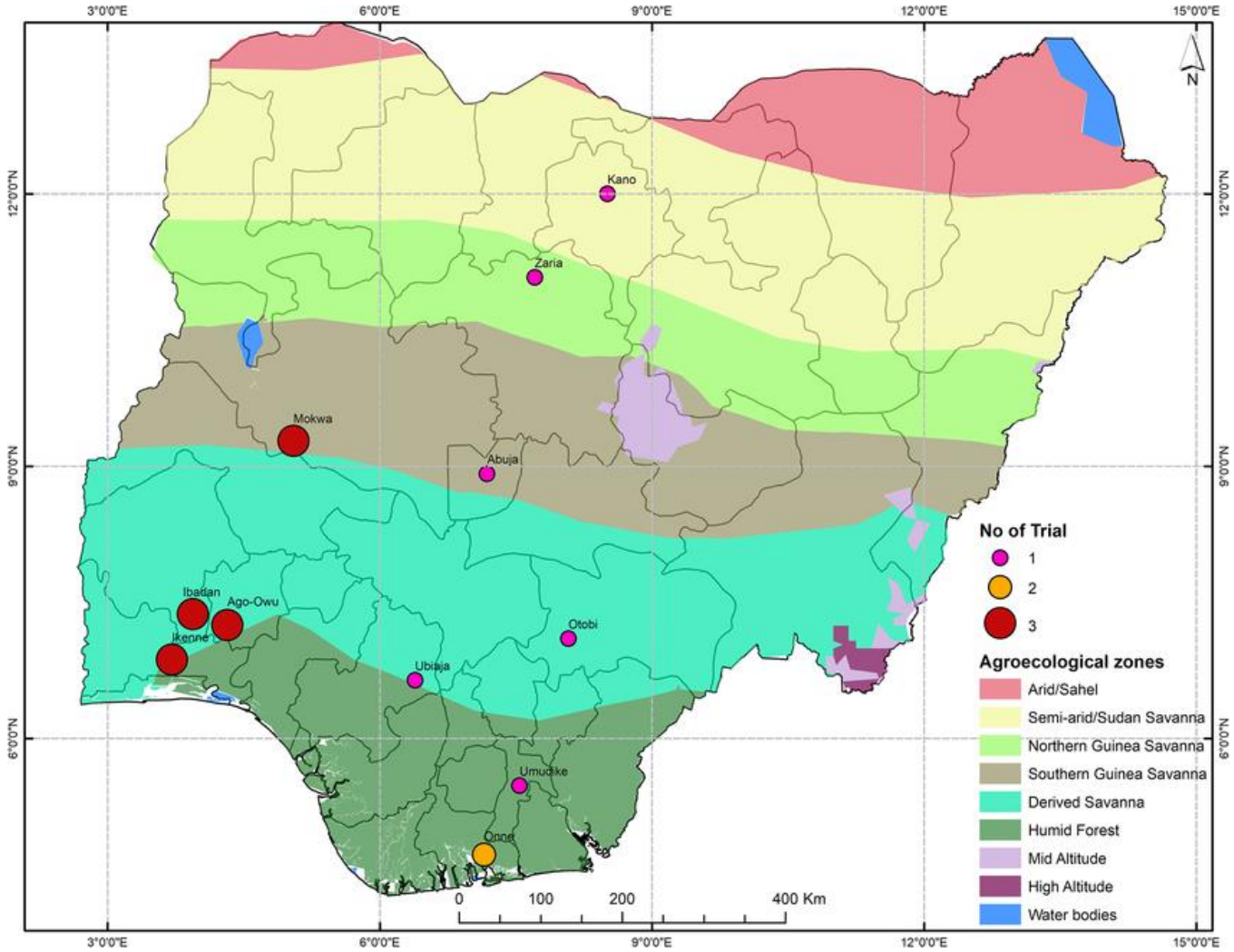

Genetic variation!
Photo by Jon Ågren


Adaptation!



Plant breeding for climate adaptation

Performance trials in diverse Environments

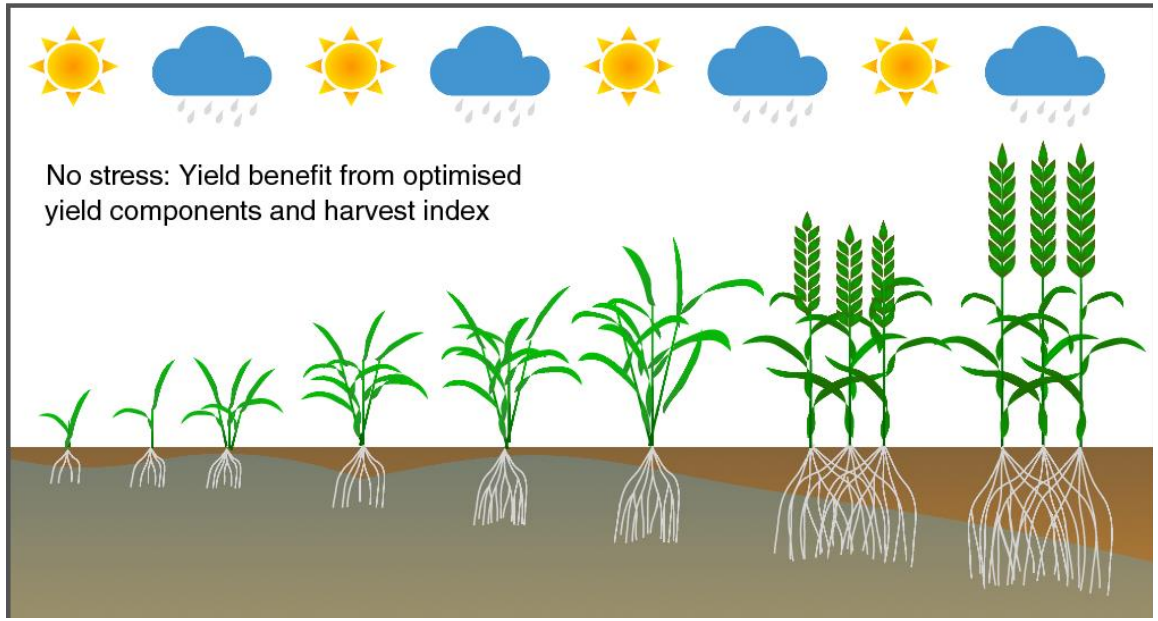


Cassava



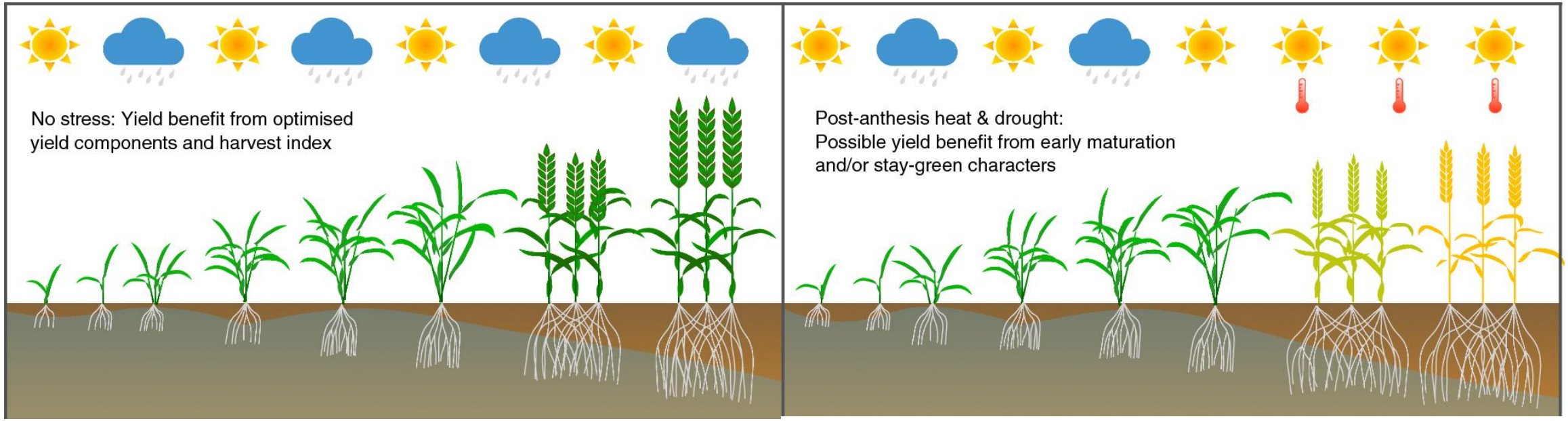
Bakare et al. 2022

Optimum traits depend on specifics about the environment



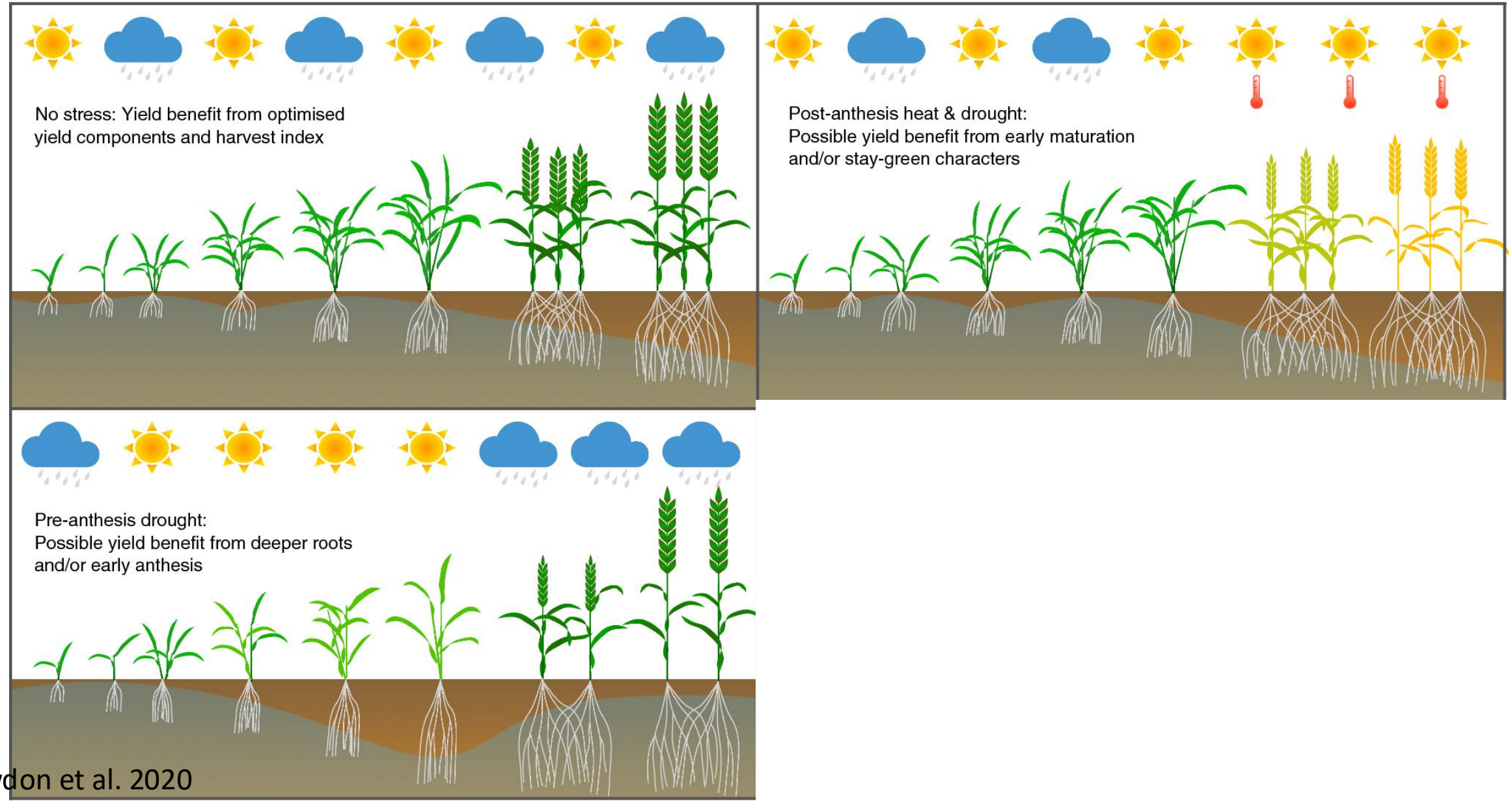
Snowdon et al. 2020

Optimum traits depend on specifics about the environment



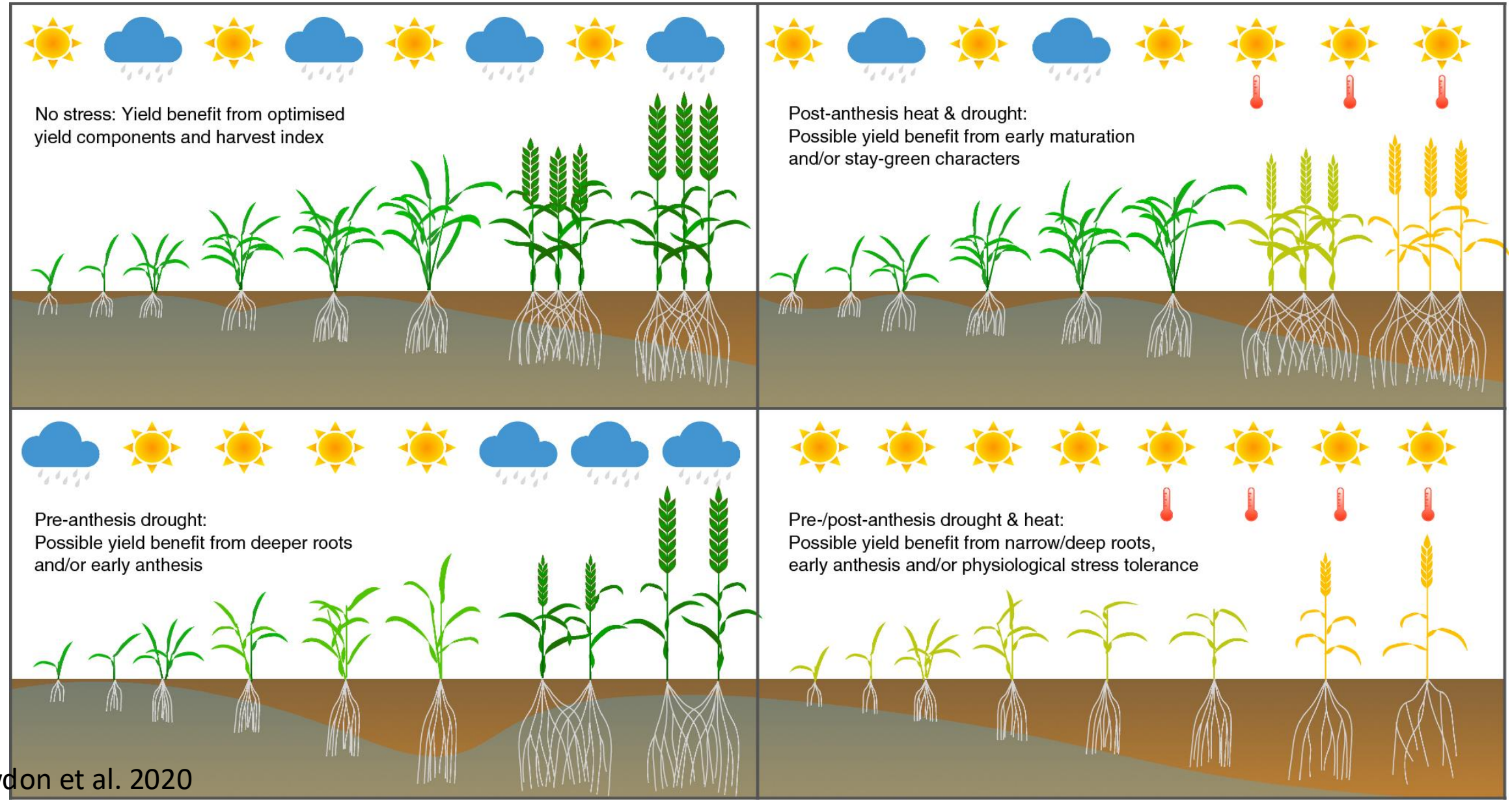
Snowdon et al. 2020

Optimum traits depend on specifics about the environment



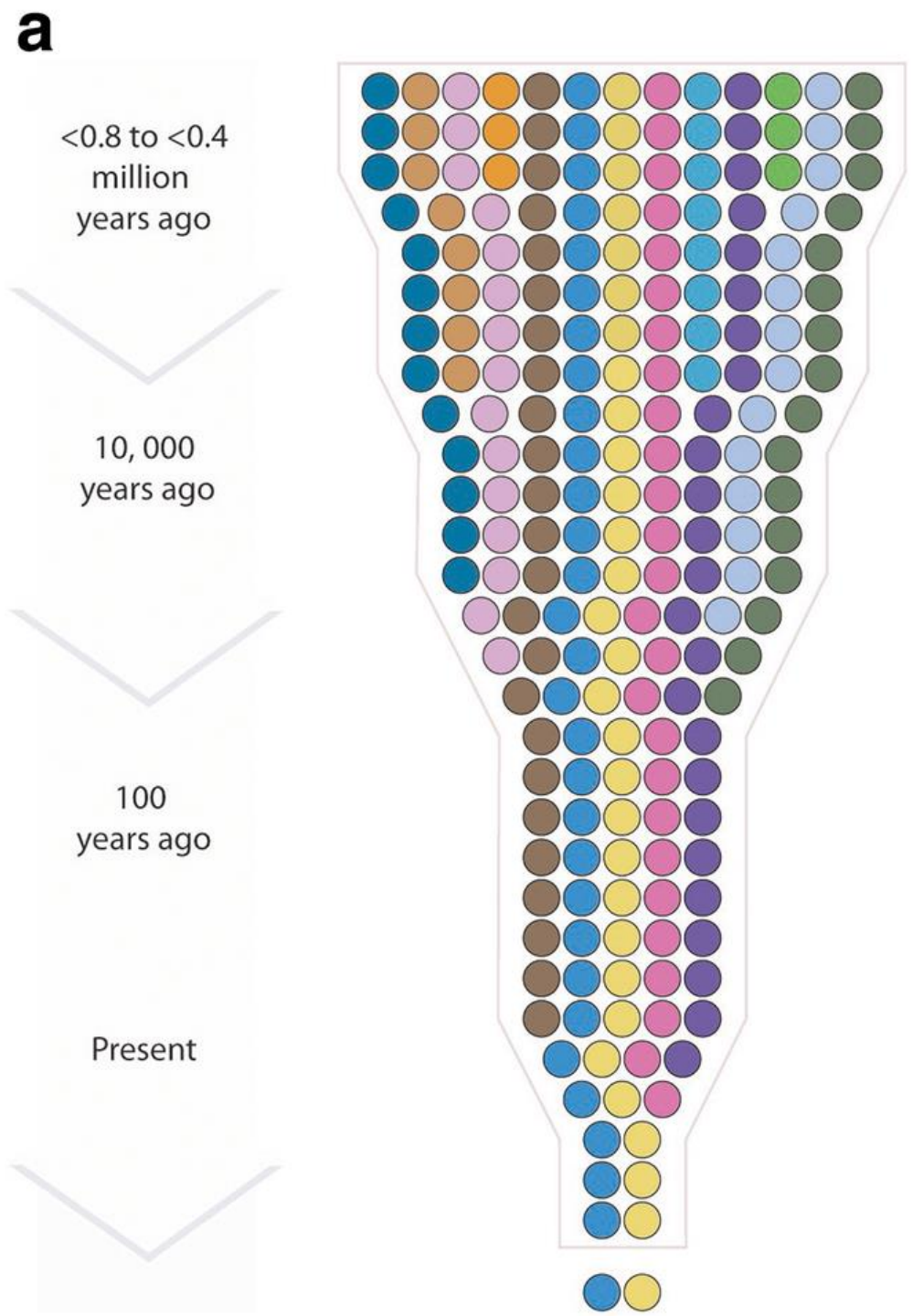
Snowdon et al. 2020

Optimum traits depend on specifics about the environment



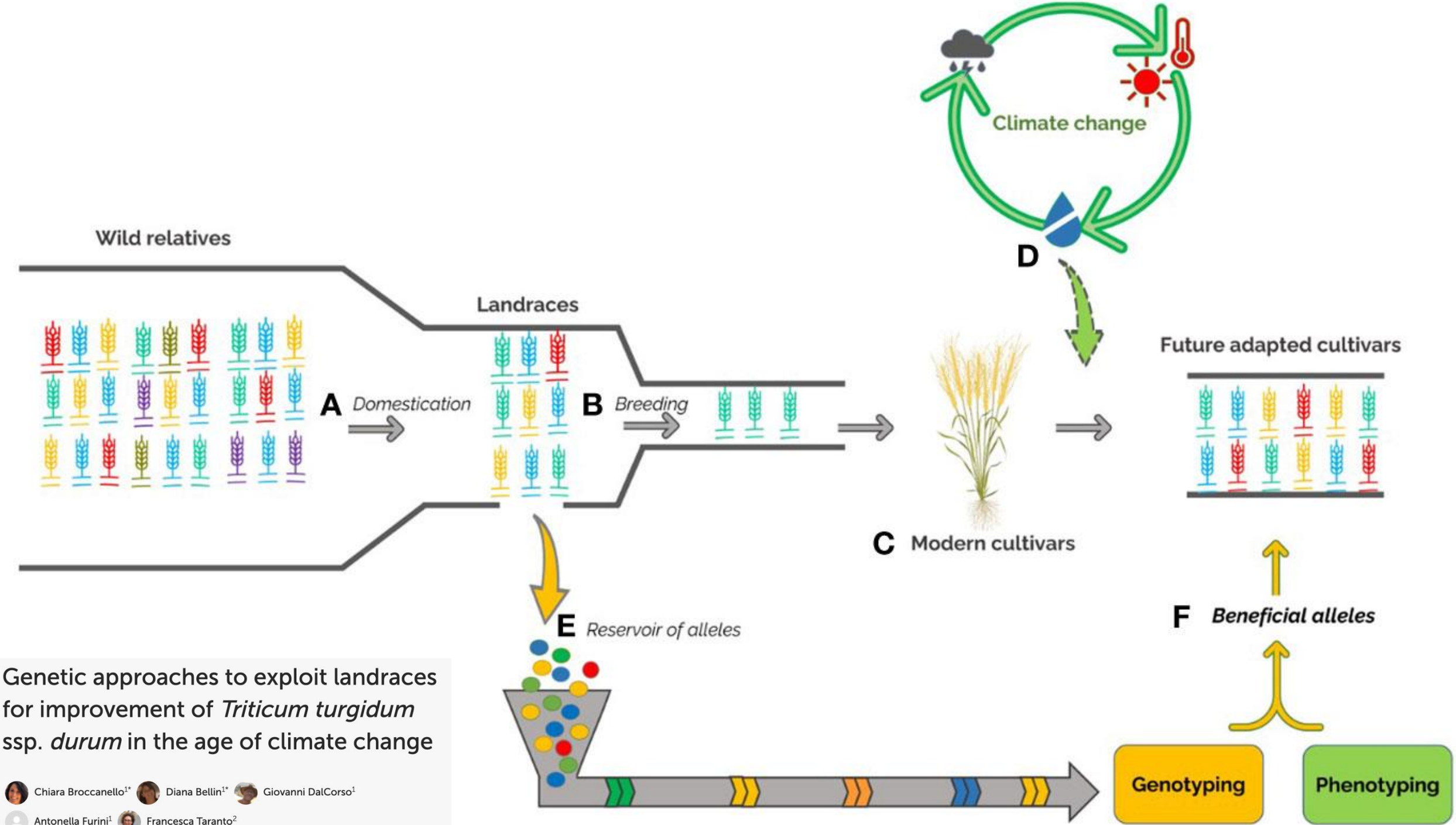
Snowdon et al. 2020

Genetic diversity





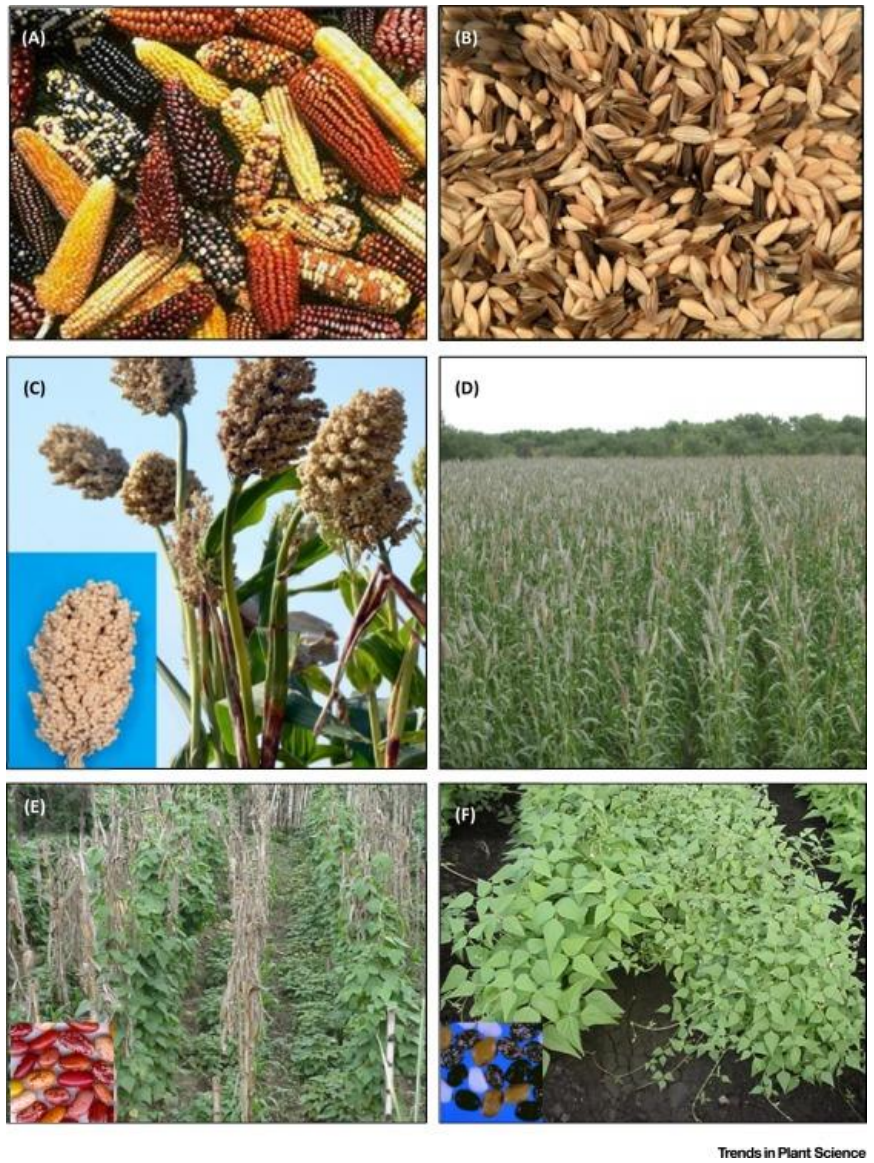
AAAS



Genetic approaches to exploit landraces for improvement of *Triticum turgidum* ssp. *durum* in the age of climate change

Chiara Broccanello[1*] Diana Bellin[1*] Giovanni DalCorso[1]
Antonella Furini[1] Francesca Taranto[2]

Using crop landraces and wild relatives to accelerate climate adaptation breeding



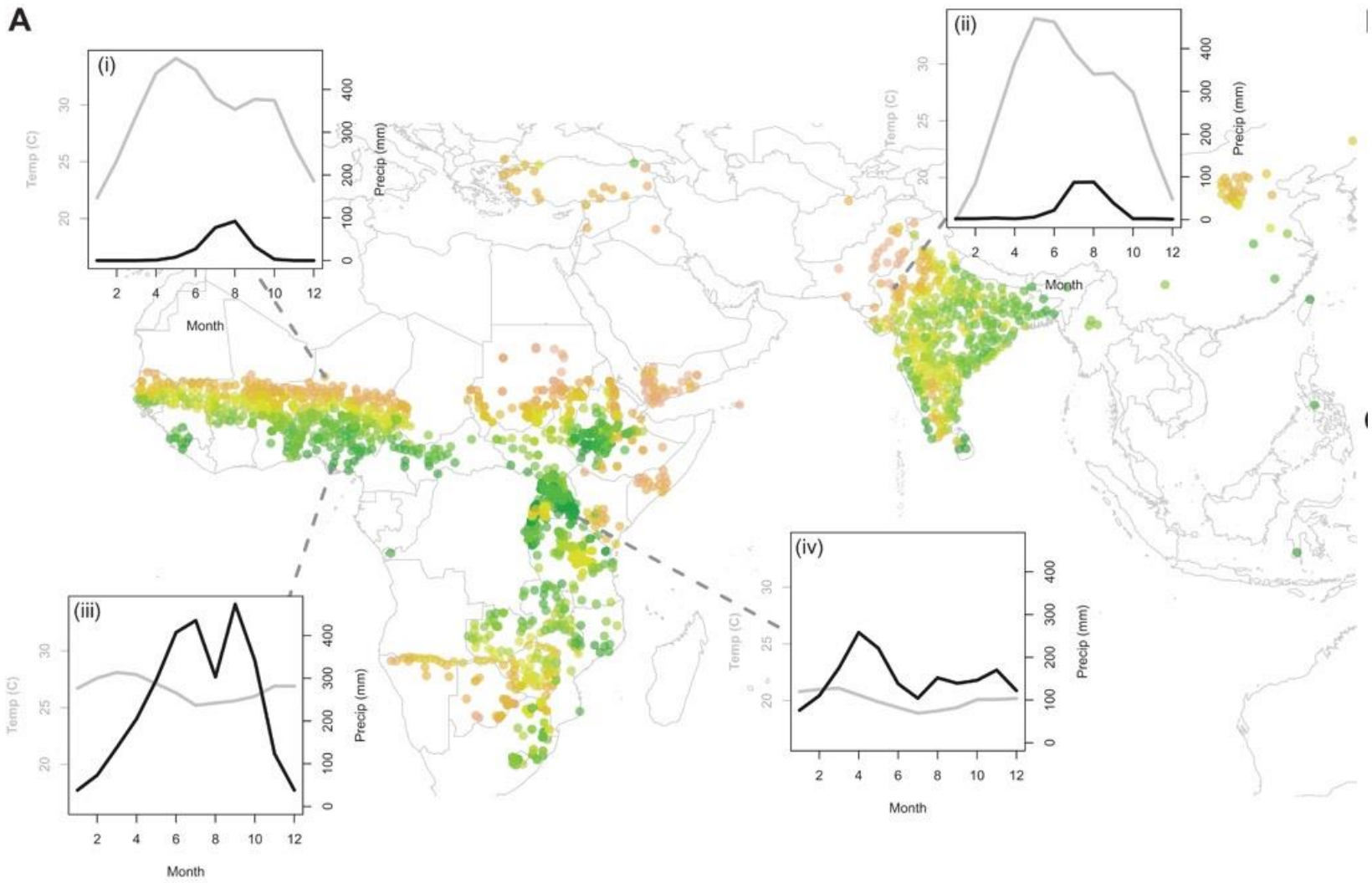
- Landrace crops are diverse plant populations native to specific regions.
- Chosen by local farmers for their resilience.
- Adapted to local climate and soil conditions.
- More genetic diversity than modern cultivars.

Landrace Germplasm for Improving Yield and Abiotic Stress Adaptation

Sangam L. Dwivedi • Salvatore Ceccarelli • Matthew W. Blair • Hari D. Upadhyaya • Ashok K. Are • Rodomiro Ortiz

Sorghum case study: Landraces come from diverse climates



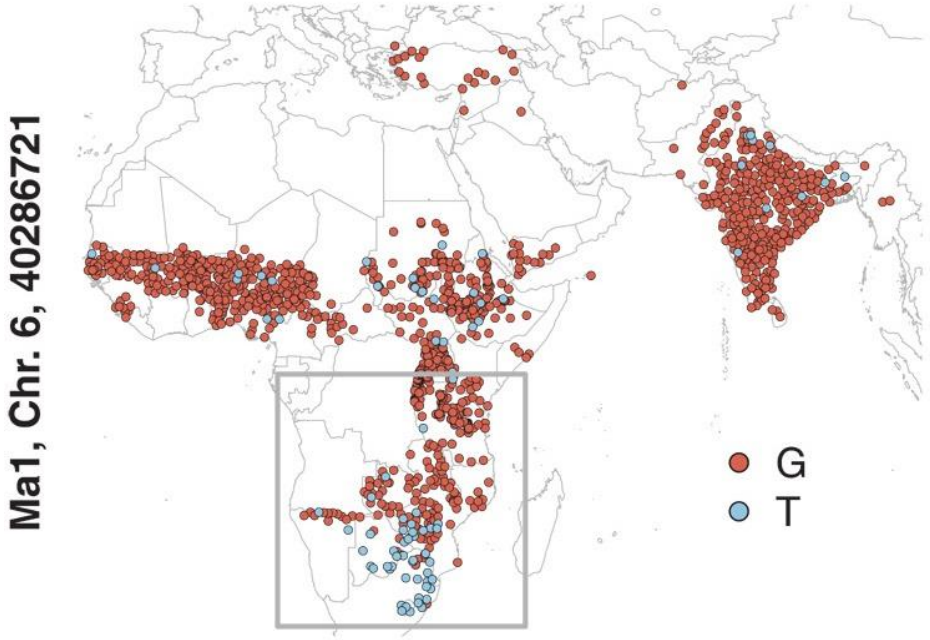


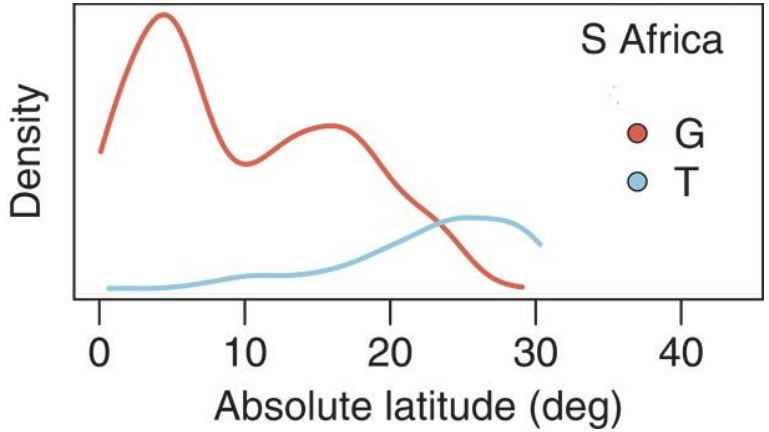
Genome-environment associations in sorghum landraces predict adaptive traits

JESSE R. LASKY, HARI D. UPADHYAYA, PUNNA RAMU, SANTOSH DESHPANDE, C. TOM HASH, JASON BONNETTE, THOMAS E. JUENGER, KATIE HYMA, CHARLOTTE ACHARYA, [...], AND GEOFFREY P. MORRIS +4 authors Authors Info & Affiliations

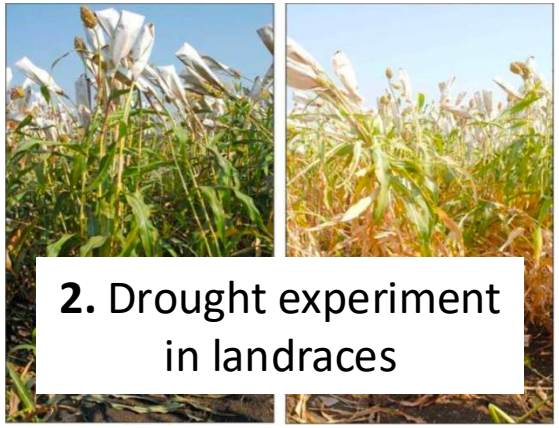
Sorghum case study:

Adaptive alleles are correlated to climate and predict stress tolerance



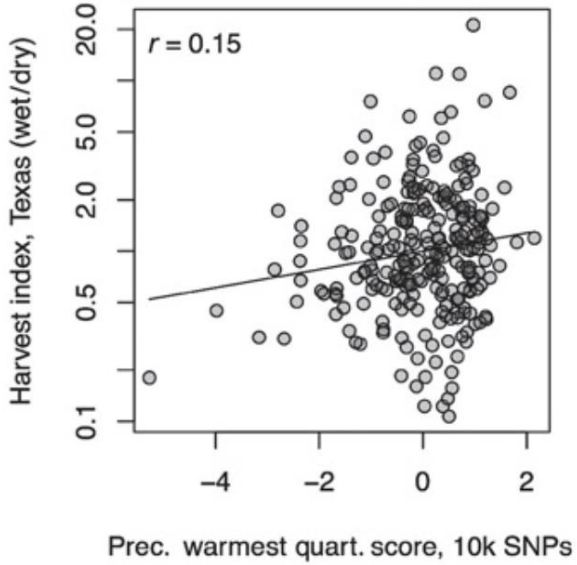


1. Identify alleles associated with wet vs dry climates



2. Drought experiment in landraces

3. Test whether those alleles predict performance under drought

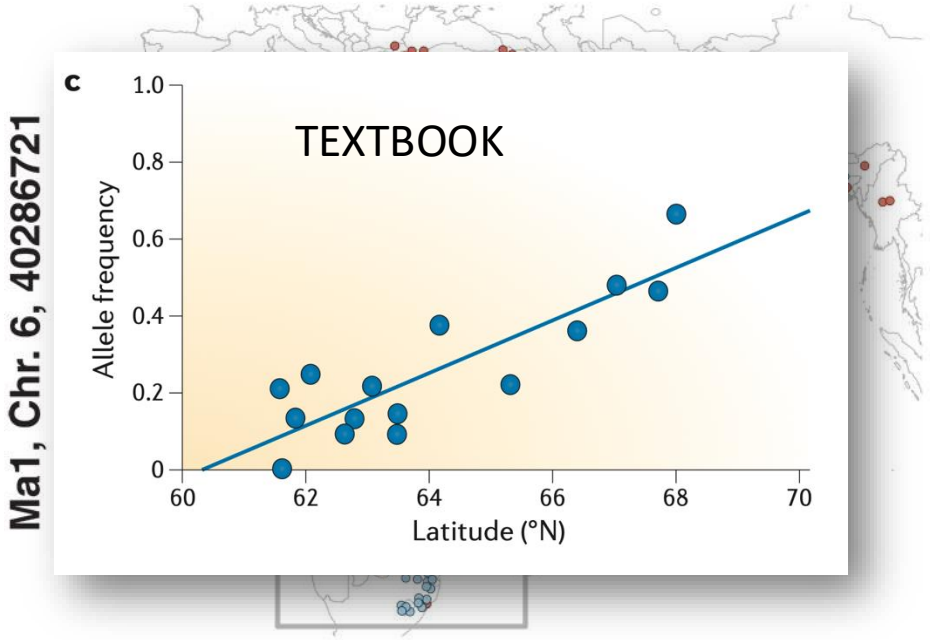


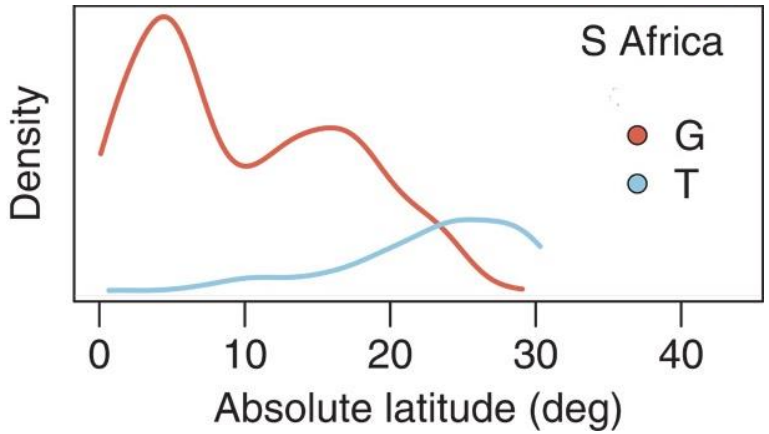
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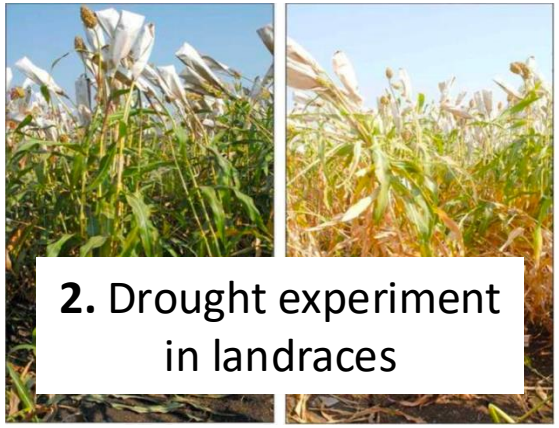
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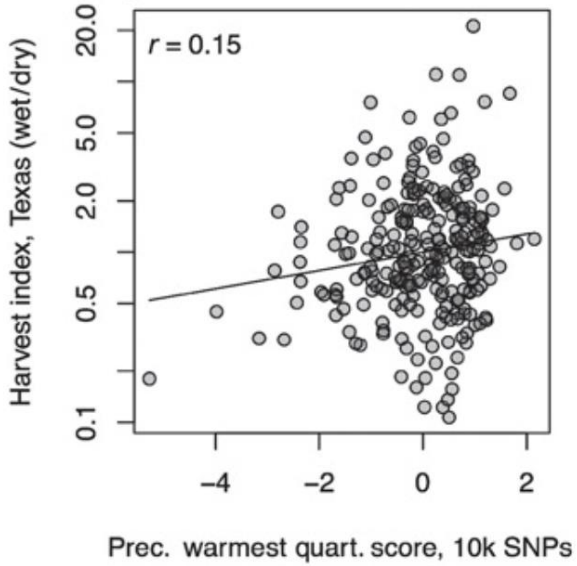


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Genome-environment associations in sorghum landraces predict adaptive traits

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Leveraging Cassava landrace genomics to identify climate-adaptive alleles for breeding





The partnership:

International team bridging genomic, climate, and breeding scientists

UC DAVIS
UNIVERSITY OF CALIFORNIA

CIAT
International Center for Tropical Agriculture
Since 1967 Science to cultivate change

UC DAVIS
DEPARTMENT OF PLANT SCIENCES

Department of
LAND, AIR AND WATER RESOURCES
University of California, Davis
Climate Change • Sustainable Agriculture
Environmental Quality • Landscape Processes

UC DAVIS
GENOME CENTER

UC DAVIS
Climate Adaptation Research Center

CGIAR
Genebank
Platform

Bioversity
International

CIAT
TRANSFORMATION
PLATFORM

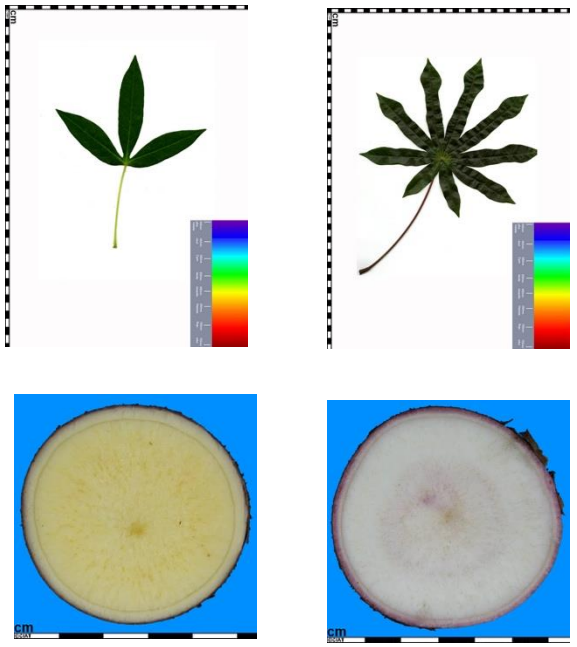
Locally adapted cassava (landraces) in CIAT Gene Bank

CIAT GeneBank

Cassava accessions

Locally adapted landraces from diverse climates in Colombia







The opportunity:

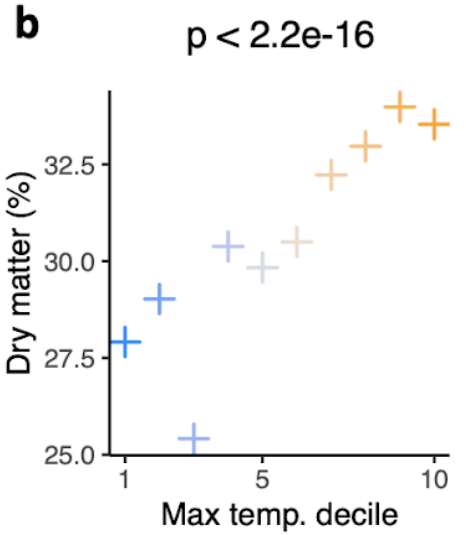
Locally adapted cassava (landraces) in CIAT Gene Bank

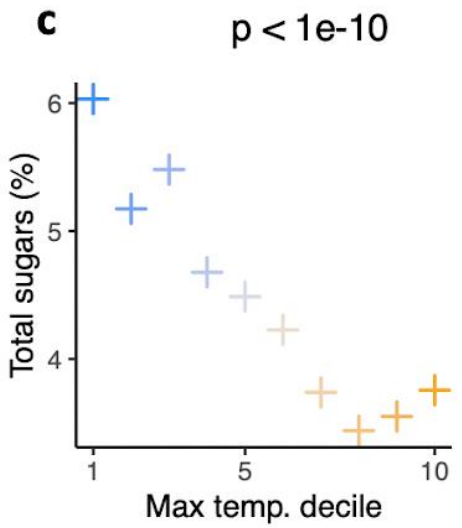
CIAT GeneBank

Cassava accessions

Locally adapted landraces from diverse climates in Colombia

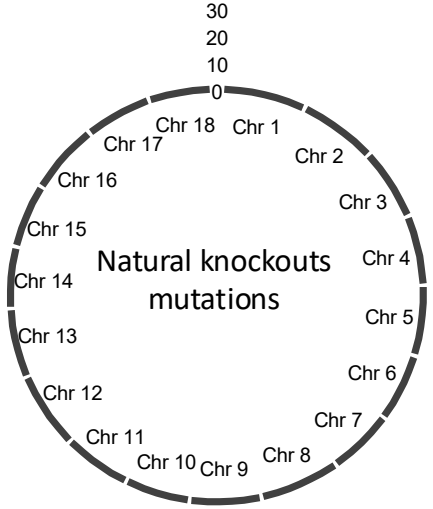


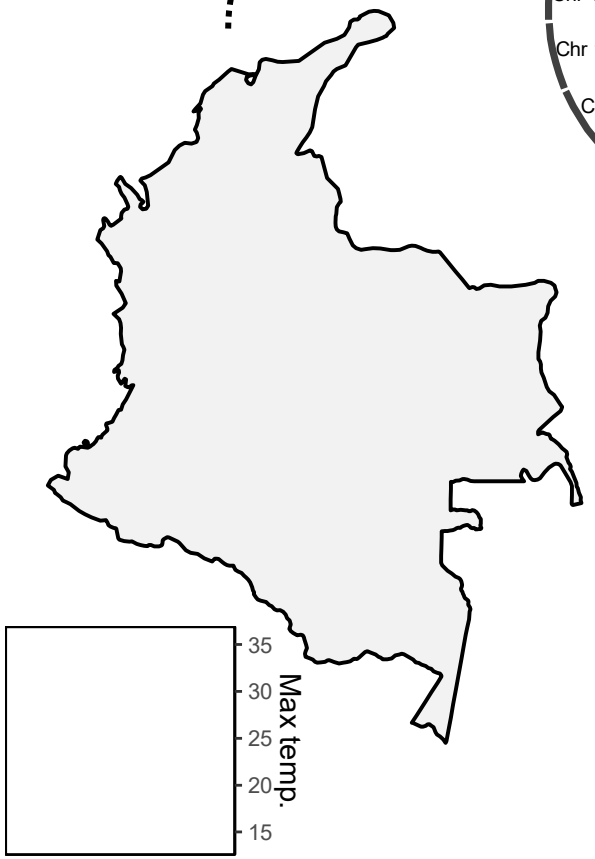




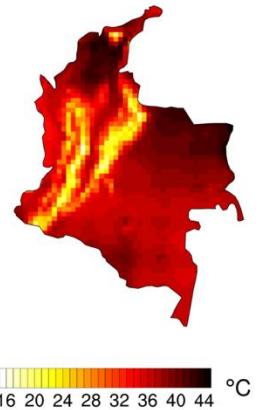
Leverage landrace genomics to engineer thermotolerant cassava

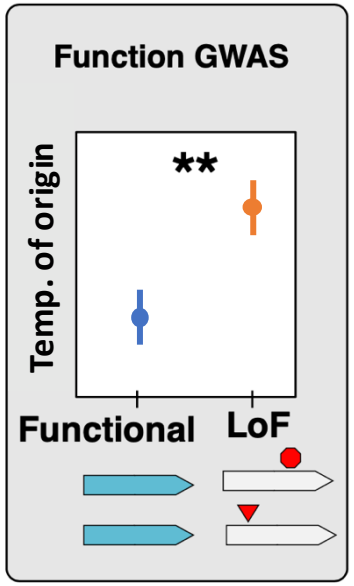
1 **Hierarchical whole genome sequencing**



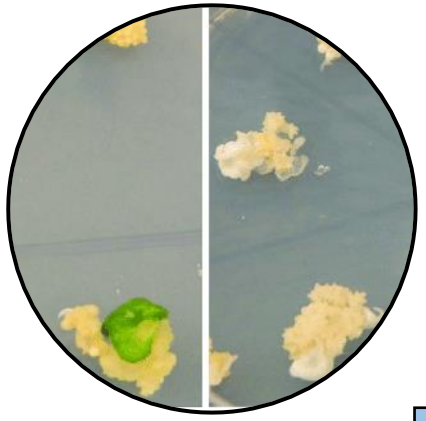


2 **Climate modeling**

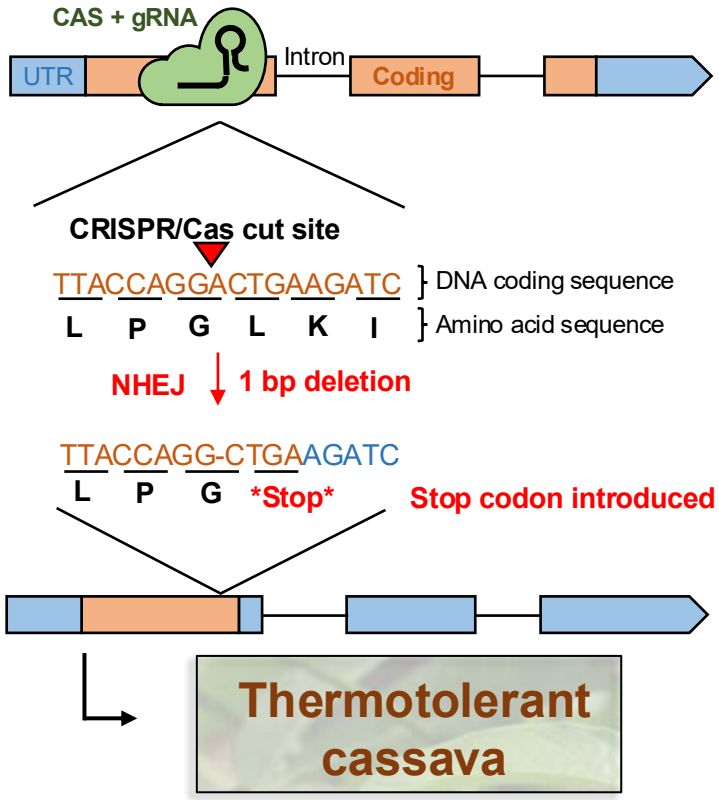




3 **Identify adaptive knockouts**



4 **Validation of candidates with CRISPR**

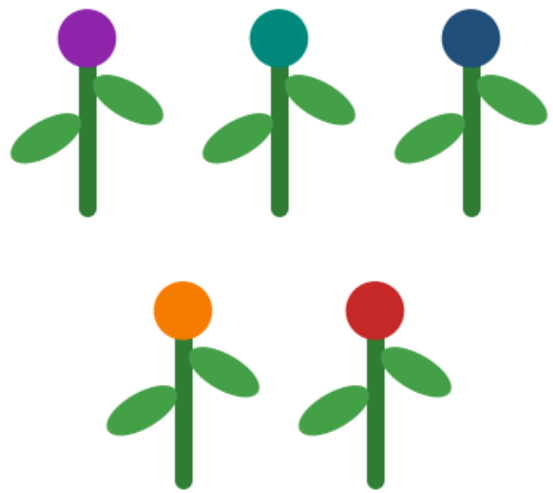


Find your final project group

1. **Look up your group**
   Canvas page: Final groups

2. **Sit together**
   move now, bring your stuff

3. **Swap contact info**
   your group's Canvas discussion



Trial & Release

Your group is a seed company with 6 secret hybrids. (a simulation)



dots = farmers

1. Plant 12 trial plots: any hybrid, any region. One plot can lie.
2. Release one hybrid. Its trial data gets published.
3. Farmers buy the best data. Local, replicated data counts most. Crowded regions get spl
4. Then the real harvest. Promised 2+ t/ha more than it yields? Those farmers get refunds
5. Everyone plants at once. Guess where rivals go. 3 rounds, then the truth.

One phone per group. QR on the next slide.

Trial & Release

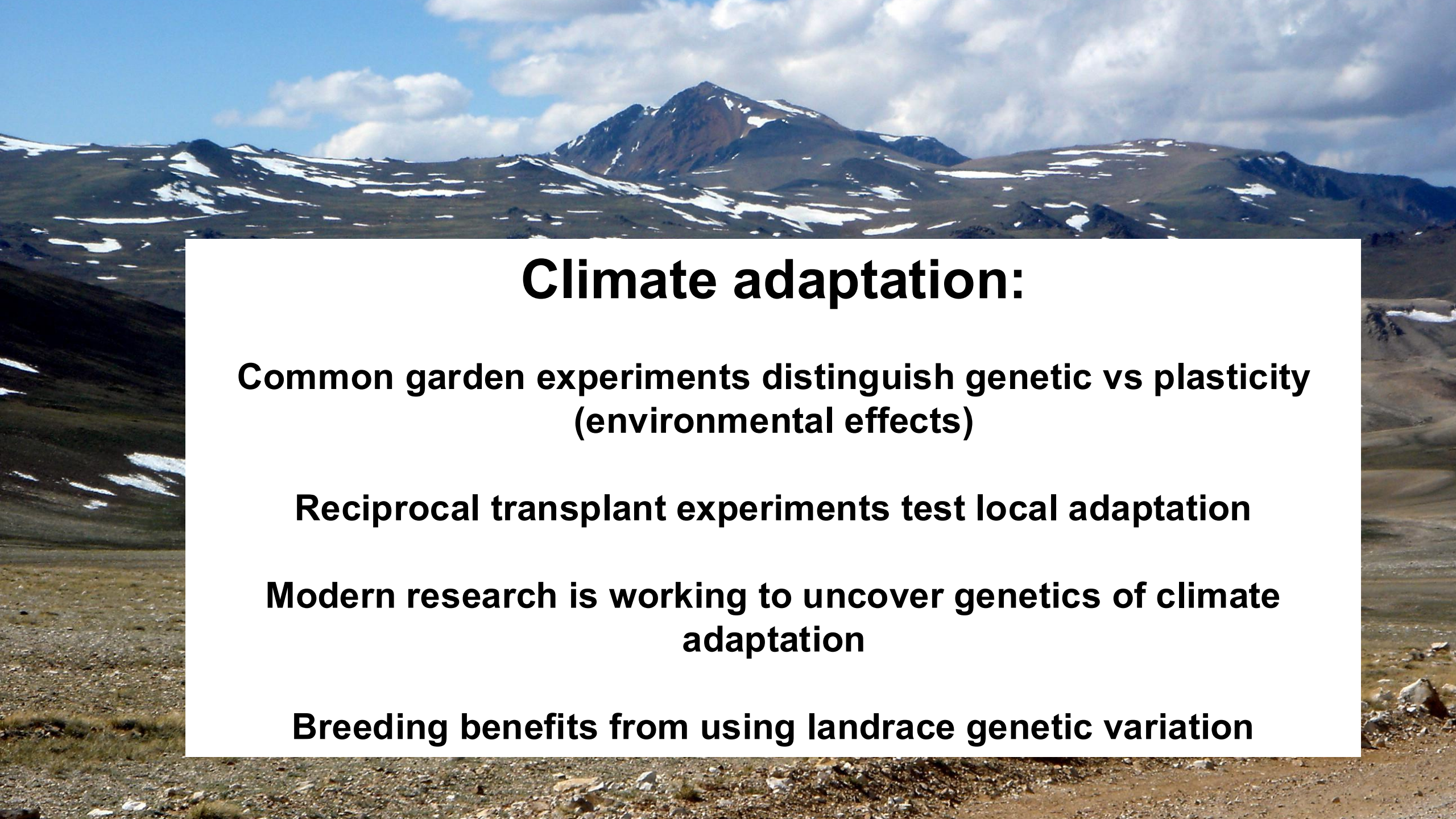
Live game: one phone per group

Scan, tap your group number.
Test. Release. Publish.
Win the farmers.

Scan with your phone

Join: greymonroe.github.io/seed-wars/trial.html?g=pls152-2026-11-05

host screen



Climate adaptation:

Common garden experiments distinguish genetic vs plasticity (environmental effects)

Reciprocal transplant experiments test local adaptation

Modern research is working to uncover genetics of climate adaptation

Breeding benefits from using landrace genetic variation